

ECEN721: Optical Interconnects Circuits and Systems Spring 2026

Lecture 11: Ring Resonator Modulator Transmitters



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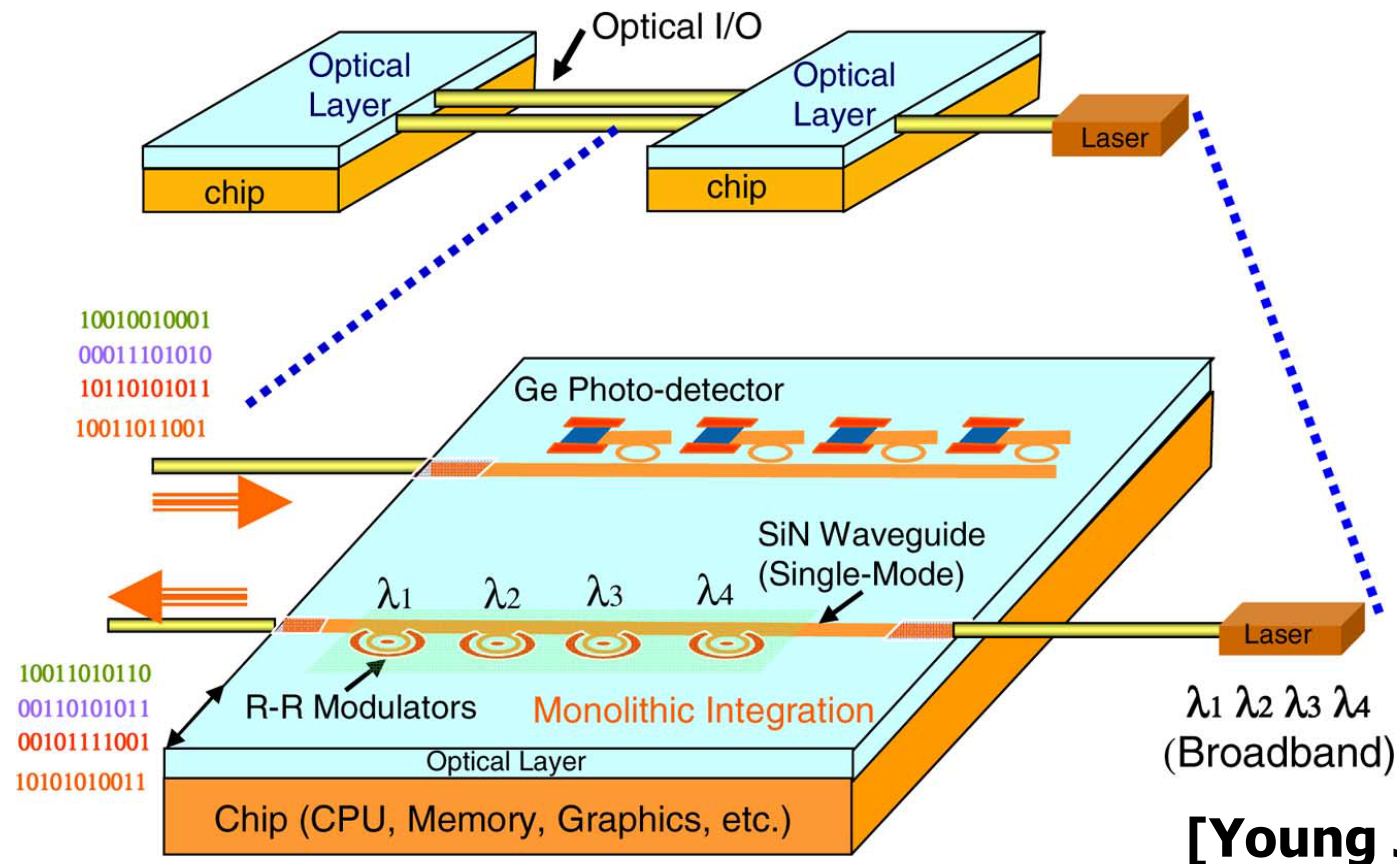
References

- Reading
 - Sackinger Chapter 8

Agenda

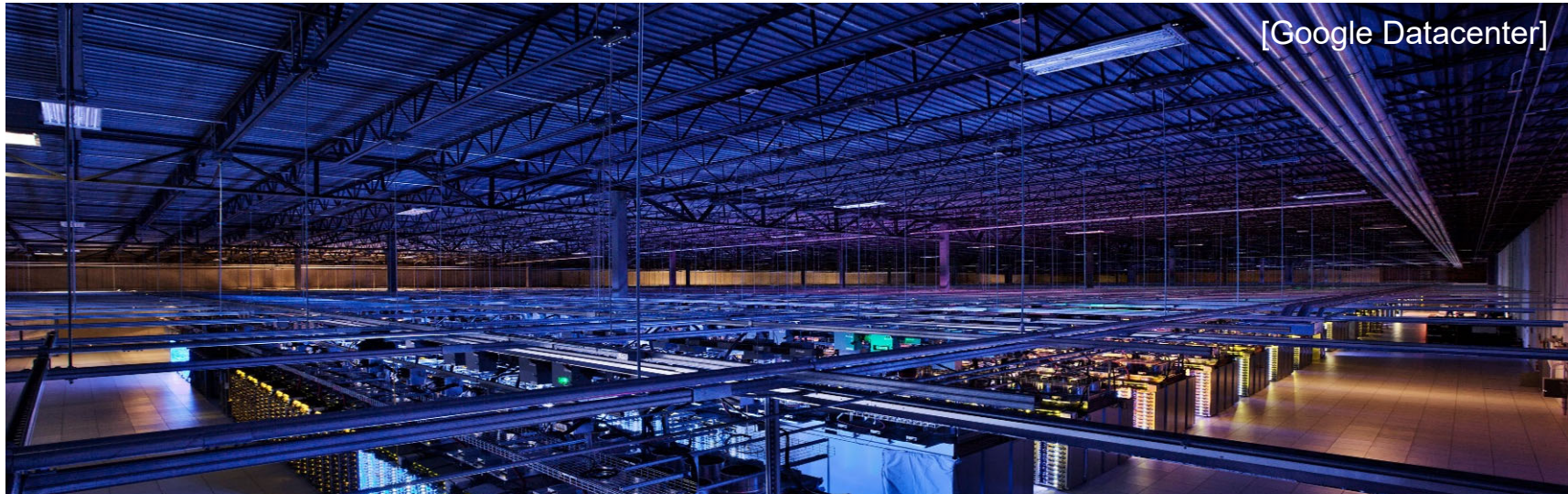
- WDM Optical Interconnect Motivation
- Silicon Photonic Modulators
- Carrier-Injection Ring Resonator Modulators
- Carrier-Depletion Ring Resonator Modulators

Wavelength-Division Multiplexing (WDM) Optical Interconnects



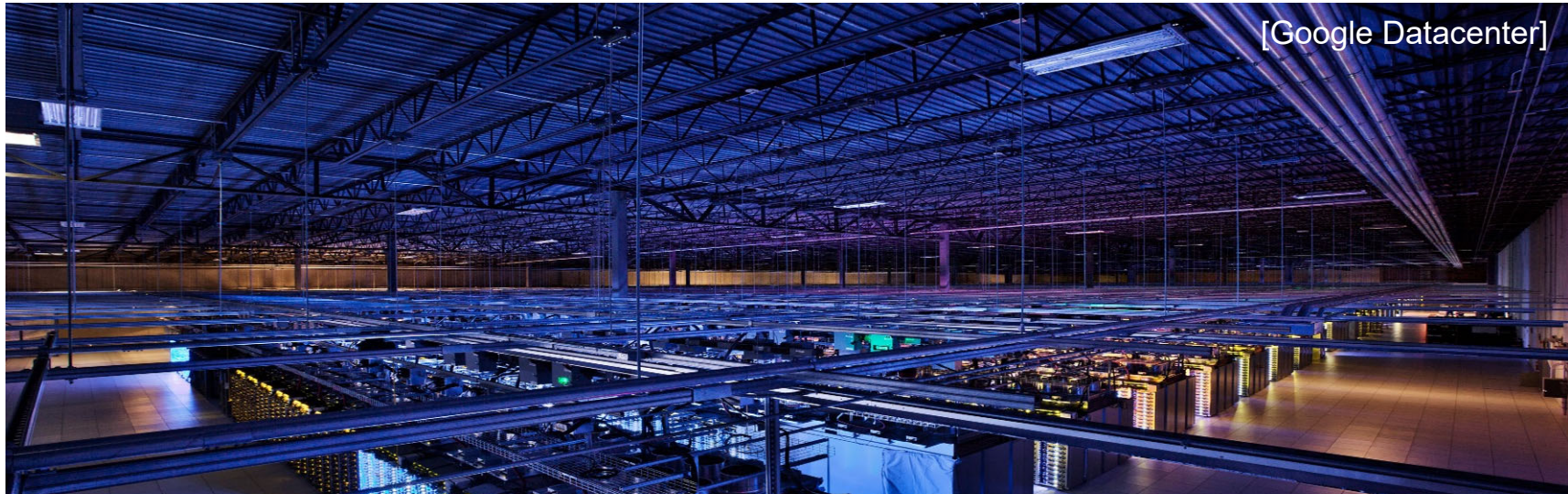
- Optical interconnects remove many channel limitations
- WDM allows for multiple high-bandwidth (10+Gb/s) signals to be packed onto one optical channel

100G Interconnects



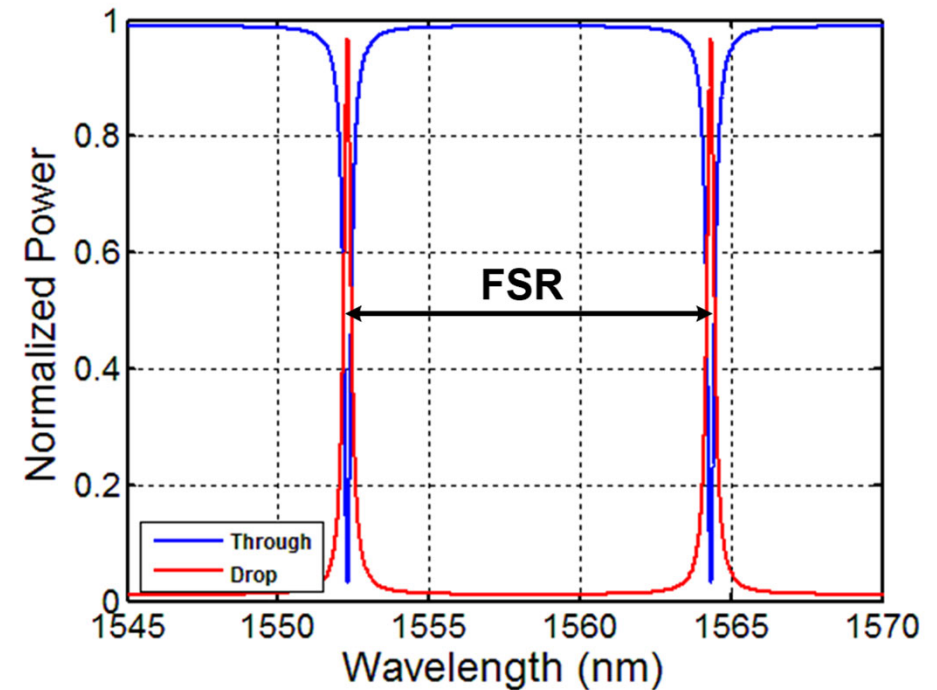
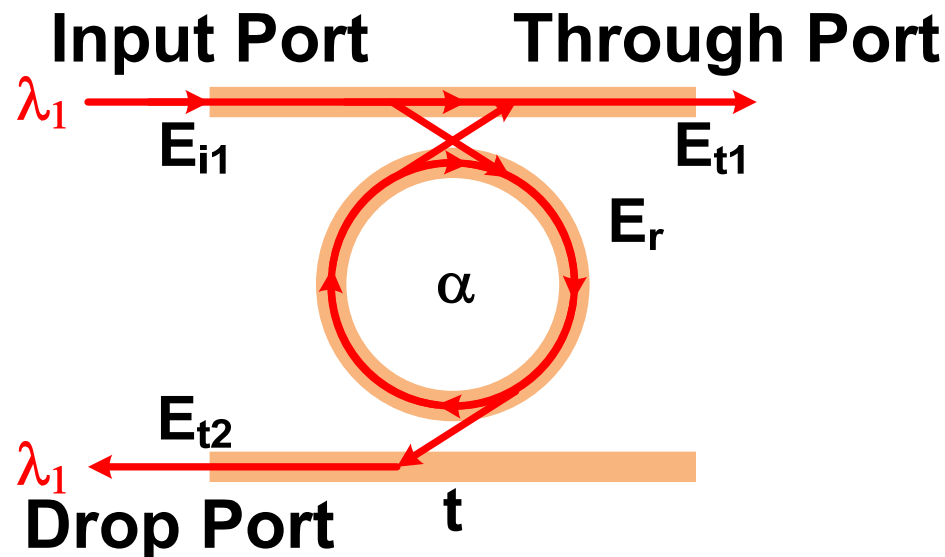
Link Distance	Transceiver Type	# of I/O Ports (25Gb/s NRZ)
0.5m	Electrical	PCB Trace × 4
1m	Electrical	Backplane × 4
10m	Electrical	Cu Cable × 4
100m	VCSEL	MMF × 4
1km	MZI / EAM	SMF × 4
> 10km	MZI / EAM	SMF × 4

100G Interconnects



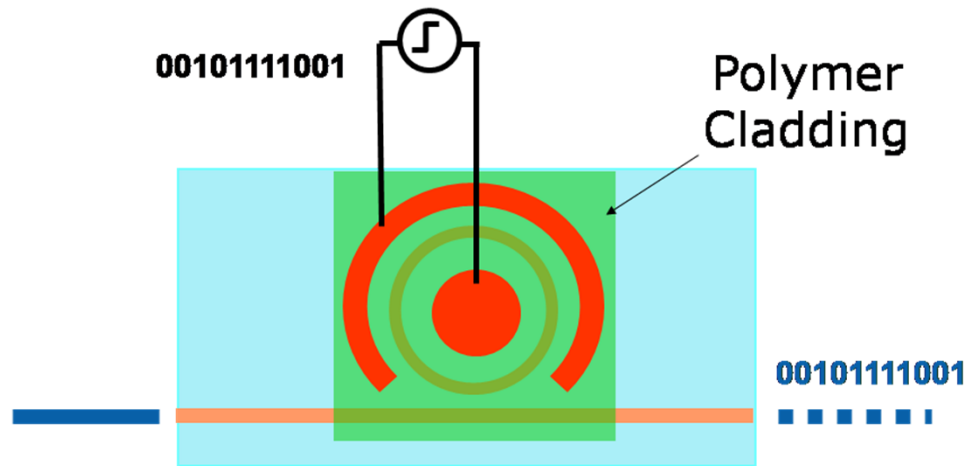
Link Distance	Transceiver Type	# of I/O Ports (25Gb/s NRZ)
0.5m	Electrical	PCB Trace × 4
1m	Electrical	Backplane × 4
10m	Ring Modulator WDM	SMF × 1
100m		
1km		
> 10km		

Ring Resonator Filter



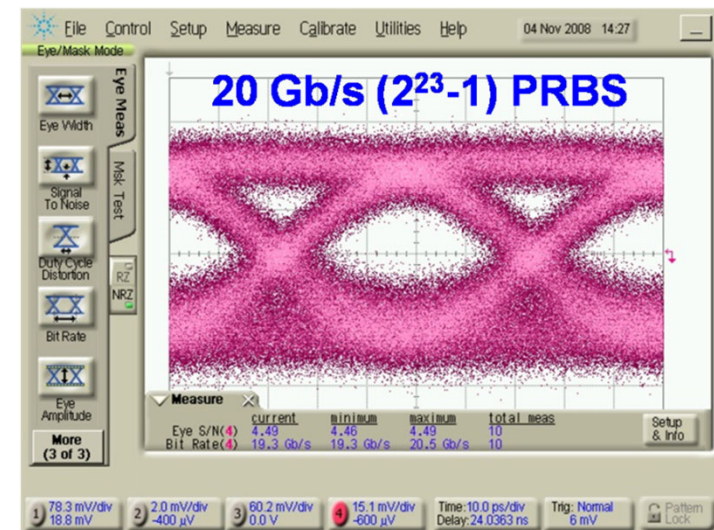
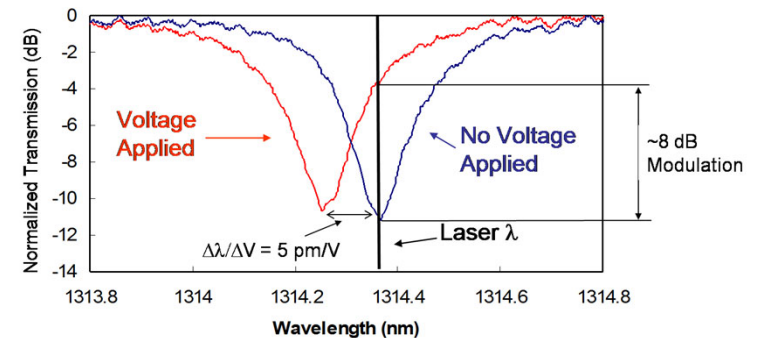
- Ring resonators display a high- Q notch filter response at the through port and a band-pass response at the drop port
- This response repeats over a free spectral range (FSR)

Ring-Resonator Modulator (RRM)



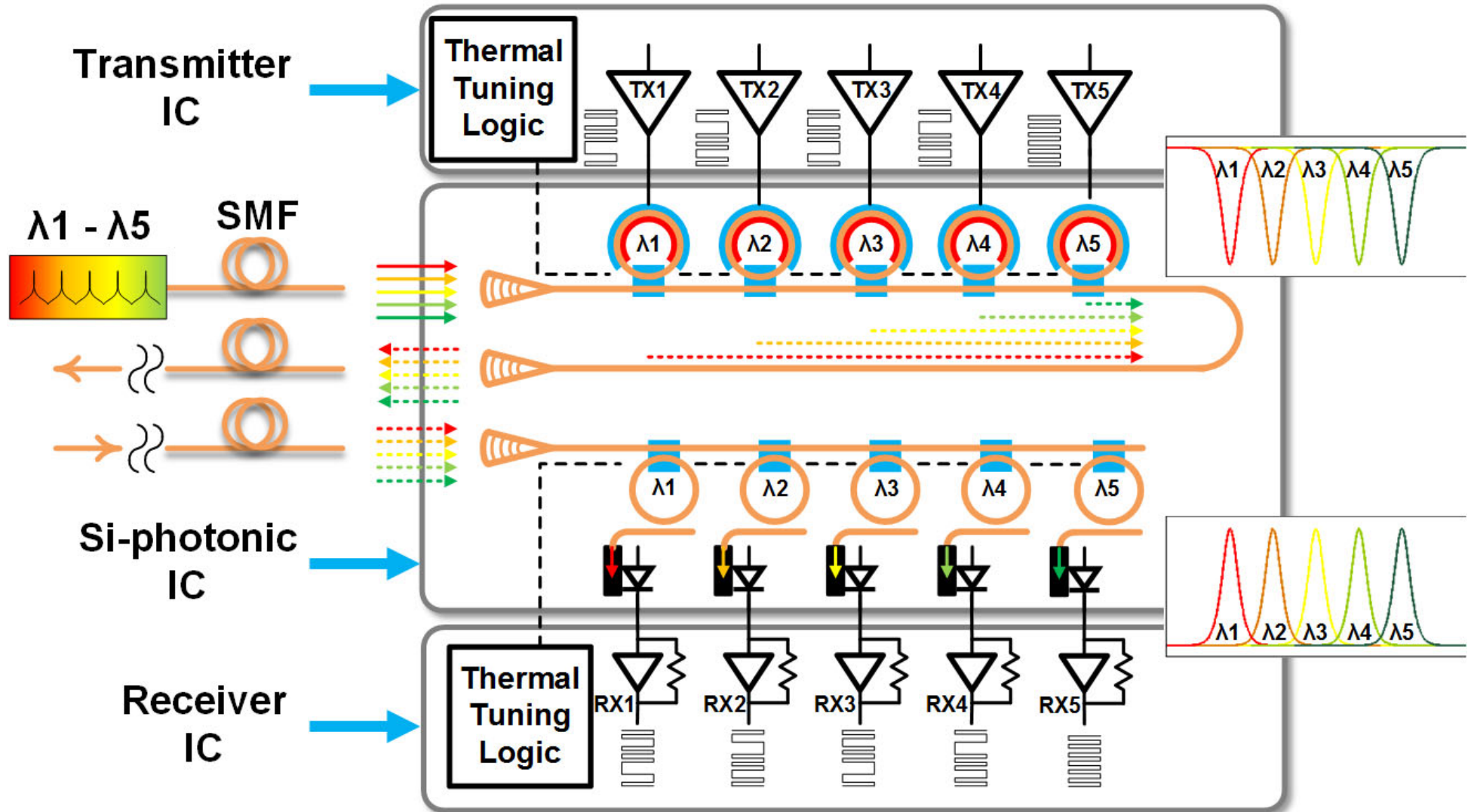
High Frequency Modulation

- Refractive devices which modulate by changing the interference light coupled into the ring with the waveguide light
- Devices are relatively small (ring diameters $< 20\mu\text{m}$) and can be treated as lumped capacitance loads ($\sim 10\text{fF}$)
- Devices can be used in WDM systems to selectively modulate an individual wavelength or as a "drop" filter at receivers



[Young ISSCC 2009]

WDM Photonic Transceiver



- High bandwidth density by combining multi-channels on a single waveguide via wavelength division multiplexing

Agenda

- WDM Optical Interconnect Motivation
- Silicon Photonic Modulators
- Carrier-Injection Ring Resonator Modulators
- Carrier-Depletion Ring Resonator Modulators

Plasma Dispersion Effect

- The change in refractive index and optical absorption coefficient is induced by free carriers in a semiconductor

$$N = n + i\alpha$$

N : complex index of refraction

n : electro-refraction

α : electro-absorption

1550 nm

$$\Delta n_{1.55\mu m} = -8.8 \times 10^{-22} \Delta N_e - 8.5 \times 10^{-18} (\Delta N_h)^{0.8}$$

$$\Delta \alpha_{1.55\mu m} = 8.5 \times 10^{-18} \Delta N_e + 6.0 \times 10^{-18} \Delta N_h [cm^{-1}]$$

1310 nm

$$\Delta n_{1.31\mu m} = -6.2 \times 10^{-22} \Delta n_e - 6.0 \times 10^{-18} (\Delta n_h)^{0.8}$$

$$\Delta \alpha_{1.31\mu m} = 6.0 \times 10^{-18} \Delta n_e + 4.0 \times 10^{-18} \Delta n_h [cm^{-1}]$$

ΔN_e : electrons density

ΔN_h : holes density

Silicon Photonic Modulators

- Carrier Accumulation
- Carrier Injection
- Carrier Depletion
- Electrooptic-polymer modulators have also been developed

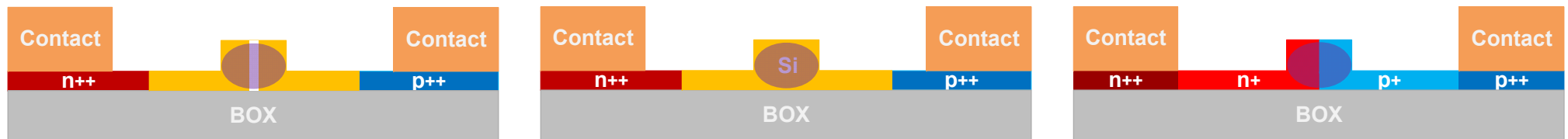


Figure of Merit	Carrier Accumulation	Carrier Injection	Carrier Depletion
Modulation Bandwidth	High +	Low -	High +
Extinction Ratio (ER)	Small -	Large +	Small -
Modulation Efficiency	High +	High +	Low -
Insertion Loss	High -	Low +	High -

MZM vs Microring Modulators

- Mach Zehnder Modulator
- Microring Modulator

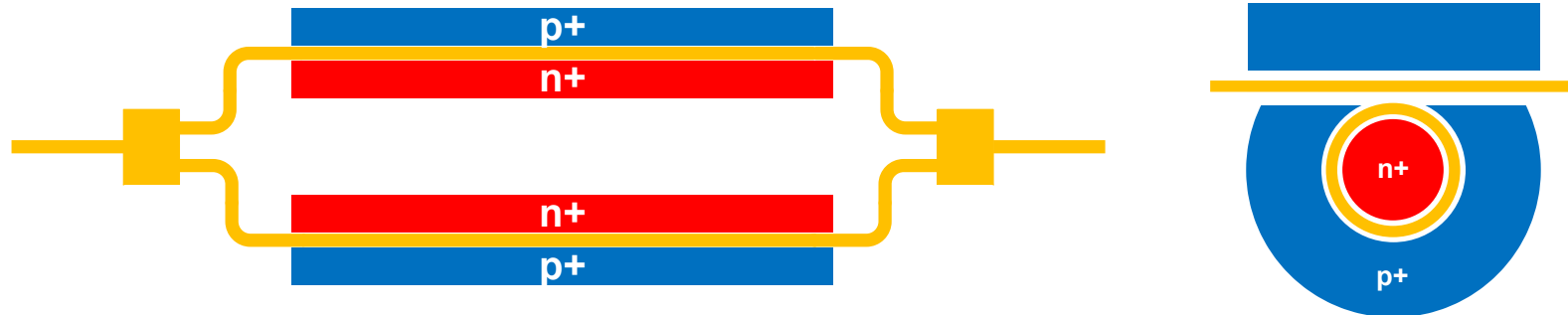
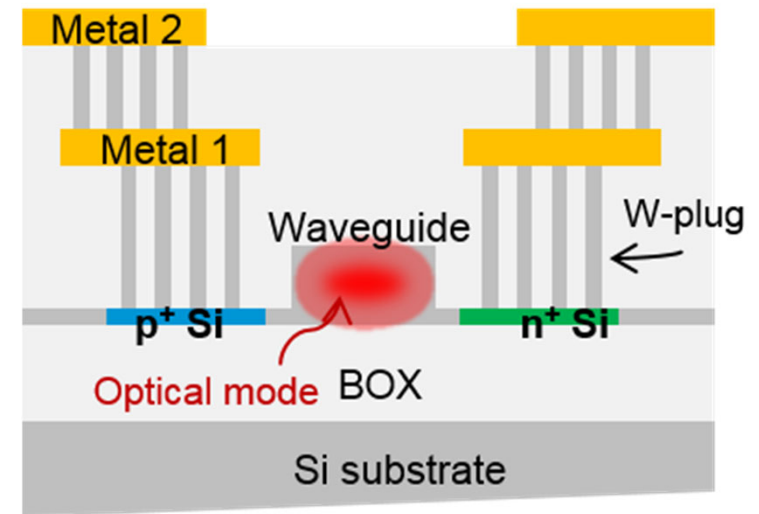
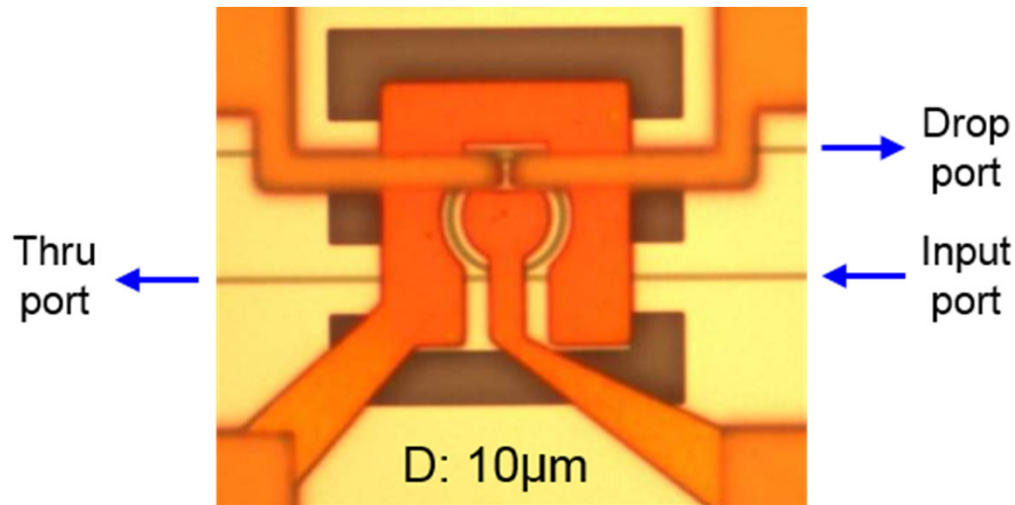


Figure of Merit	Mach Zehnder	Microring
Footprint	Large -	Small +
Extinction Ratio (ER)	Small -	Large +
Insertion Loss	High -	Low +
Wavelength Sensitivity	Low +	High -

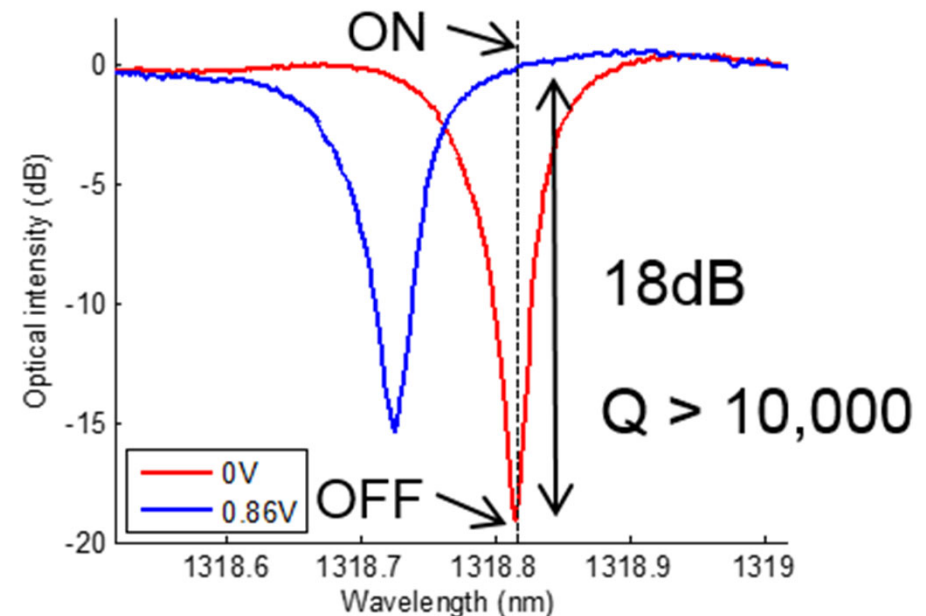
Agenda

- WDM Optical Interconnect Motivation
- Silicon Photonic Modulators
- Carrier-Injection Ring Resonator Modulators
 - Device Operation and Modeling
 - High-Speed Driver
 - Wavelength Stabilization Loop
- Carrier-Depletion Ring Resonator Modulators

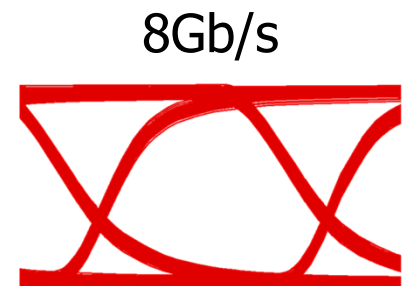
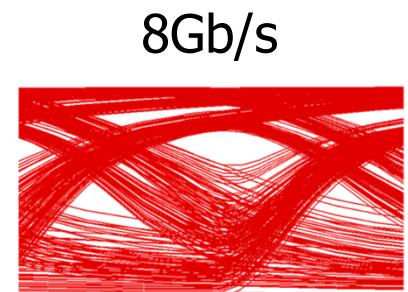
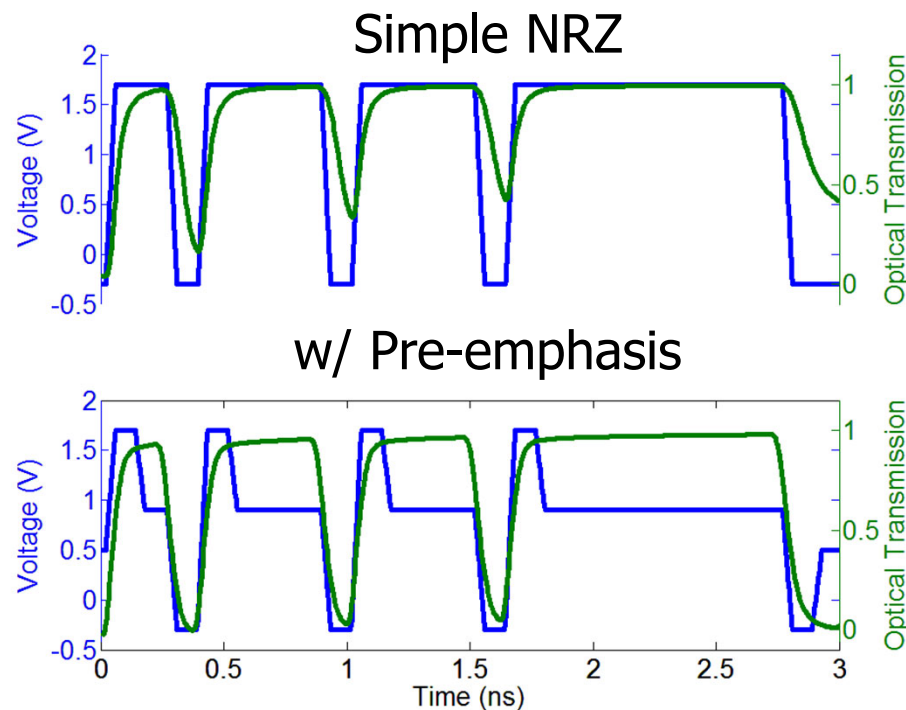
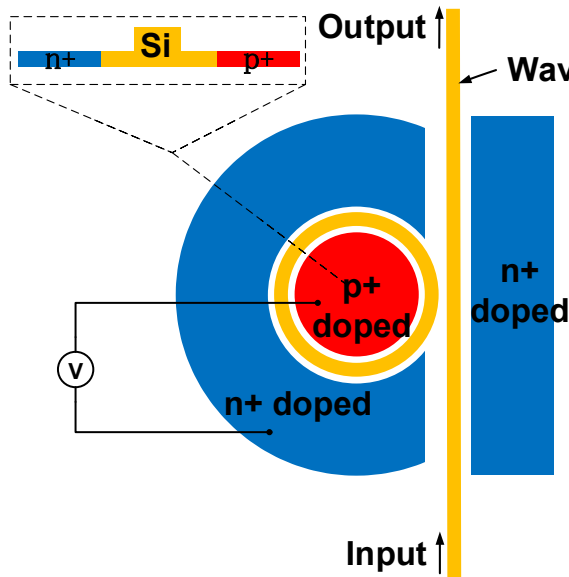
Carrier-Injection Microring Modulator



- Ring waveguide is doped as a PIN junction
- Forward-bias voltage injects carriers, changing refractive index for optical modulation via the free carrier plasma dispersion effect



C-I Ring Modulator Speed Limitations



- Speed is limited by long minority carrier lifetimes
- Pre-emphasis signaling significantly improves data rates

Carrier-Injection Ring Modulator Modeling

- Previous Related Modeling Work
 - P-I-N diode model (only electrical dynamics) [Strollo TPE 1997]
 - Ring resonator model (only optical dynamics) [Smy JOSA B 2011]
 - Carrier-injection ring modulator model (lack accurate large-signal behavior) [Xu OE 2007, Wu OE 2015]
- Carrier-Injection Ring Modulator Model
 - Capture nonlinear electrical and optical dynamics

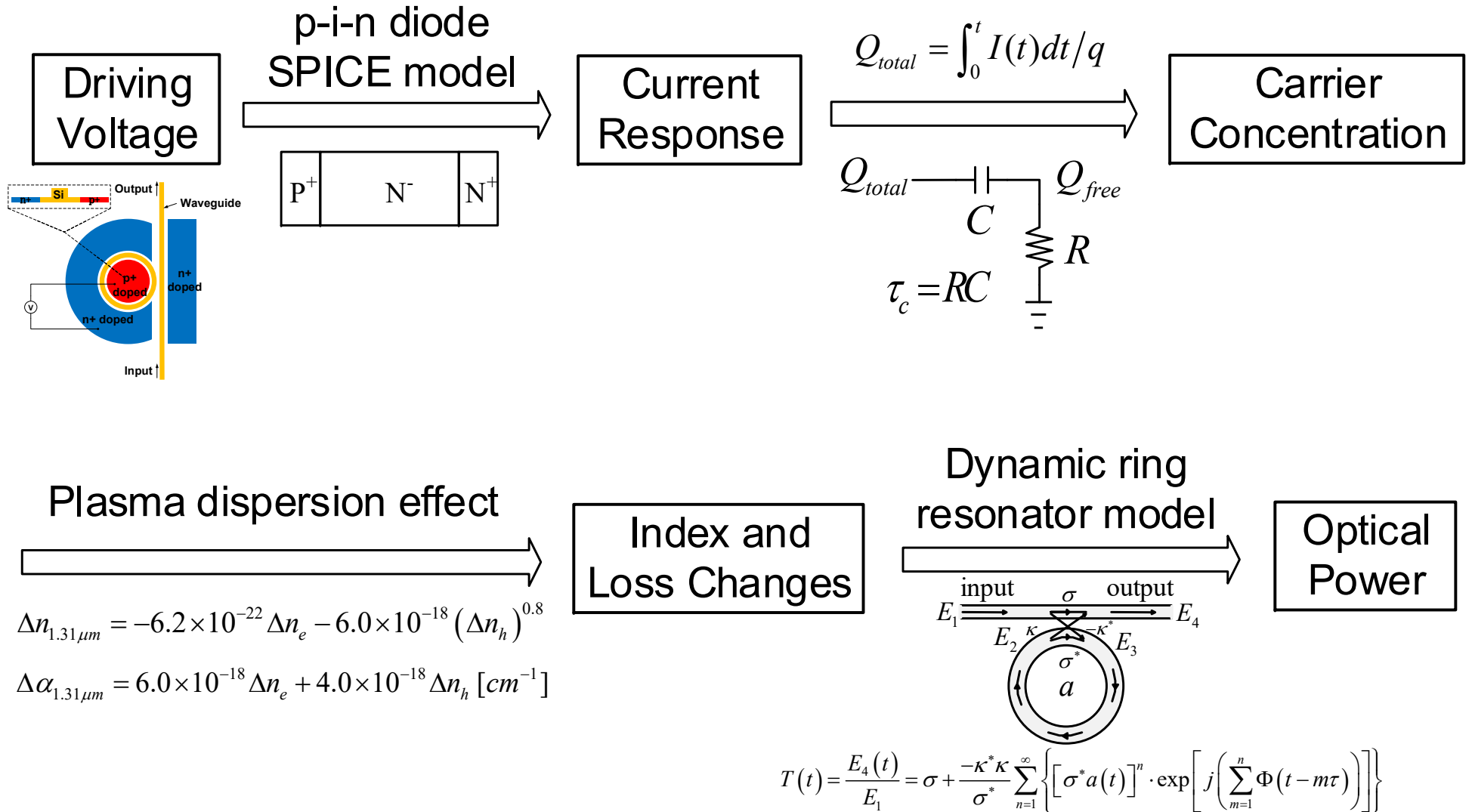
[**Binhao Wang**, Cheng Li, Chin-Hui Chen, Kunzhi Yu, Marco Fiorentino, Raymond Beausoleil, and Samuel Palermo, "Compact Verilog-A Modeling of Carrier-Injection Microring Modulators for Optical Interconnect Transceiver Circuitry Design," accepted in Journal of Lightwave Technology]

[**Binhao Wang**, Cheng Li, Chin-Hui Chen, Kunzhi Yu, Marco Fiorentino, Raymond Beausoleil, and Samuel Palermo, "Compact Verilog-A Modeling of Silicon Carrier-Injection Ring Modulators," IEEE Optical Interconnects Conference (OIC), 2015]

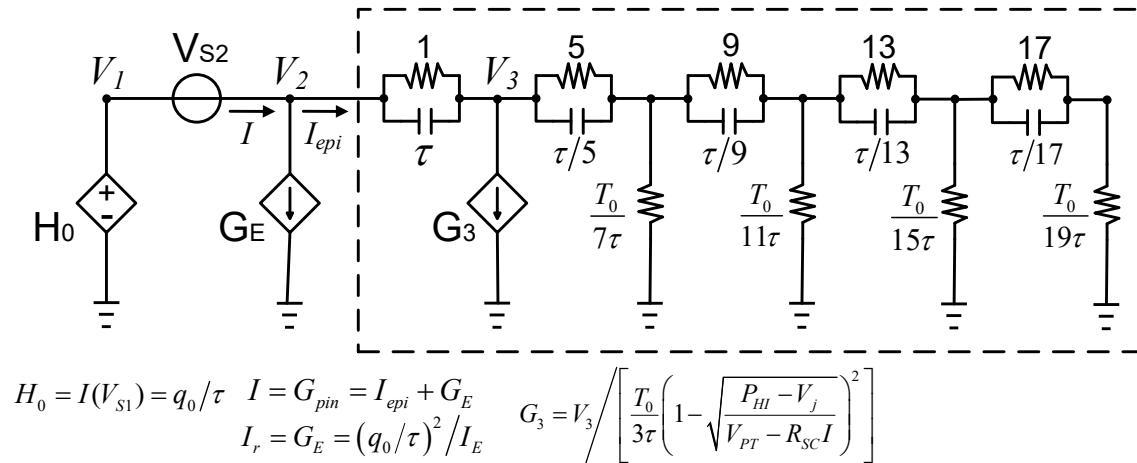
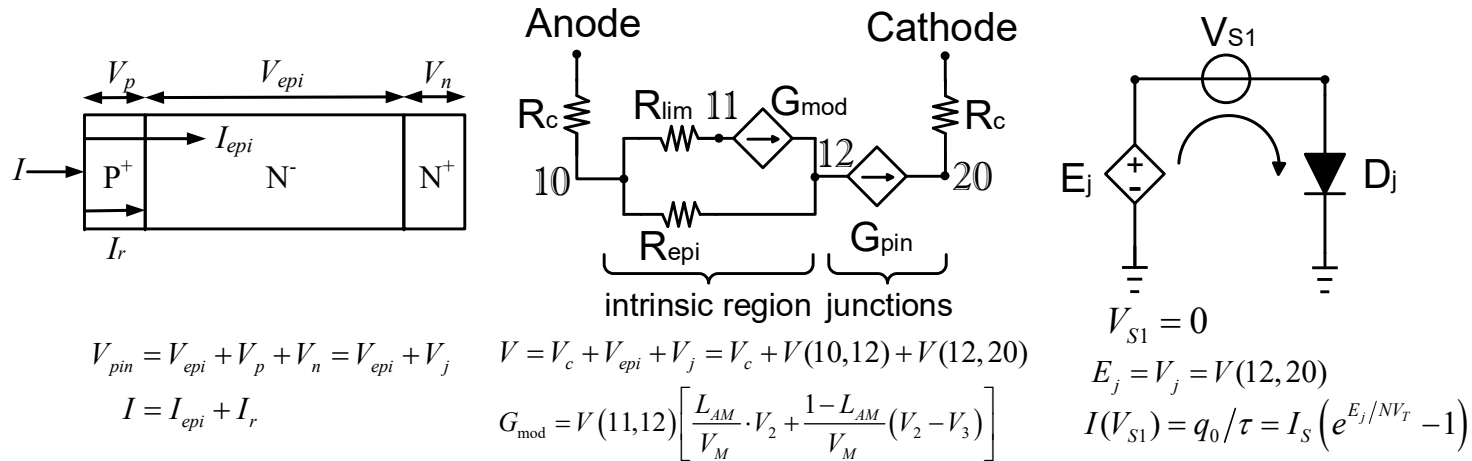
[Cheng Li, Chin-Hui Chen, **Binhao Wang**, Samuel Palermo, Marco Fiorentino, and Raymond Beausoleil, "Design of an Energy-Efficient Silicon Microring Resonator-Based Photonic Transmitter," IEEE Design & Test of Computers, 31, 46-54, 2014]

[Cheng Li, Rui Bai, Ayman Shafik, Ehsan Zhian Tabasy, **Binhao Wang**, Geng Tang, Chao Ma, Chin-Hui Chen, Zhen Peng, Marco Fiorentino, Raymond Beausoleil, Patrick Chiang, and Samuel Palermo, "Silicon Photonic Transceiver Circuits with Microring Resonator Bias-Based Wavelength Stabilization in 65-nm CMOS," IEEE Journal of Solid-State Circuits (JSSC), 49, 1419-1436, 2014]

Model Flow Chart



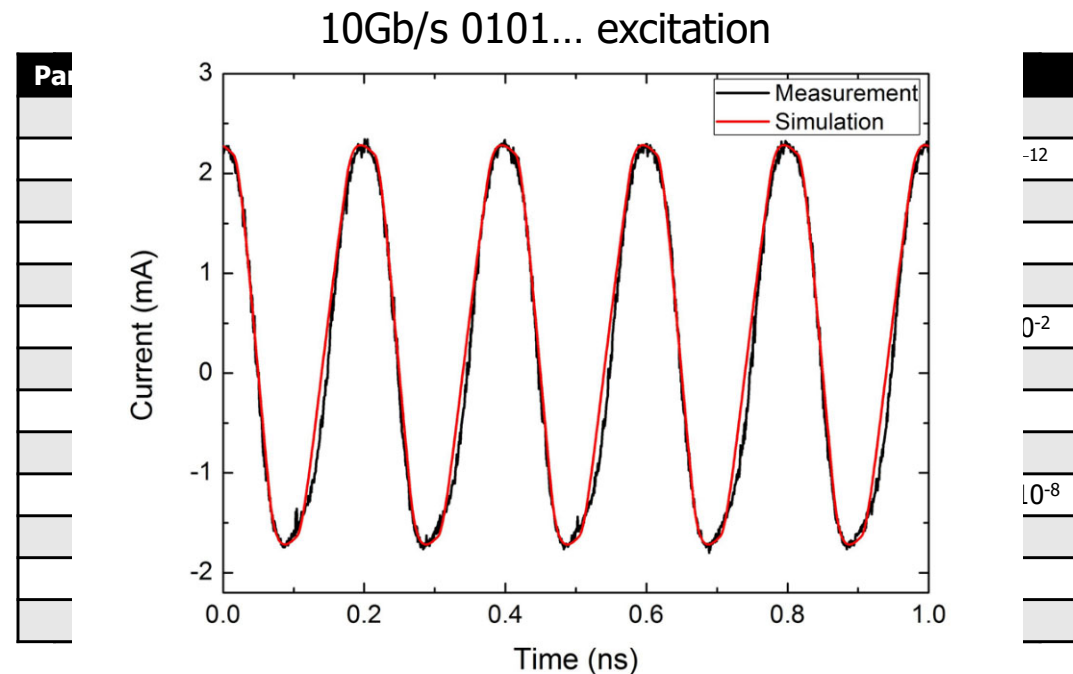
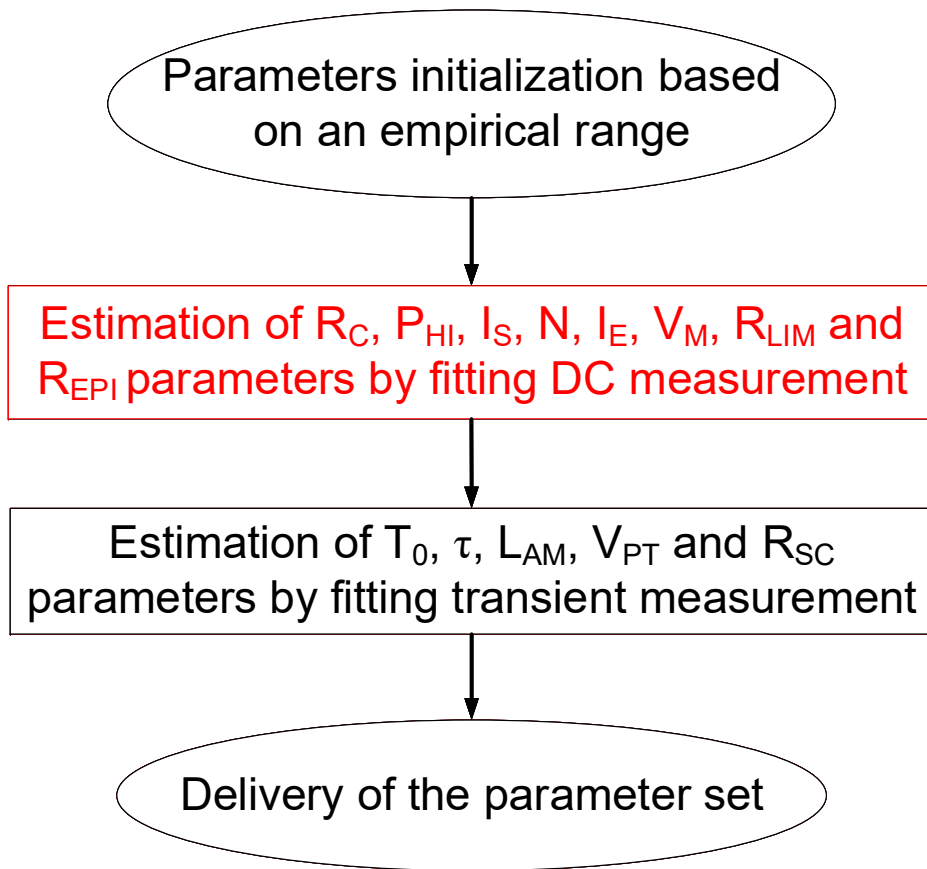
Electrical Modeling - PIN Dynamics



[Strollo TPE 1997]

- I-V dynamics are based on a moment-matching approximation of the ambipolar diffusion equation

Electrical Modeling - Parameter Extraction

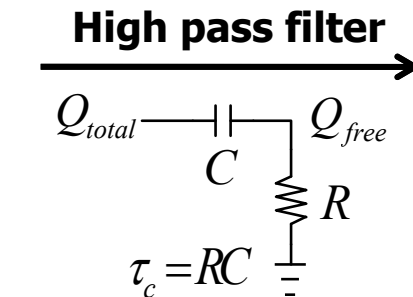
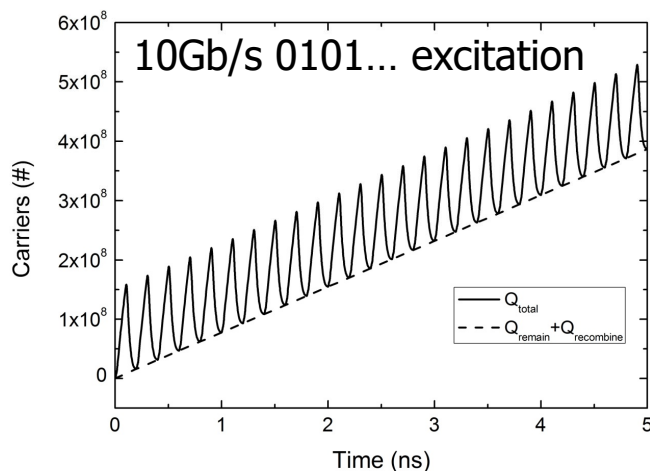


Optical Modeling - Carrier Dynamics

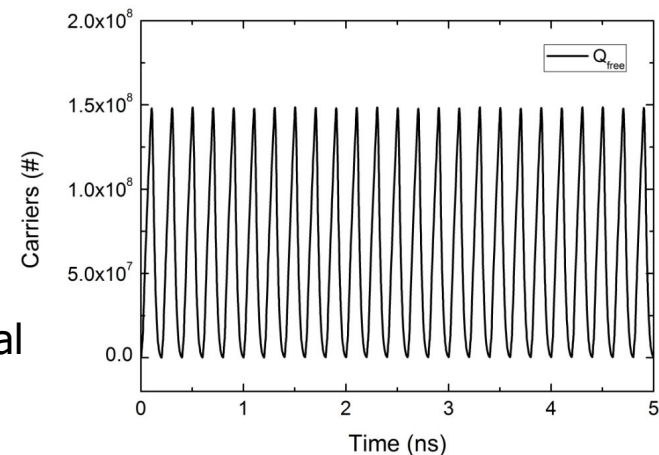
- Carrier Concentration

$$Q = \int_0^t I(t) dt / q \quad I_{total}(t) = I_{free}(t) + I_{remain}(t) + I_{recombine}(t)$$

$$Q_{total}(t) = Q_{free}(t) + Q_{remain}(t) + Q_{recombine}(t)$$



The time constant is equal to the carrier lifetime

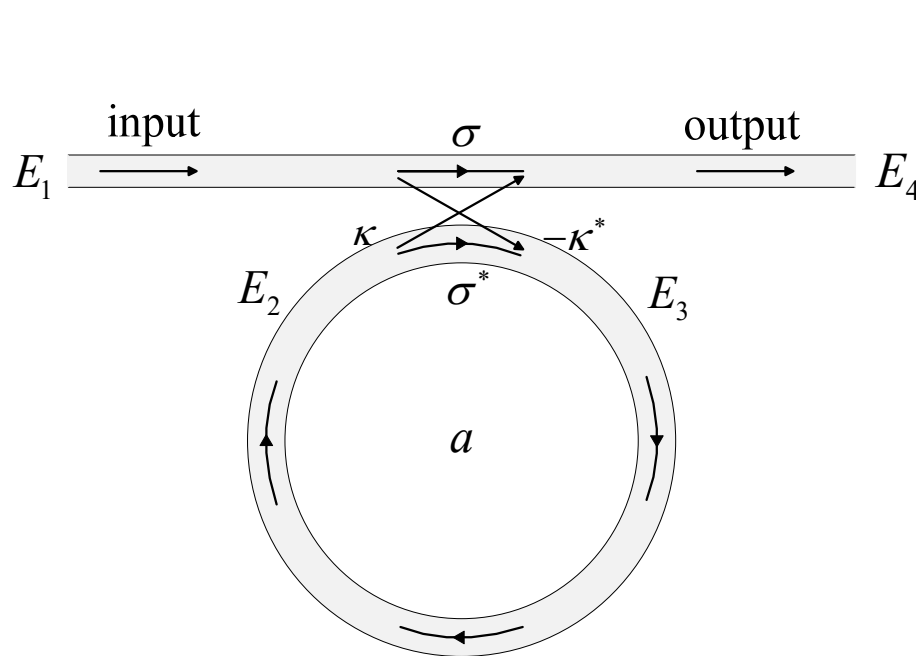


- Index and Loss Changes

$$\Delta n_{1.31\mu m} = -6.2 \times 10^{-22} \Delta n_e - 6.0 \times 10^{-18} (\Delta n_h)^{0.8}$$

$$\Delta \alpha_{1.31\mu m} = 6.0 \times 10^{-18} \Delta n_e + 4.0 \times 10^{-18} \Delta n_h [cm^{-1}]$$

Optical Modeling - Optical Dynamics



$$T(t) = \frac{E_4(t)}{E_1} = \sigma + \frac{-\kappa^* \kappa}{\sigma^*} \sum_{n=1}^{\infty} \left\{ \left[\sigma^* a(t) \right]^n \cdot \exp \left[j \left(\sum_{m=1}^n \Phi(t - m\tau) \right) \right] \right\}$$

$$\begin{aligned} T(t) &= \sigma + \frac{-\kappa^* \kappa}{\sigma^*} \sum_{n=1}^{\infty} \left\{ \left[\sigma^* a(t) \right]^n \cdot \exp [jn\Phi(t)] \right\} \\ &= \frac{\sigma - a(t) \cdot \exp [j(\Phi(t))]}{1 - \sigma^* a(t) \cdot \exp [j(\Phi(t))]} \end{aligned}$$

$$\text{where } \Phi(t) = \frac{2\pi}{\lambda} n_{\text{eff}}(t) L$$

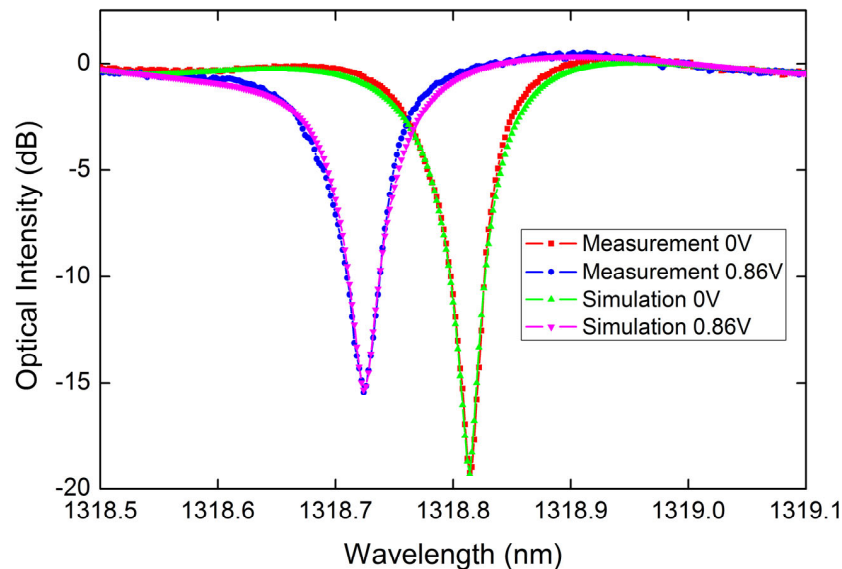
$$\sigma^2 + \kappa^2 = 1$$

[Loannidis OL 1988]

- Consider the ring's cumulative phase shift
- Capture non-linear optical dynamics

Optical Modeling - Optical Dynamics

- Parameter Extraction

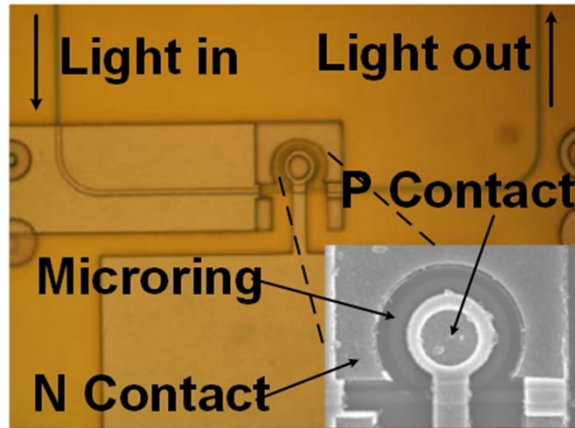


Parameter	Unit	Description	Value
σ	-	Transmission coefficient	0.9944
a	-	Loss coefficient	0.9931
n_{eff}	-	Effective index	2.5188
r	μm	Ring radius	5

- Large extinction ratio (ER) $\sim 20\text{dB}$
- High modulation efficiency $\sim 560\text{pm/V}$

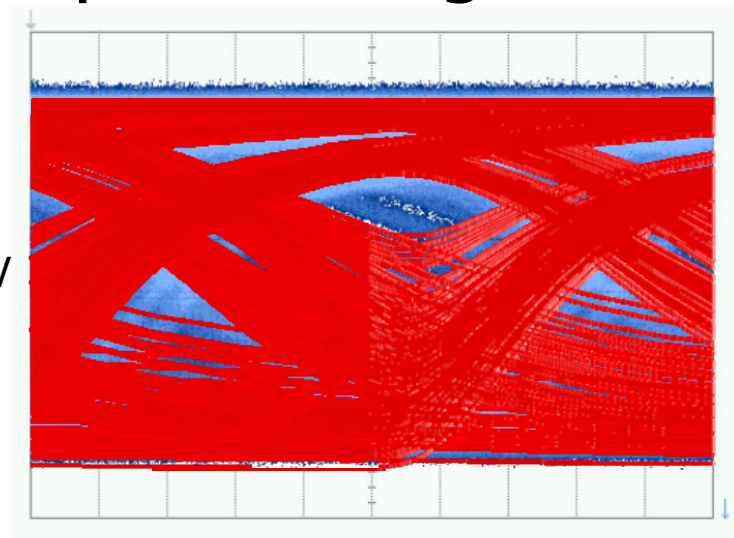
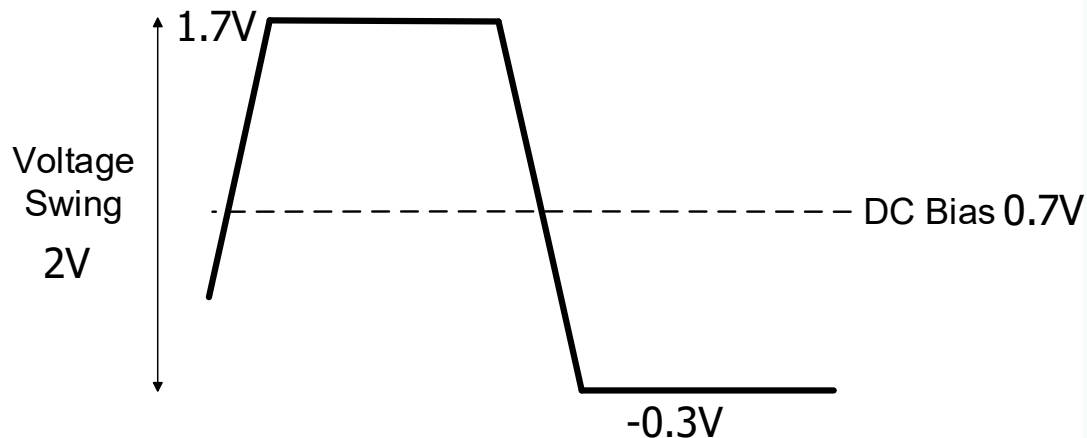
Simple NRZ Signaling

- Ring modulator under test



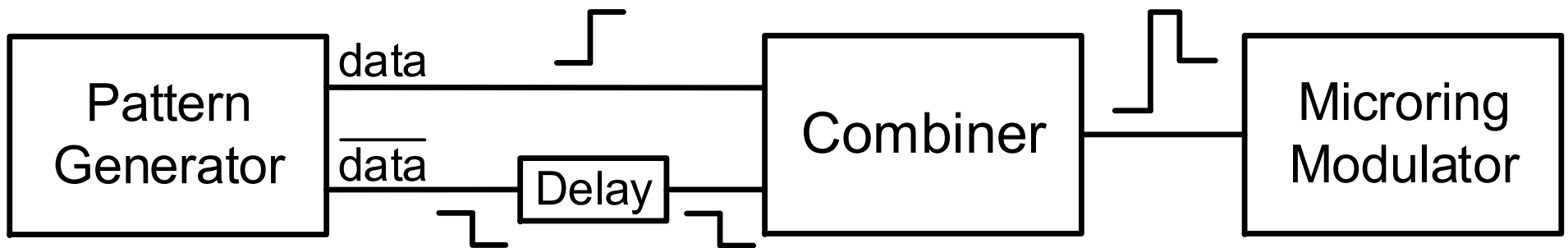
Parameter	Description
Coupling waveguide	350nm (W), 250nm (H), 50nm (slab)
Ring waveguide	450nm (W), 250nm (H), 50nm (slab)
Gap	250nm
Radius	5um
P+ doping	BF ²⁺ , 5e14 cm ⁻² , 10 KeV, Tilt 8°, Twist 27°
N+ doping	As, 5e14 cm ⁻² , 10 KeV, Tilt 8°, Twist 27°

- 8Gb/s eye diagrams with simple NRZ signal



Pre-Emphasis Signaling

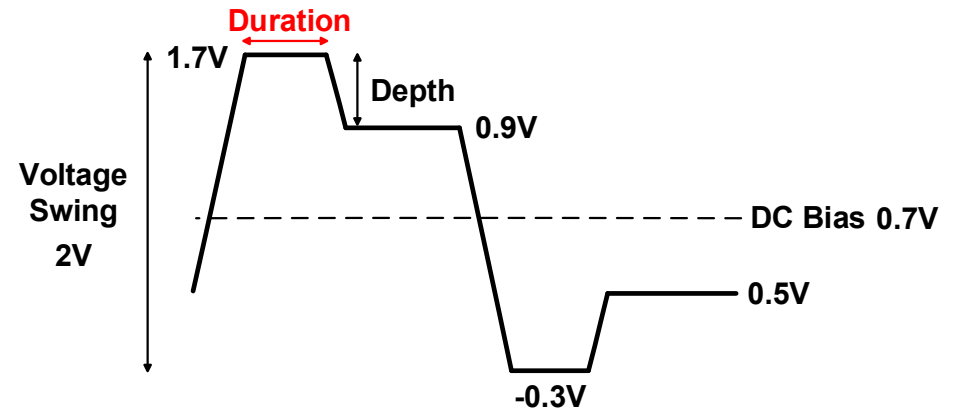
- Pre-emphasis NRZ signal generation



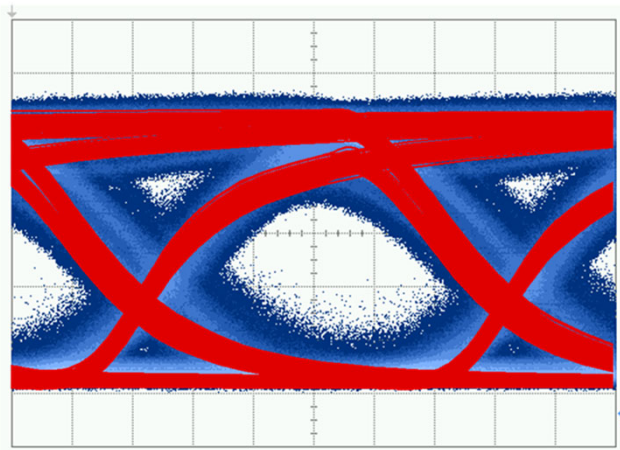
- Proposed model utilized to study the impact of key pre-emphasis parameters
 - Pulse duration
 - Pulse depth
 - DC bias

Pre-Emphasis Optimization - Duration

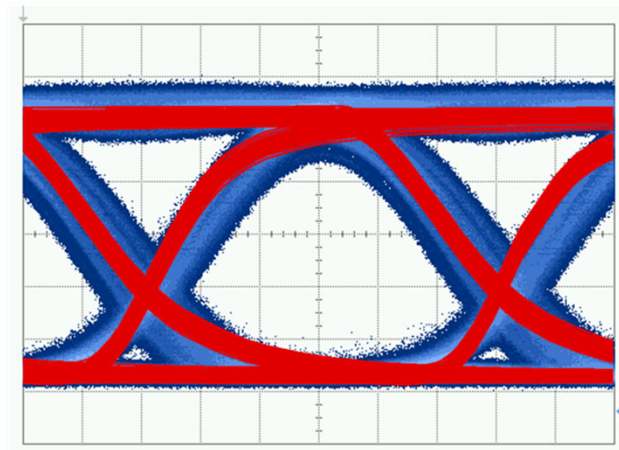
- **Pulse Duration**
- Pulse Depth = 0.8V
- DC Bias = 0.7V



40ps



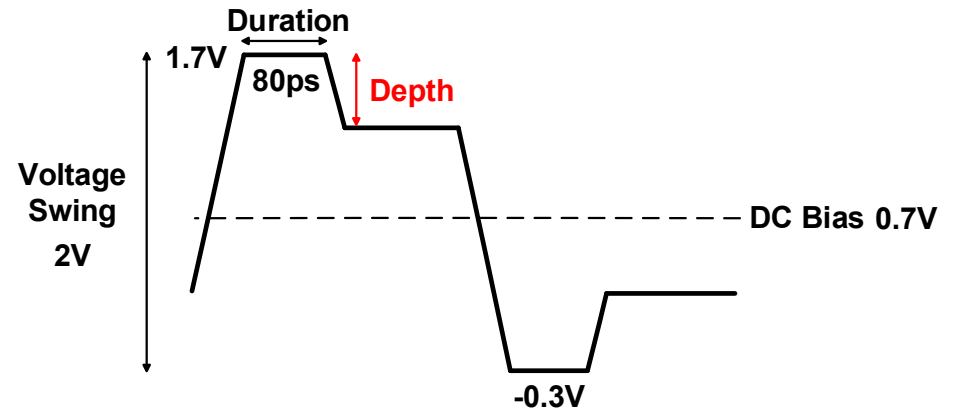
80ps



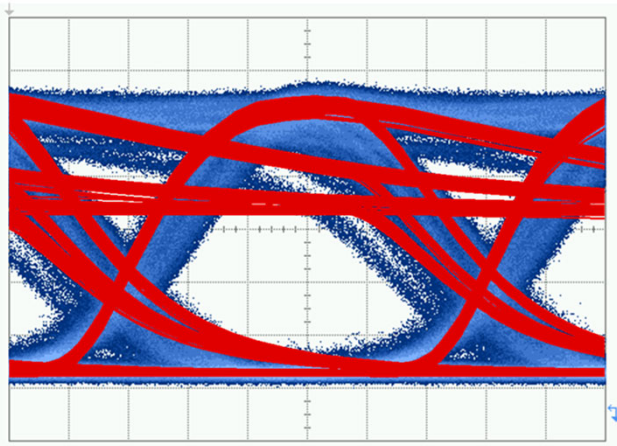
- 40ps pulse duration allows the eye to partially open
- 80ps pulse duration provides optimal eye opening

Pre-Emphasis Optimization - Depth

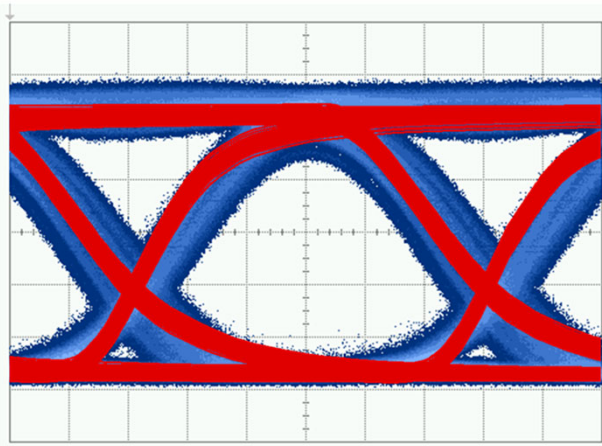
- Pulse Duration = 80ps
- **Pulse Depth**
- DC Bias = 0.7V



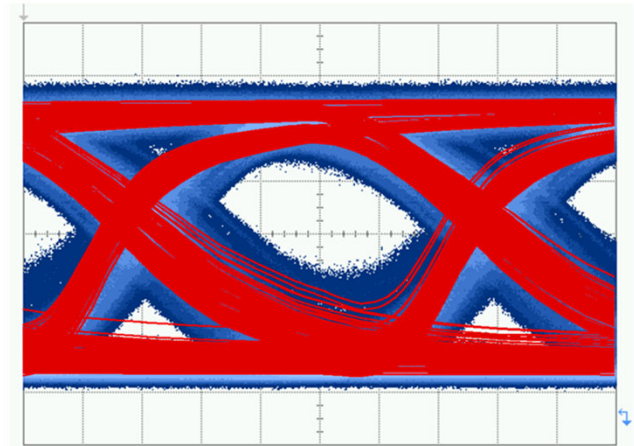
0.9V



0.8V



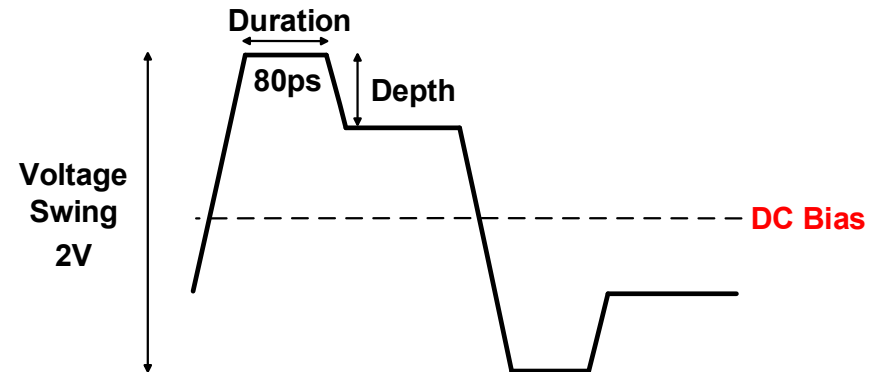
0.7V



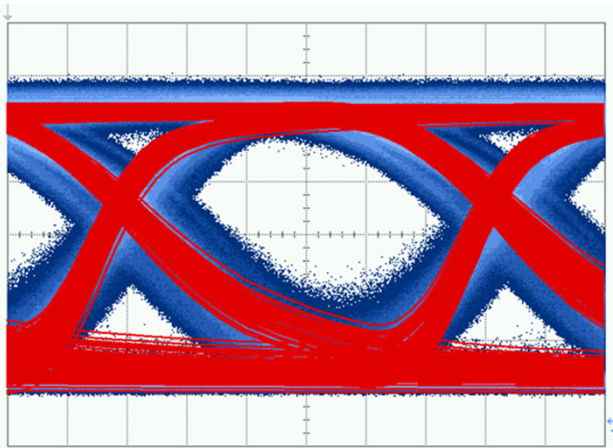
- 0.9V pulse depth results in low amount of charge for logic "1"
- 0.7V pulse depth produces excessive charge for logic "1"

Pre-Emphasis Optimization - DC Bias

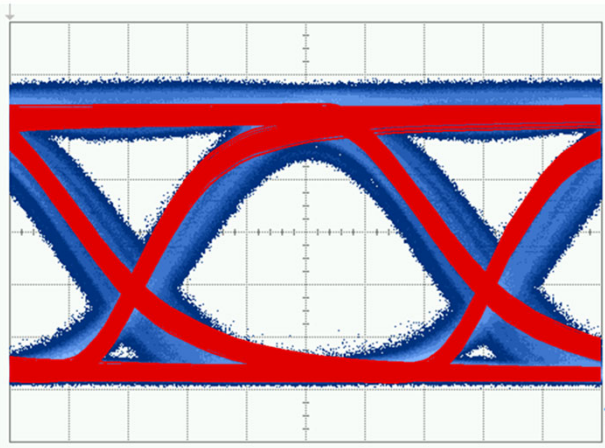
- Pulse Duration = 80ps
- Pulse Depth = 0.8 V
- DC Bias



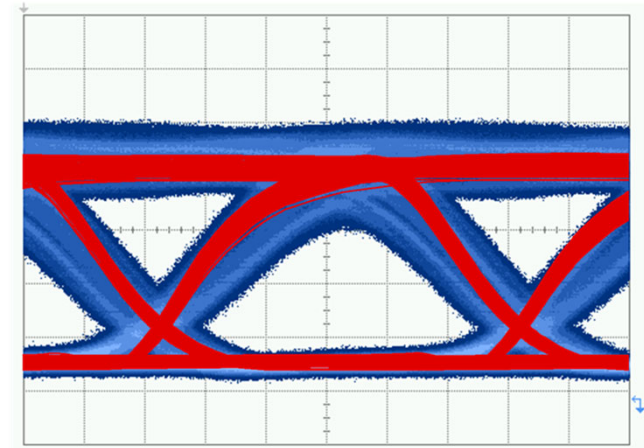
0.75V



0.7V

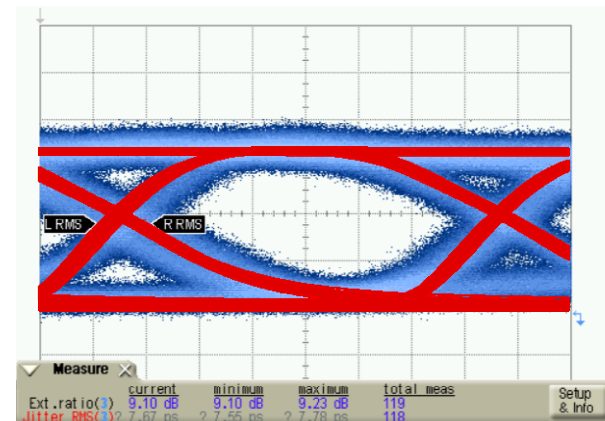
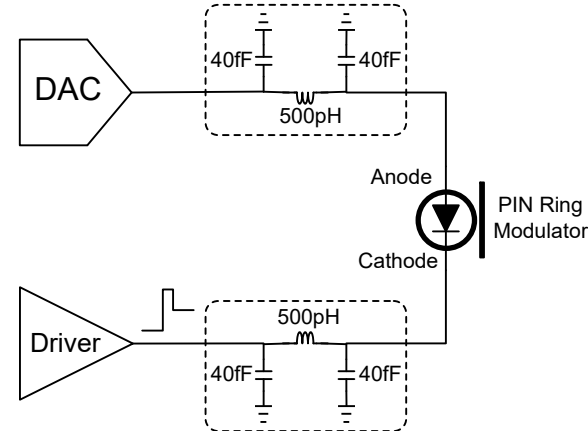
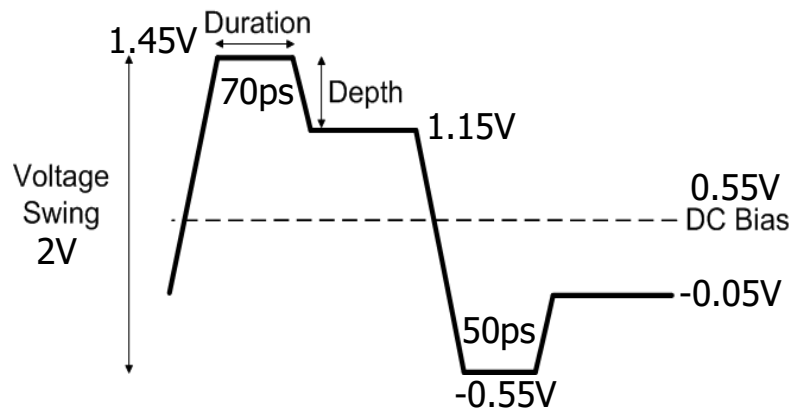
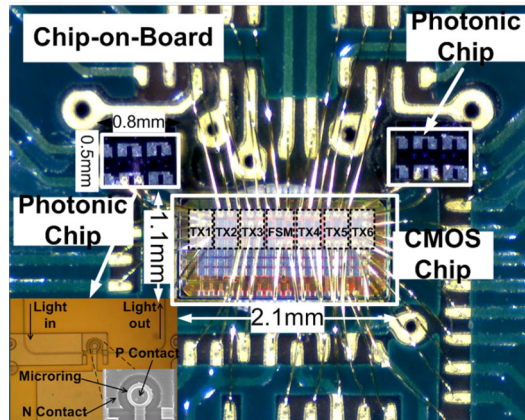


0.65V



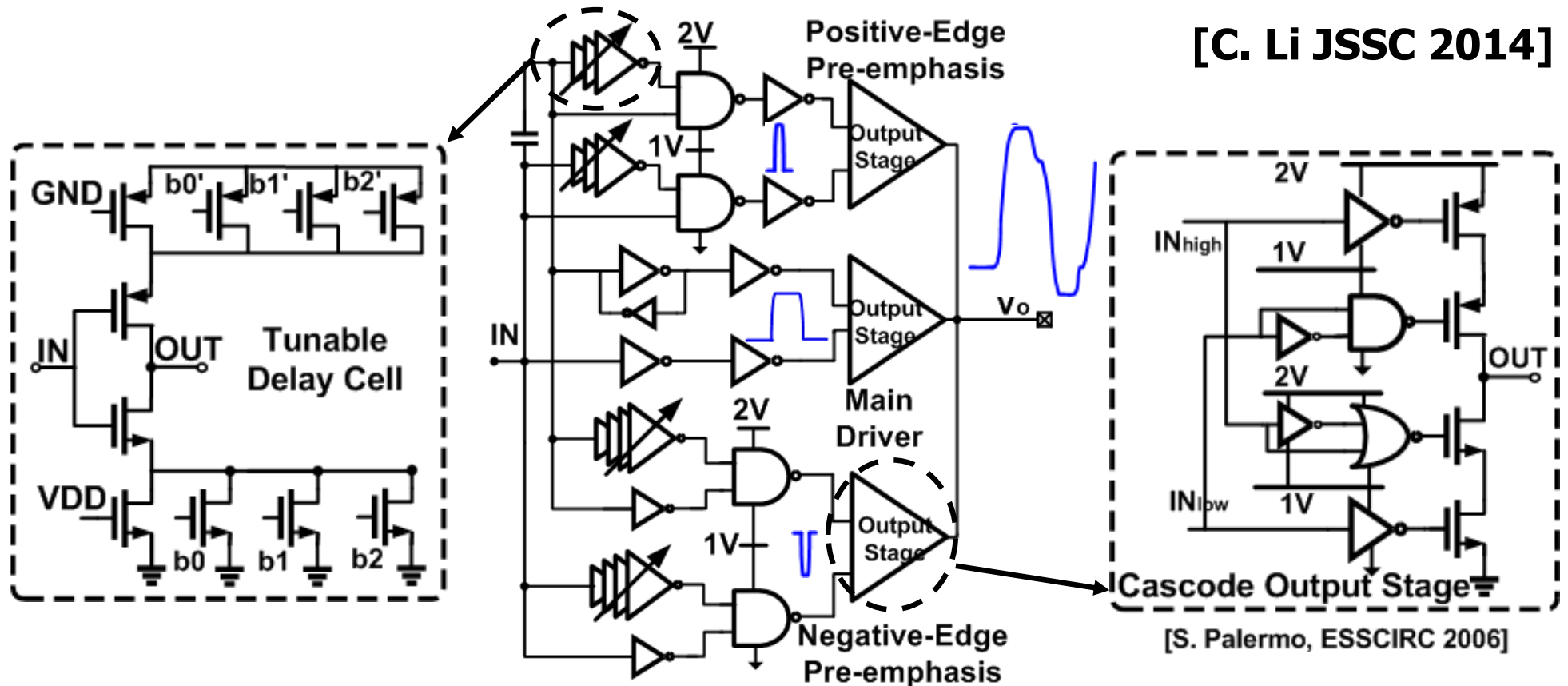
- 0.75V DC bias produces excessive charge for logic "1"
- 0.65V DC bias results in slower carrier injection for logic "1"

Co-Simulation with CMOS Driver



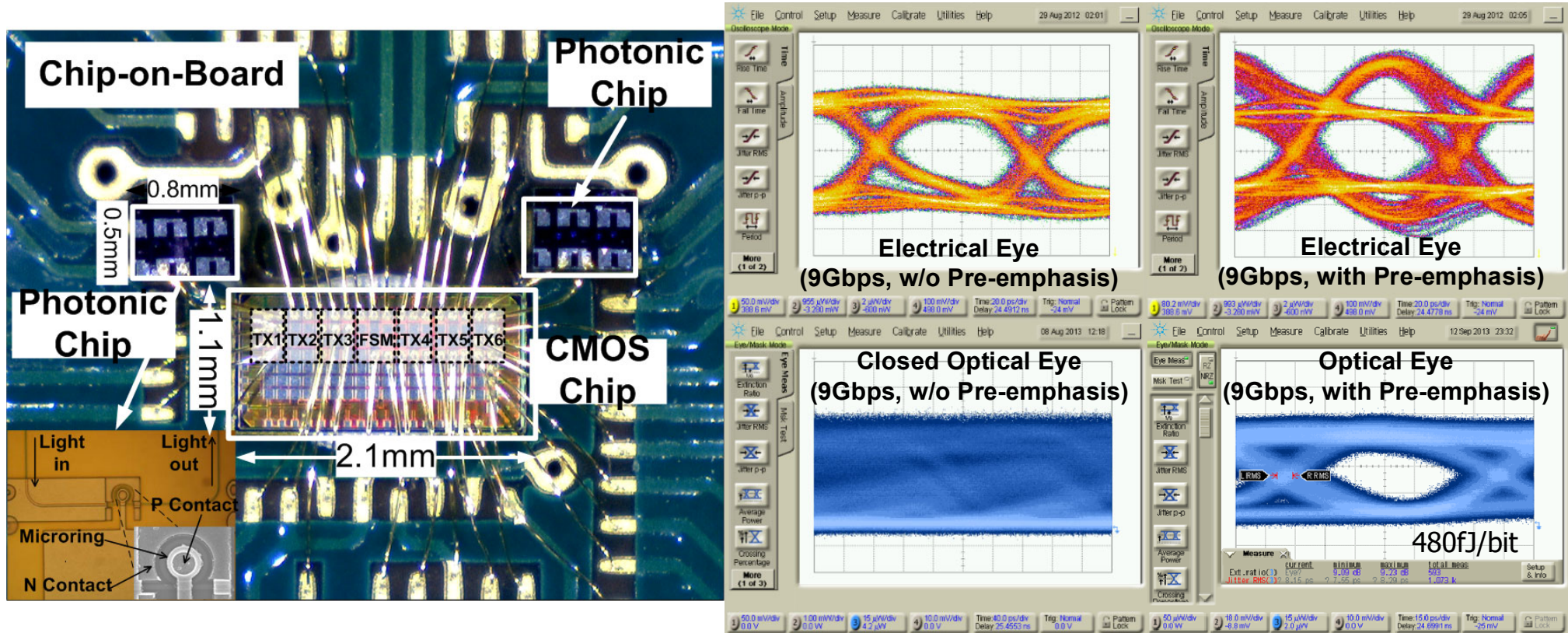
- Hybrid-integrated CMOS and silicon photonics prototype
- Optical transmitter co-simulation schematic
- Asymmetric pulse duration pre-emphasis setting
- 9Gb/s measured and co-simulated eye diagrams

High-Swing Pre-Emphasis Driver



- Dual-edge pre-emphasis with pulse width controlled by tunable delay cells (30ps~60ps)
- Cascode output stage used to meet high modulation swing requirement

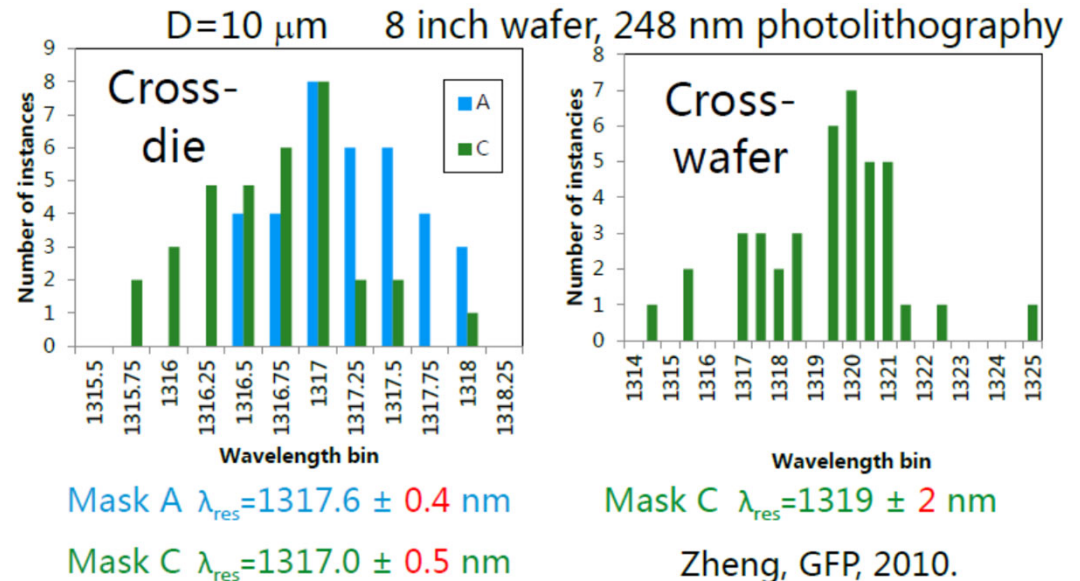
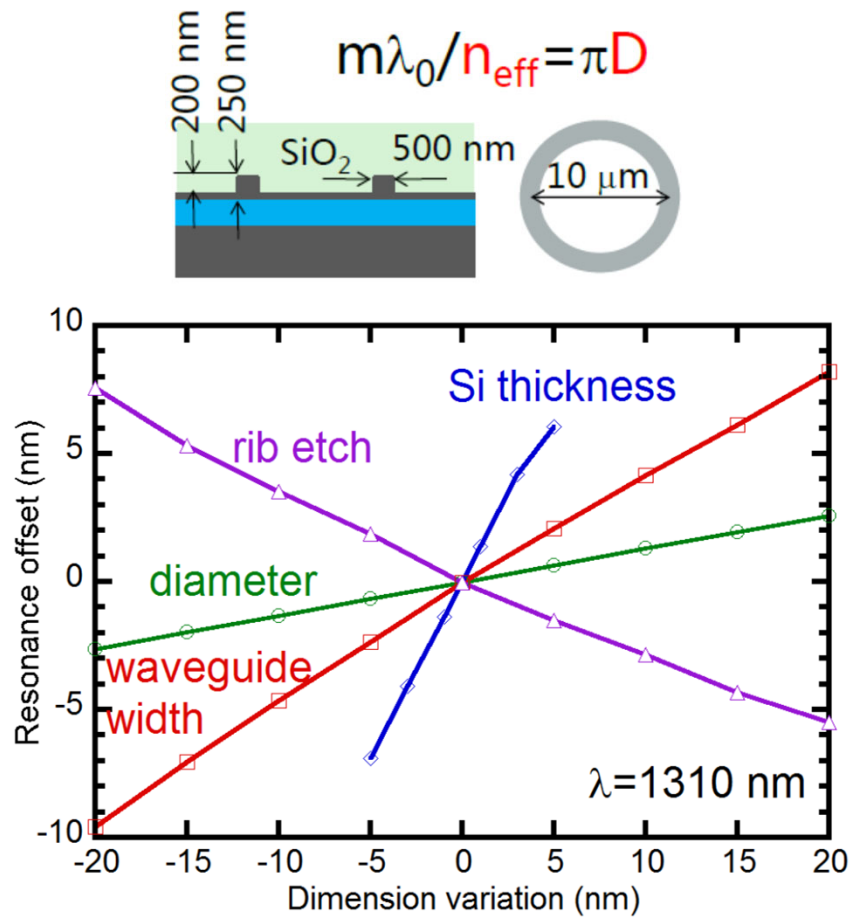
Optical Transmitter Assembly



* C. Li et. al. IEEE Design & Test, 2014

- GP 65nm CMOS 5-channel TX prototype
- 130nm SOI carrier-injection ring resonator modulators

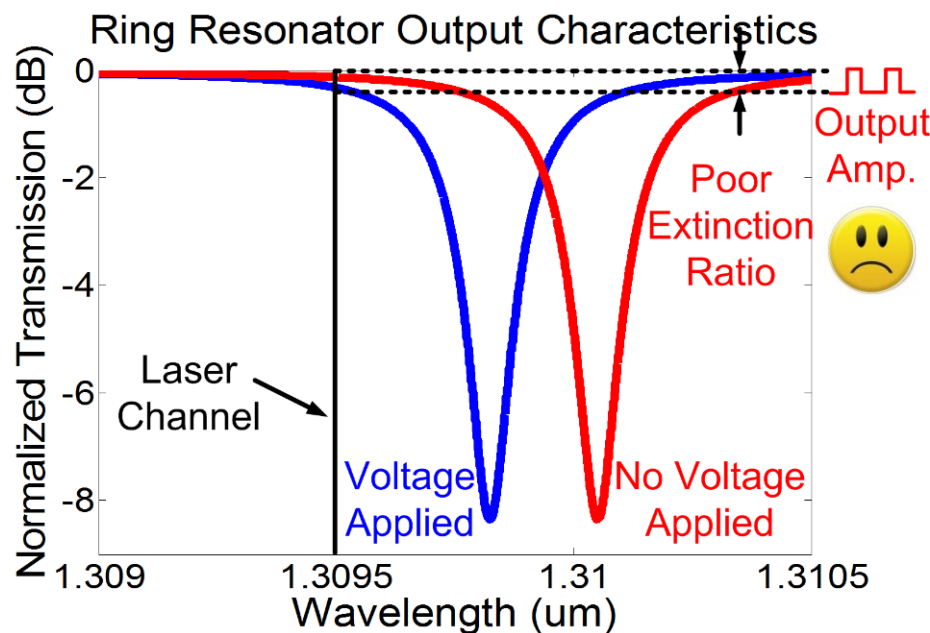
Resonant Wavelength Sensitivity



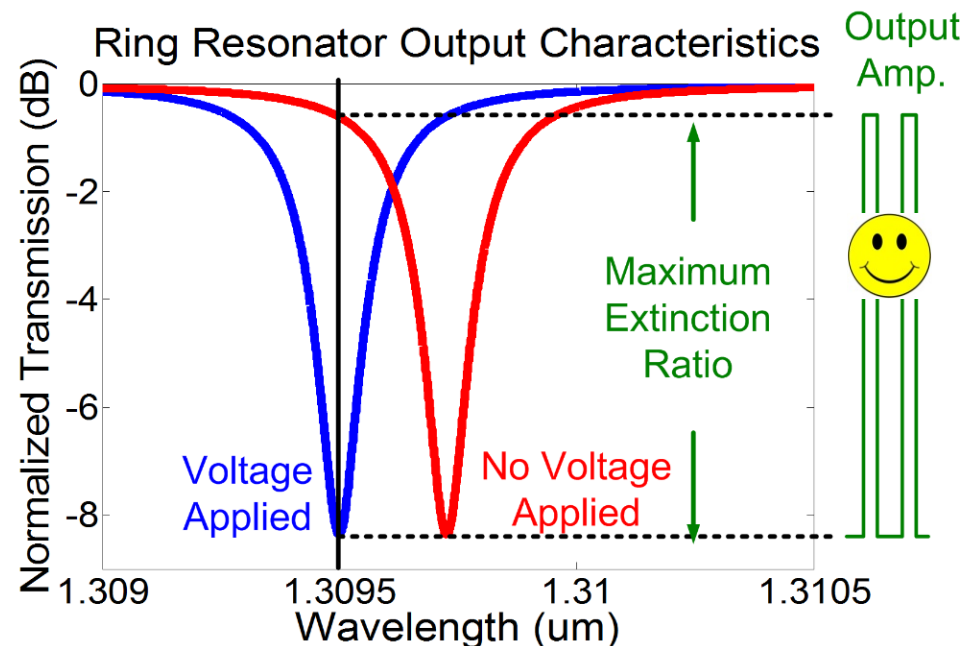
- Ring's resonance wavelength is sensitive to fabrication variations and temperature fluctuations
- Requires tuning schemes to compensate wavelength drifts

Extinction Ratio Impact

- Ring devices resonance wavelength can shift with fabrication and temperature variations
- Tuning schemes necessary to stabilize resonance wavelength

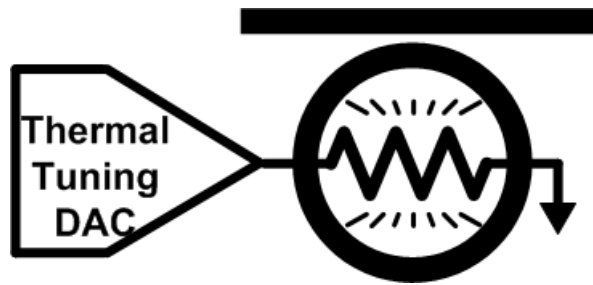


Before Tuning



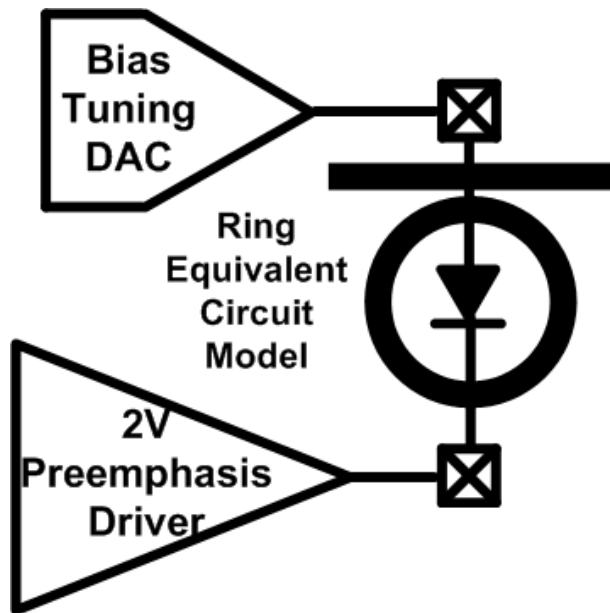
After Tuning

Bias vs Thermal Tuning



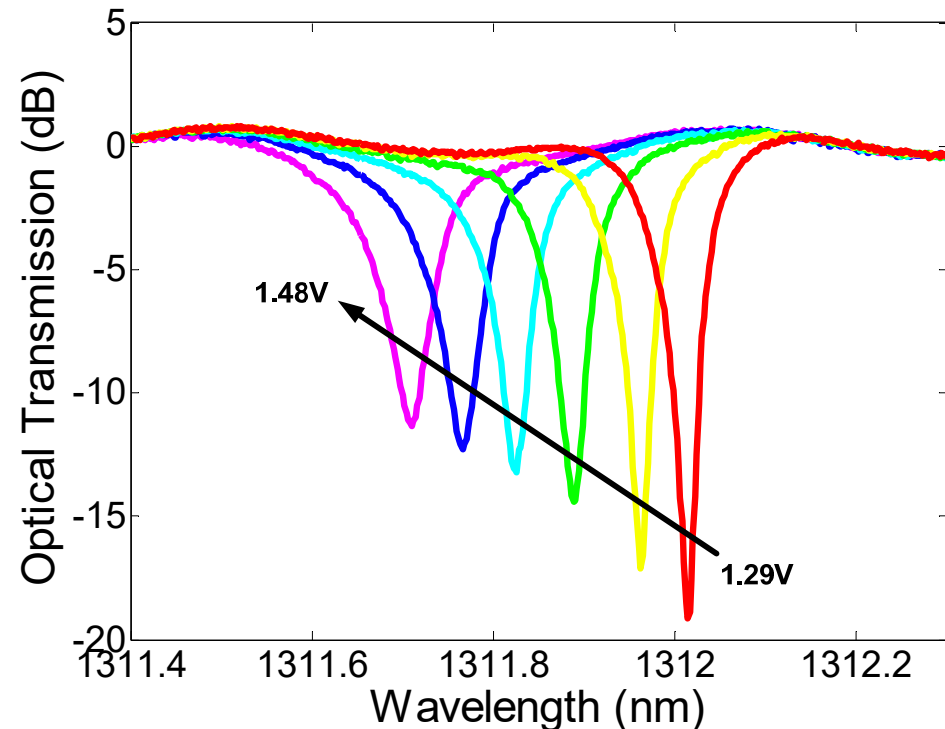
Thermal Tuning

	Bias Tuning	Thermal Tuning
Speed	Fast ($\sim\mu\text{s}$) 😊	Slow ($\sim\text{ms}$) ☹️
Direction	Blue shift	Red shift
Power	Low 😊	High ☹️
Range	Narrow ☹️	Wide 😊

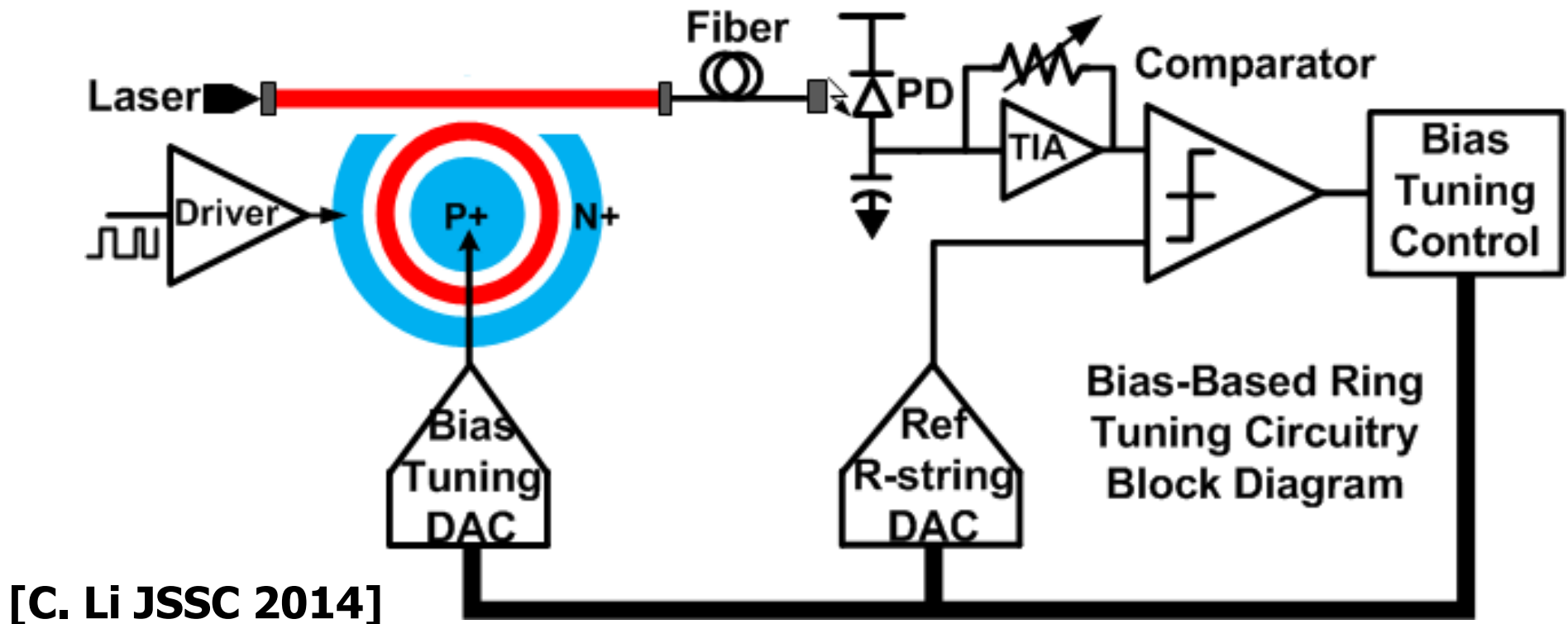


Bias Tuning

Ring Spectrum vs Bias Voltage

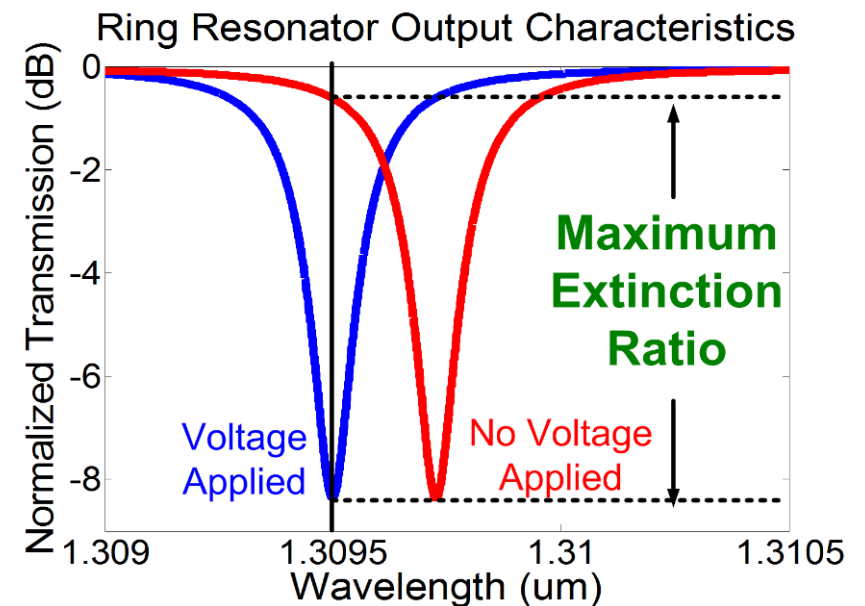
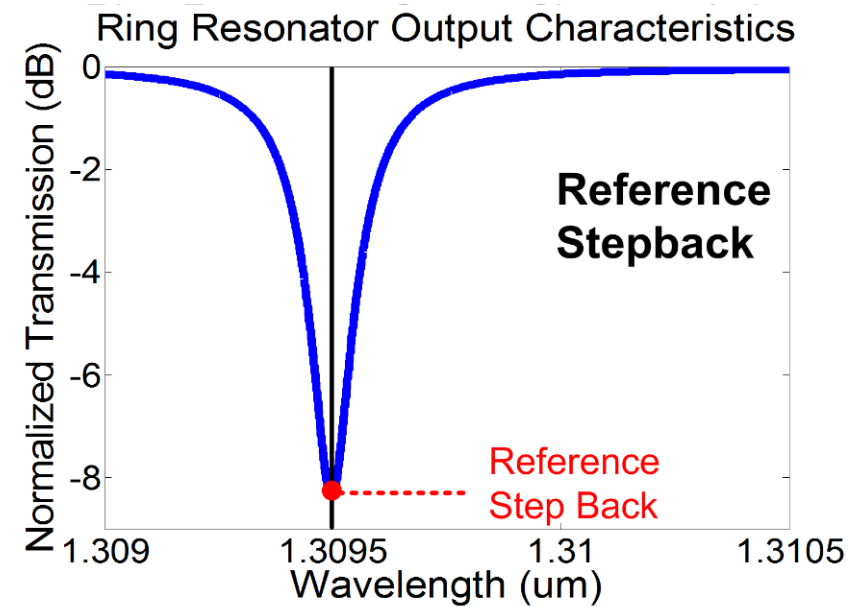
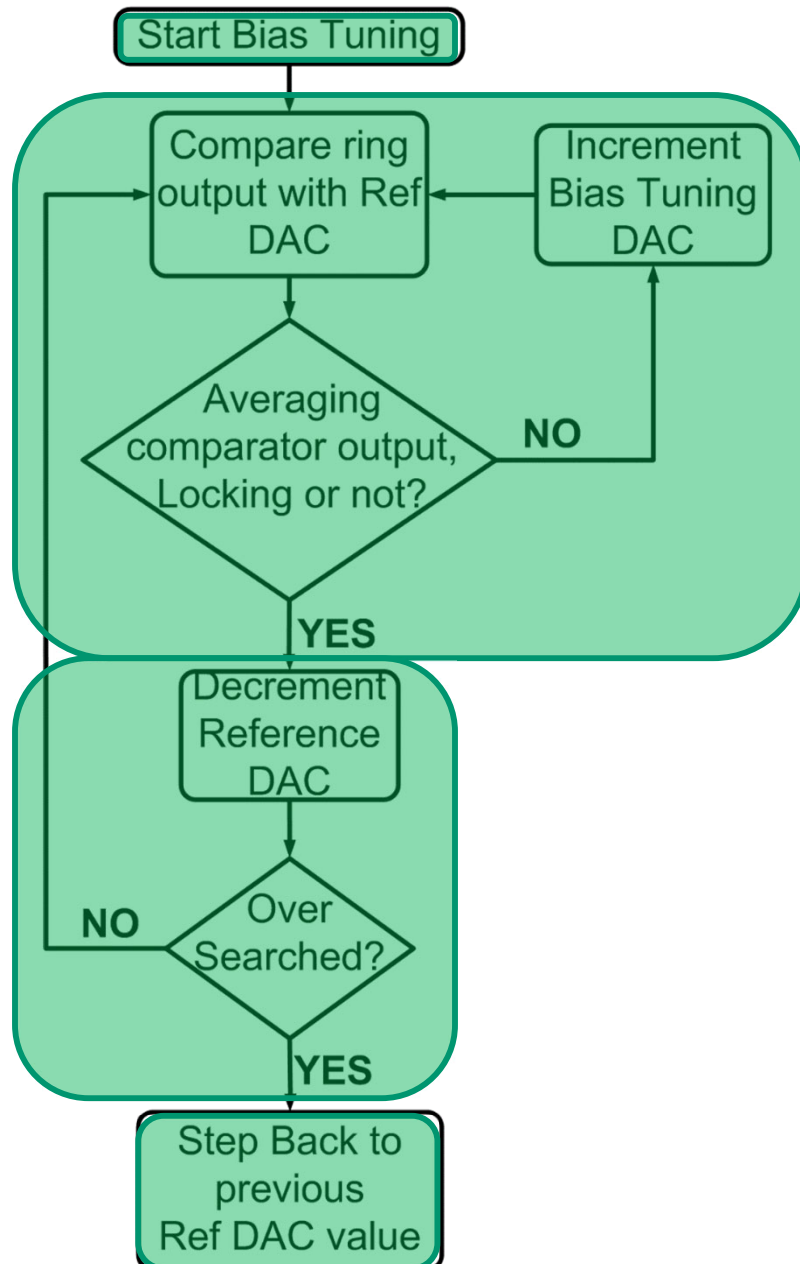


Automatic Tuning Loop



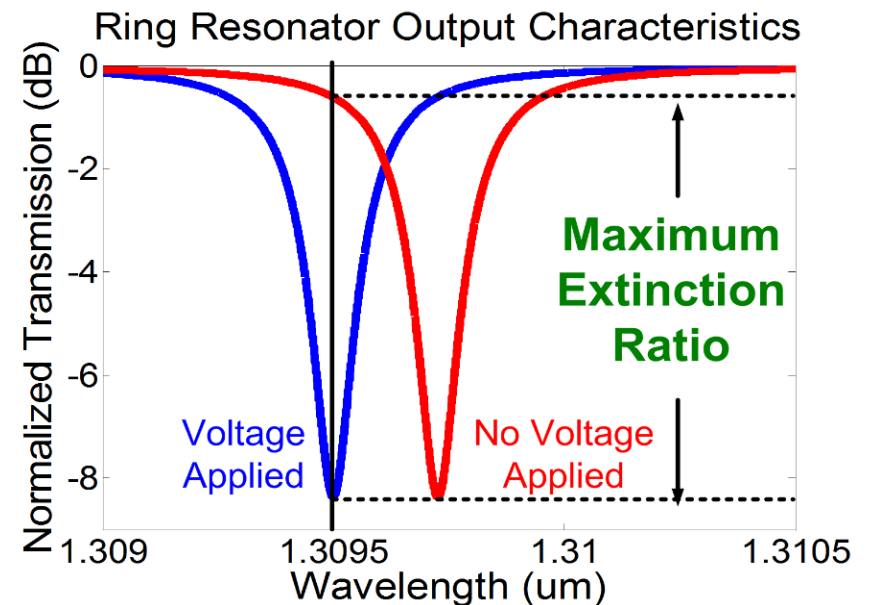
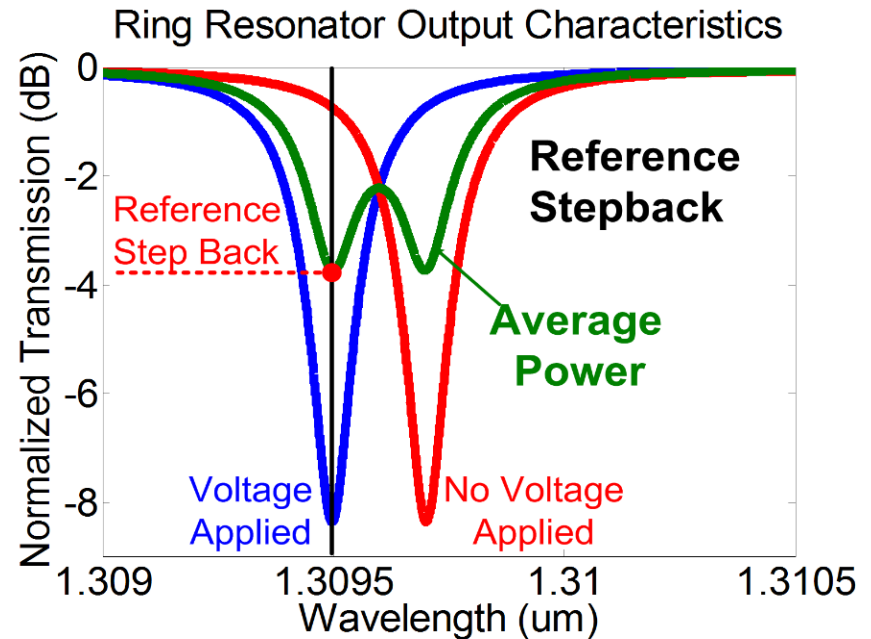
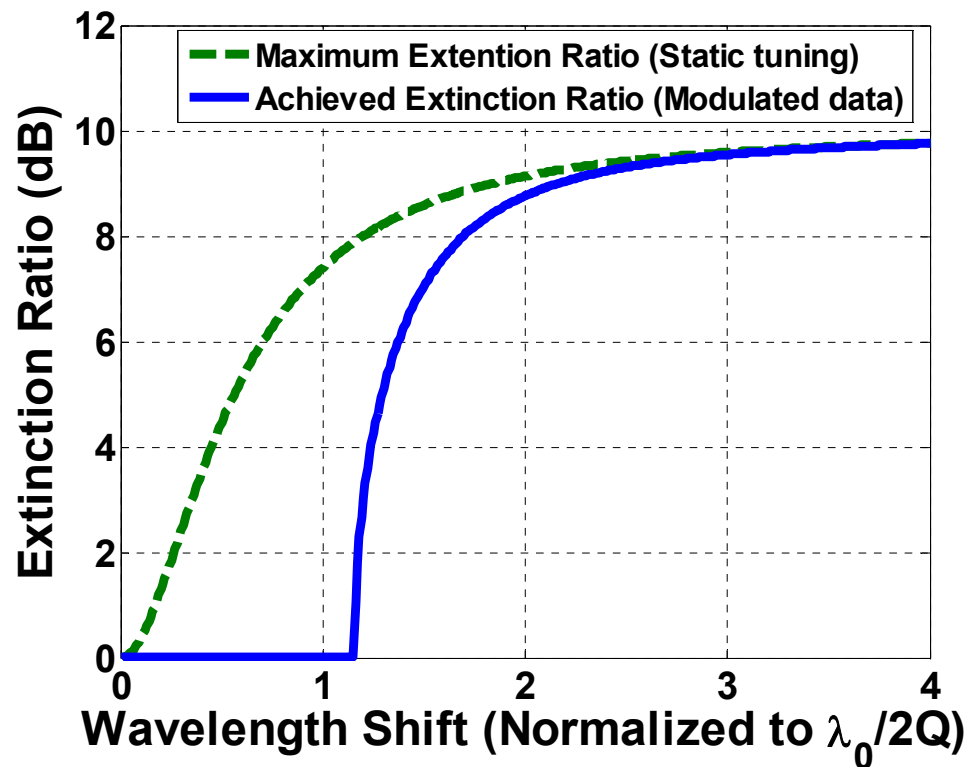
- Automatic tuning loop sets ring output power to a DAC-generated reference level corresponding to the ring's resonance point
- Also applicable for thermal tuning

Static Tuning Mode

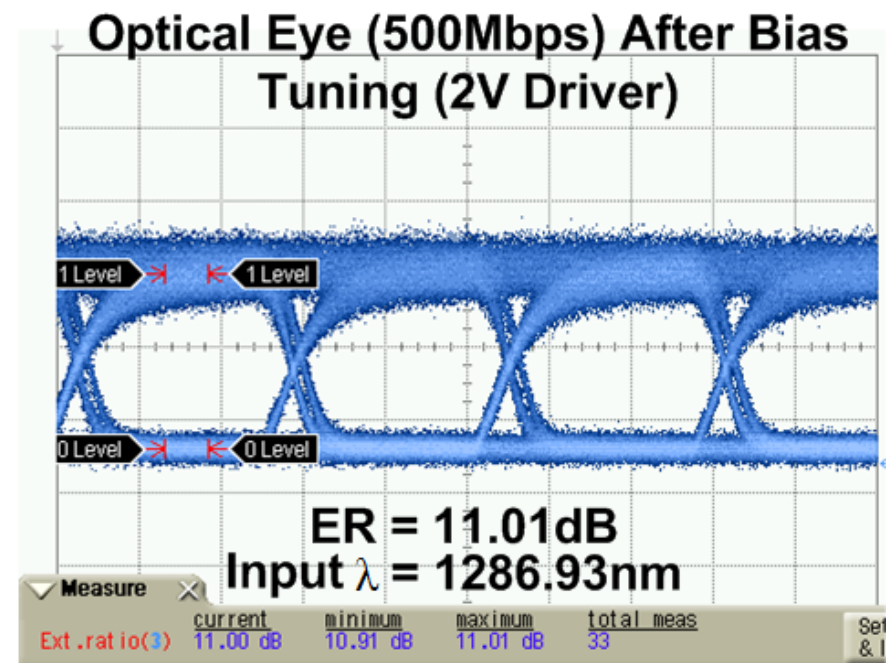
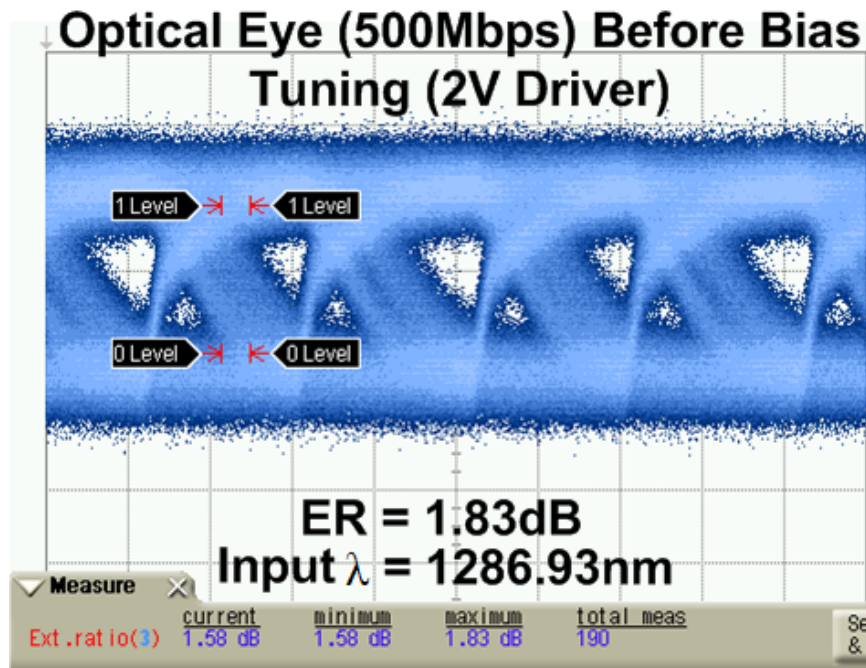


Dynamic Tuning Mode

- Lock to average power
- $\Delta \lambda_{shift} \geq FWHM$ to guarantee ER



Bias-Based Tuning Measurements



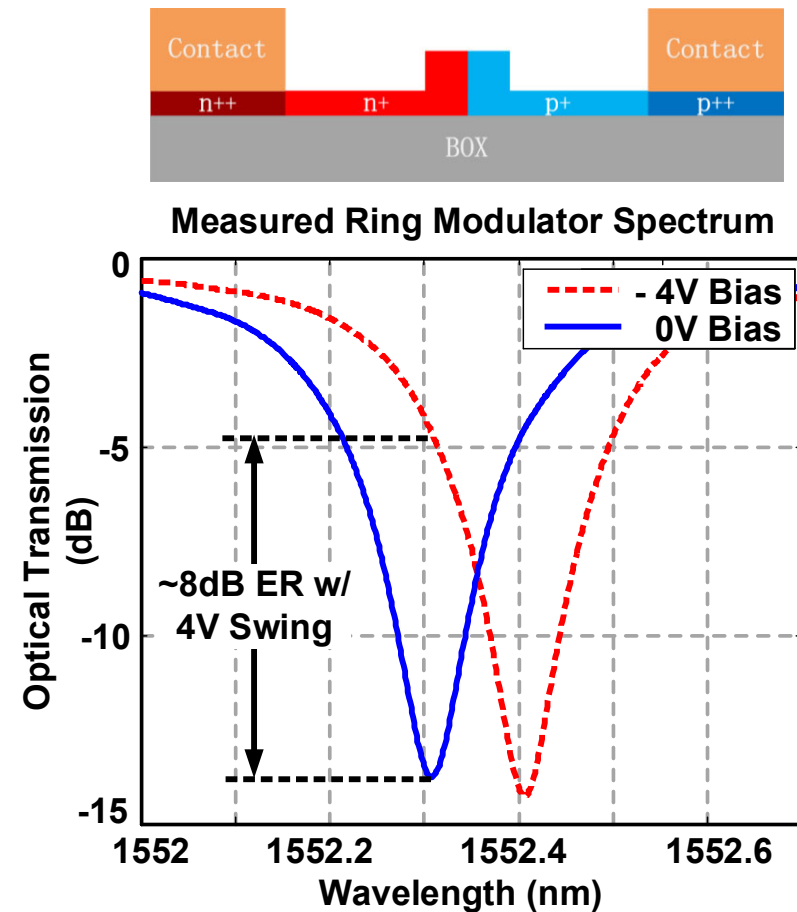
- Extinction ratio dramatically improved after bias-based tuning
- $340\mu\text{W}$ for a tuning range of 0.28nm

Agenda

- WDM Optical Interconnect Motivation
- Silicon Photonic Modulators
- Carrier-Injection Ring Resonator Modulators
- Carrier-Depletion Ring Resonator Modulators
 - Device Operation and Modeling
 - High-Speed Driver
 - Wavelength Stabilization Loop

Carrier-Depletion Ring Modulator Challenge I: Output Swing & Biasing

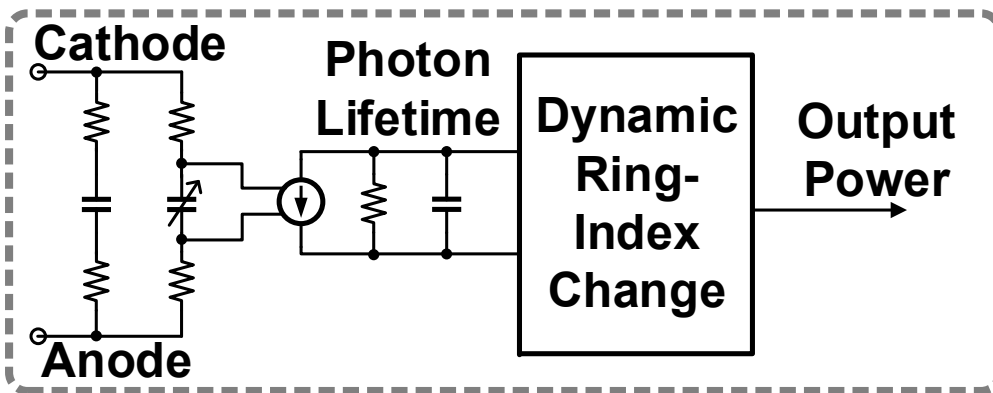
	ISSCC 2013	This Work
Ring Type	Injection	Depletion
Doping Profile	PIN	Lateral PN
Q	8000	5000
Tunability (pm/V)	350	25
Data Rate	9Gb/s	25Gb/s
Swing for >7dB ER	< 2V _{pp}	> 4V _{pp}



- High-speed depletion-mode ring modulator requires:
 - Large swing: >4V
 - Negative DC-bias: -2V

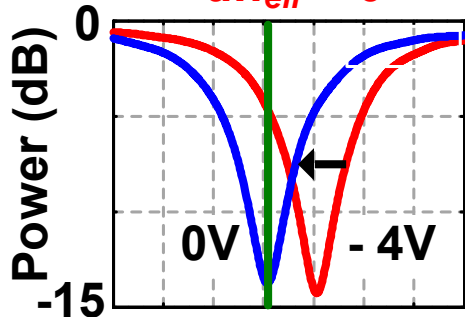
Carrier-Depletion Ring Modulator Challenge II: Nonlinear Dynamics

Depletion Ring E-O Model



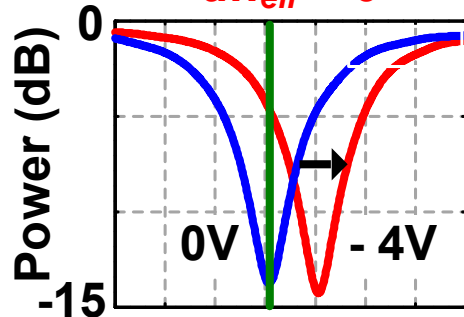
Falling Edge

$$dn_{eff} > 0$$

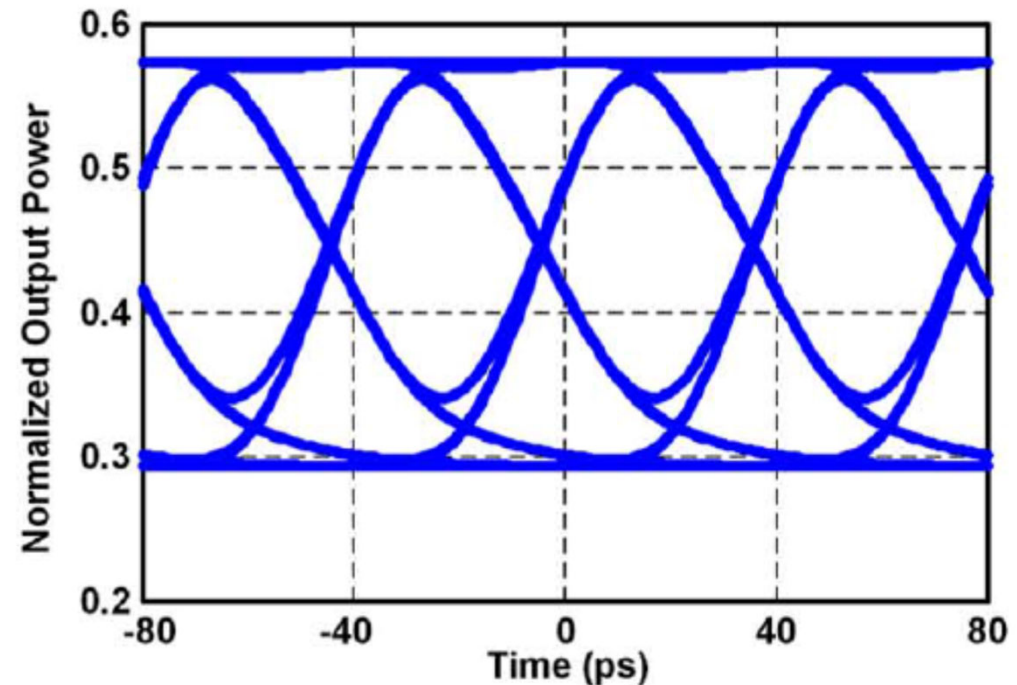


Rising Edge

$$dn_{eff} < 0$$



Simulated 25Gb/s Optical Eye Diagram



- Dynamic change of n_{eff} → unequal rise/fall times
- Asymmetric equalization for non-linearity cancellation

Carrier-Depletion Ring Modulator Modeling

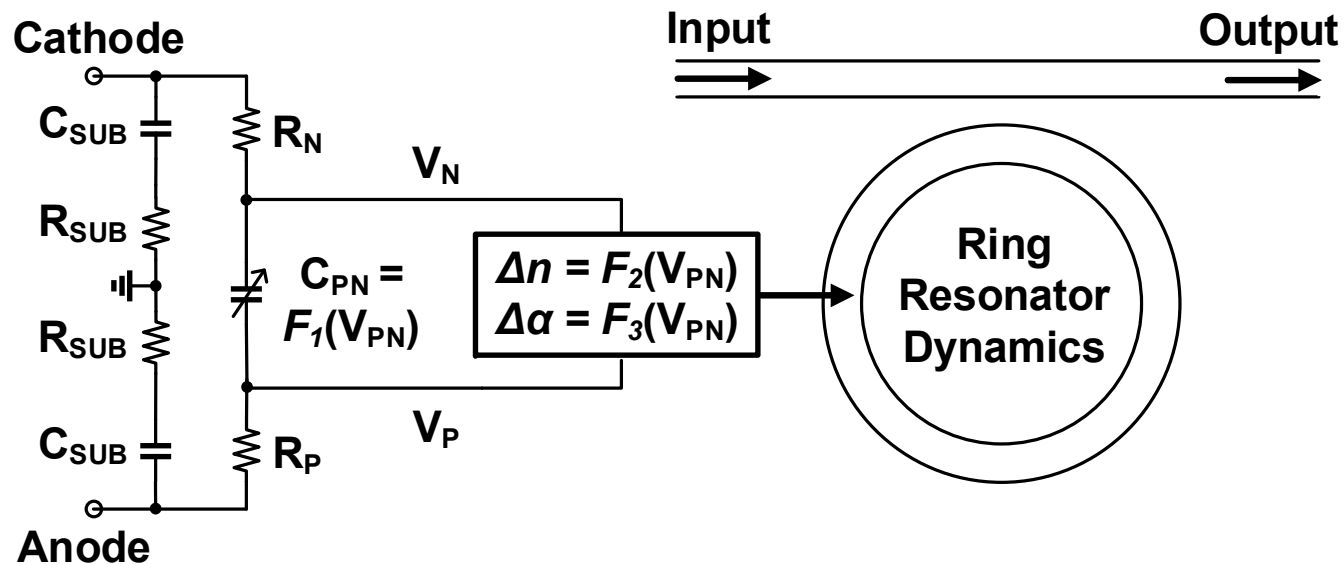
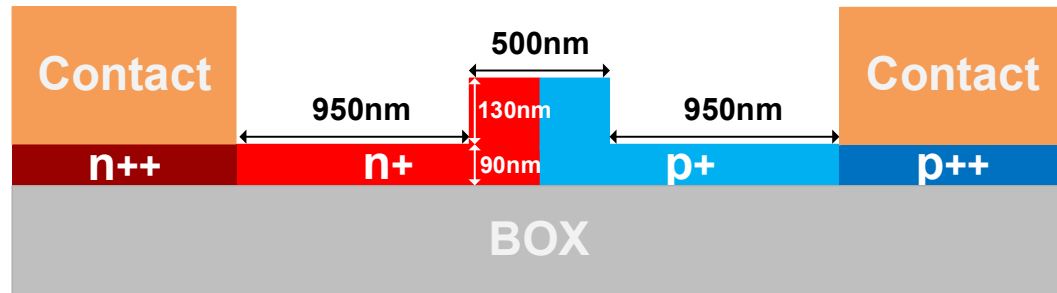
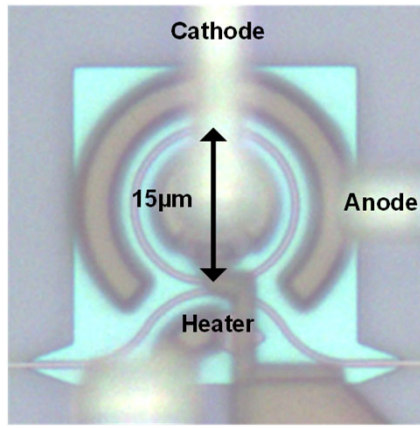
- Previous Related Modeling Work
 - Ring resonator model (only optical dynamics) [Smy JOSA B 2011]
 - Carrier-depletion ring modulator model (lack electrical dynamics) [Zhang JSTQE 2010, Buckwalter JSSC 2012, Ban OIC 2015]
- Carrier-Depletion Ring Modulator Model
 - Capture nonlinear electrical and optical dynamics

[Ashkan Roashan-Zamir, **Binhao Wang**, Shashank Telaprolu, Kunzhi Yu, Cheng Li, M. Ashkan Seyed, Marco Fiorentino, Raymond Beausoleil, and Samuel Palermo, "A 40Gb/s PAM4 Silicon Microring Resonator Modulator Transmitter in 65nm CMOS," accepted in IEEE Optical Interconnects Conference (OIC), 2016]

[Hao Li, Zhe Xuan, Cheng Li, Alex Titriku, Kunzhi Yu, **Binhao Wang**, Nan Qi, Ayman Shafik, Marco Fiorentino, Michael Hochberg, Samuel Palermo, and Patrick Yin Chiang, "A 25Gb/s, 4.4V Swing, AC-Coupled Ring Modulator-Based WDM Transmitter with Wavelength Stabilization in 65nm CMOS," IEEE Journal of Solid-State Circuits (JSSC), 50, 3145-3159, 2015]

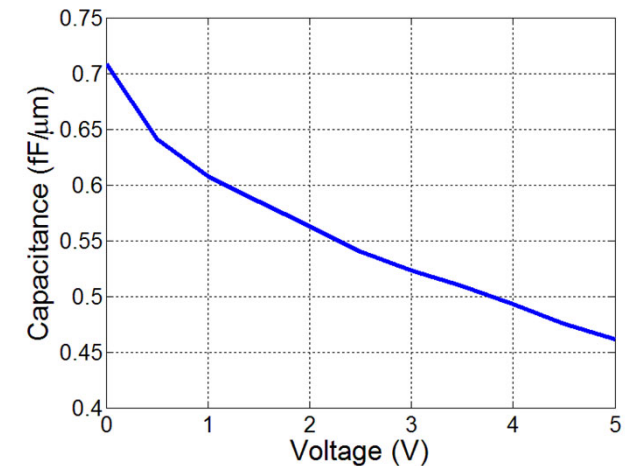
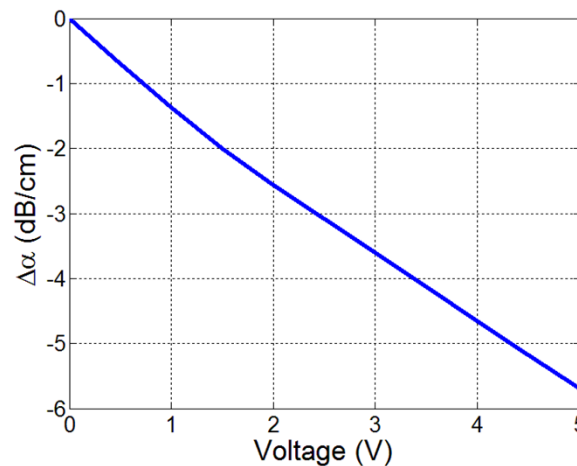
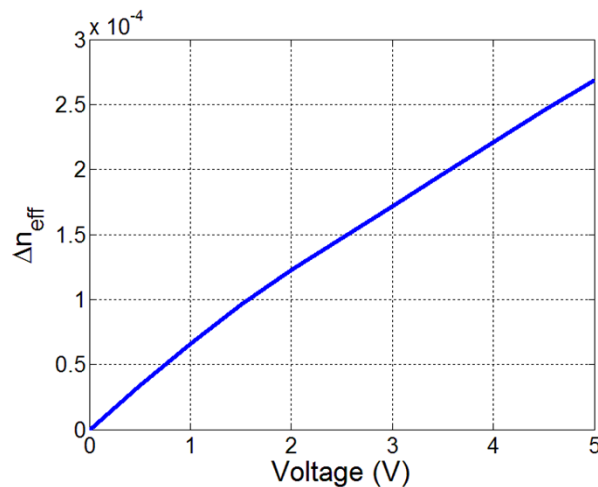
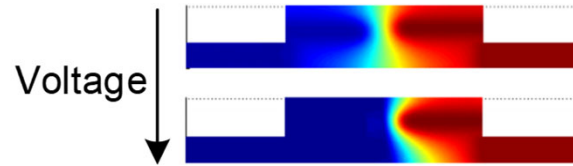
[Hao Li, Zhe Xuan, Cheng Li, Alex Titriku, Kunzhi Yu, **Binhao Wang**, Nan Qi, Ayman Shafik, Marco Fiorentino, Michael Hochberg, Samuel Palermo, and Patrick Yin Chiang, "A 25Gb/s, 4.4V Swing, AC-Coupled, Si-Photonic Microring Transmitter with 2-Tap Asymmetric FFE and Dynamic Thermal Tuning in 65nm CMOS," IEEE International Solid-State Circuits Conference (ISSCC), 2015]

Carrier-Depletion Ring Modulator Model



Electrical Modeling - Parameter Extraction

- Lumerical

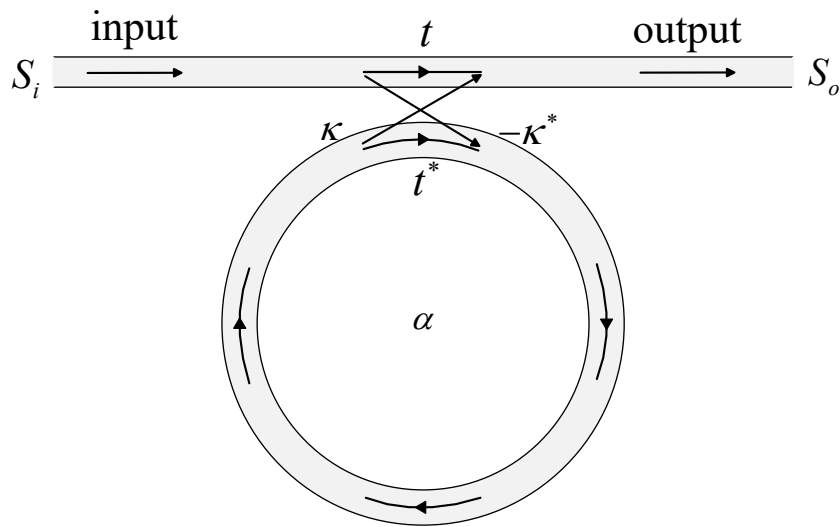


- Polynomial Curve Fitting

$$f(V) = a_0 + a_1V + a_2V^2 + a_3V^3 + a_4V^4$$

Parameter	Unit	a_0	a_1	a_2	a_3	a_4
Δn_{eff}	-	-4.3×10^{-7}	7.3×10^{-5}	8.0×10^{-6}	1.1×10^{-6}	5.2×10^{-8}
$\Delta \alpha$	dB/cm	0.01	1.5	0.17	-2.3×10^{-2}	1.0×10^{-3}
C	fF/μm	0.71	-0.14	5.5×10^{-2}	-1.2×10^{-2}	1.0×10^{-3}

Optical Modeling - Optical Dynamics



$$\frac{\partial A}{\partial t} = \left(2\pi c j \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right) - \frac{1}{\tau} \right) A + j\mu S_i$$

$$S_o = S_i + j\mu A$$

where $1/\tau = 1/\tau_c + 1/\tau_l$

$$\mu^2 = \kappa^2 v_g / 2\pi R = 2/\tau_c$$

$$\tau_l = 1 / \left(v_g e^{2\pi R(\alpha_0 + 0.75\Delta\alpha)} \right)$$

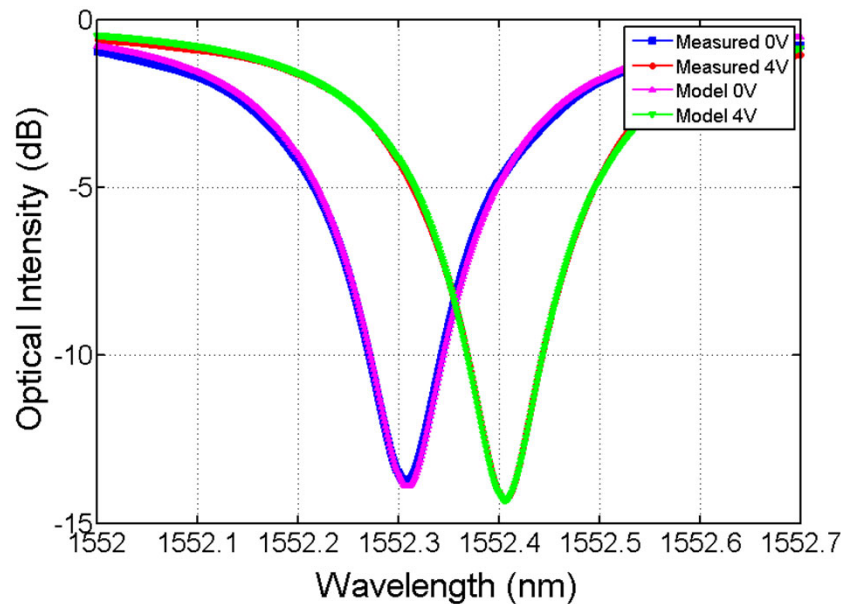
$$2\pi(n_0 + \Delta n)R = m\lambda_0$$

[Little JLT 1997]

- Capture non-linear optical dynamics
- Require less memory and computation time

Optical Modeling - Optical Dynamics

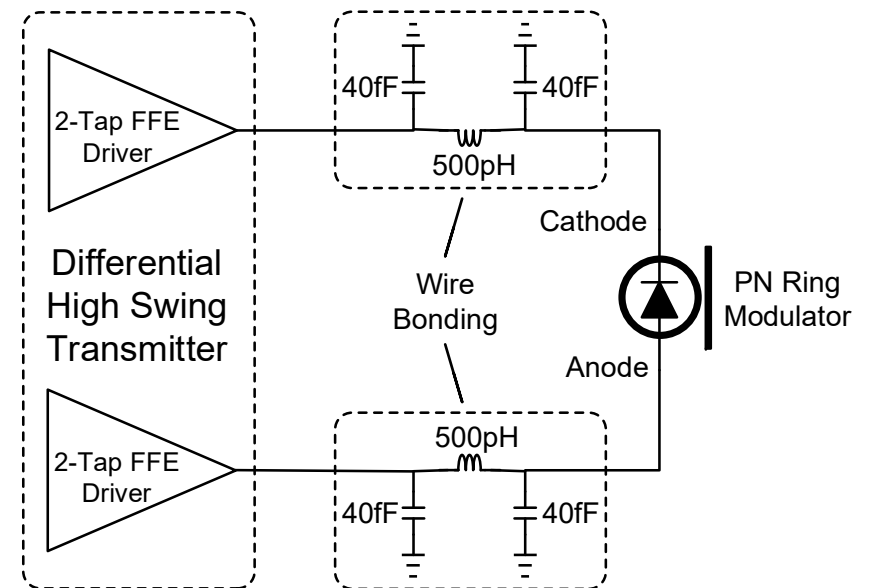
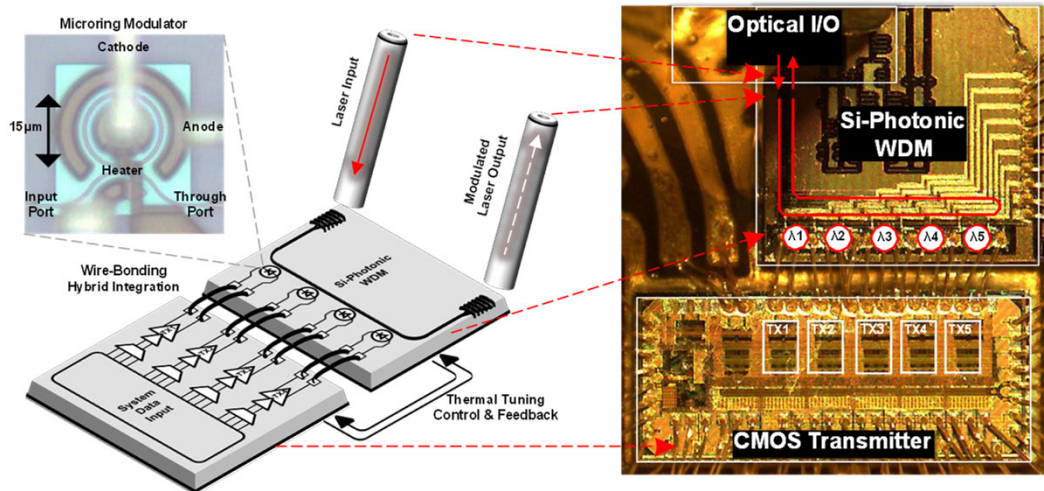
- Parameter Extraction



Parameter	Unit	Description	Value
τ	ps	Amplitude decay time	9.07
κ	-	Coupling ratio	0.188
m	-	Mode number	28
r	μm	Ring radius	7.5

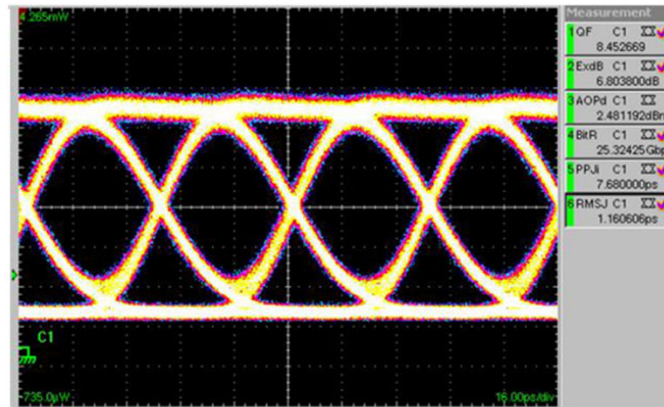
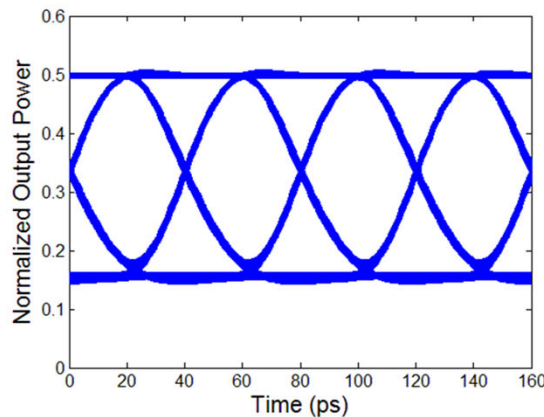
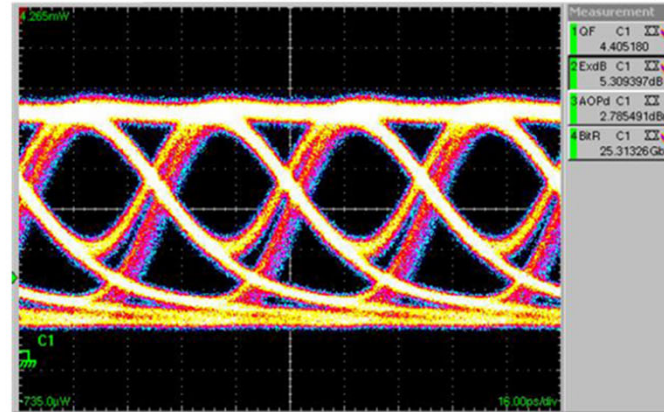
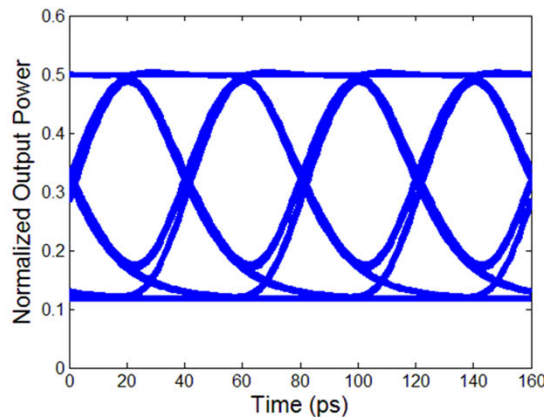
- Extinction ratio (ER) $\sim 10\text{dB}$
- Modulation efficiency $\sim 25\text{pm/V}$

Co-Simulation with CMOS Driver



- Optical transmitter prototype assembly
- Optical transmitter co-simulation schematic

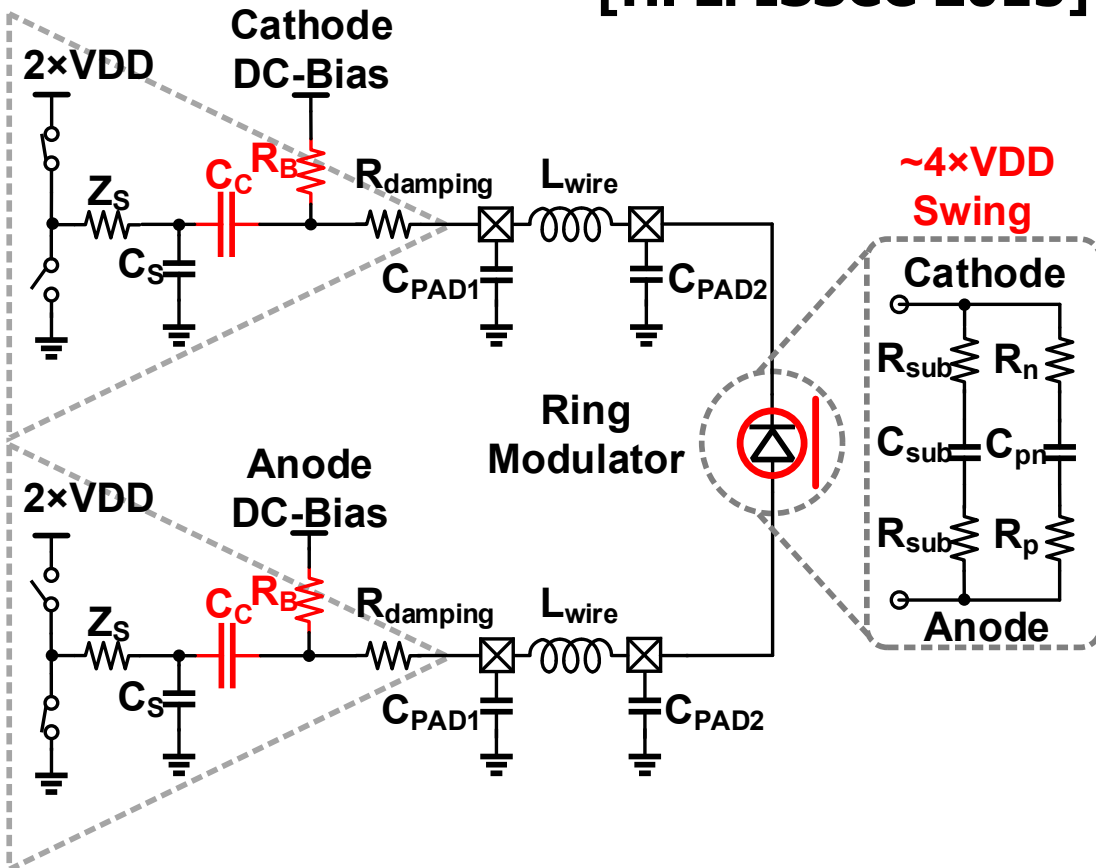
Measured and Co-Simulated 25Gb/s Eye Diagrams



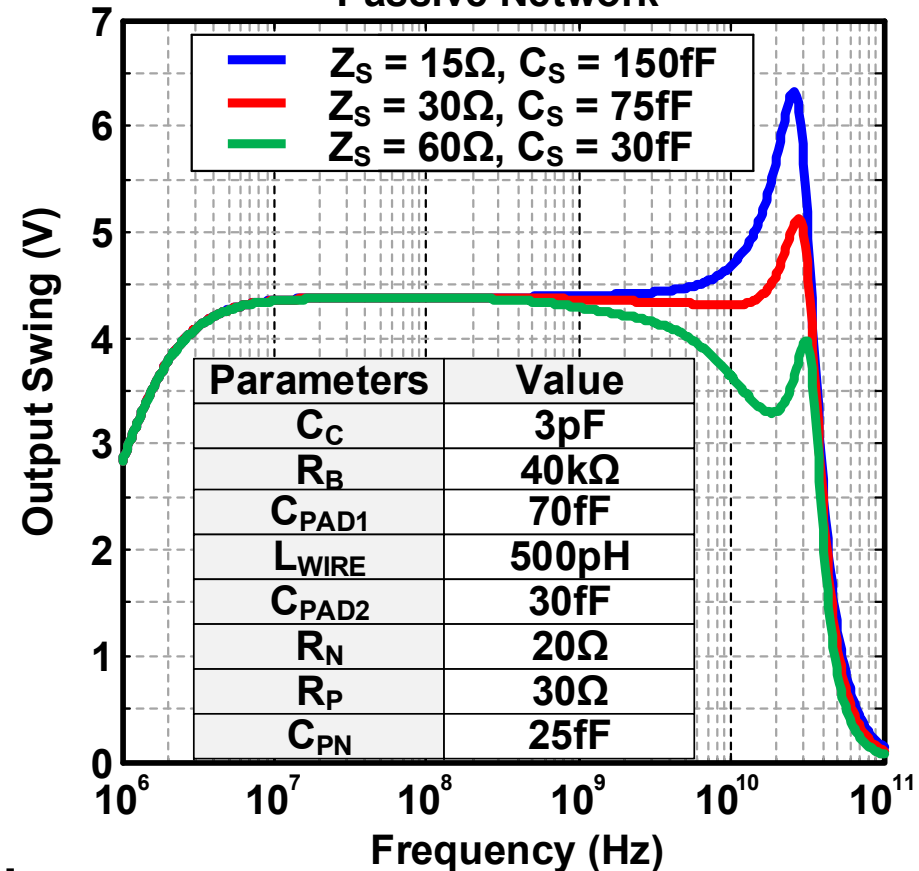
- Asymmetrical ISI is due to the device nonlinearity
- It is compensated by an optimized nonlinear equalizer

Proposed AC-Coupled Differential Driver

[H. Li ISSCC 2015]



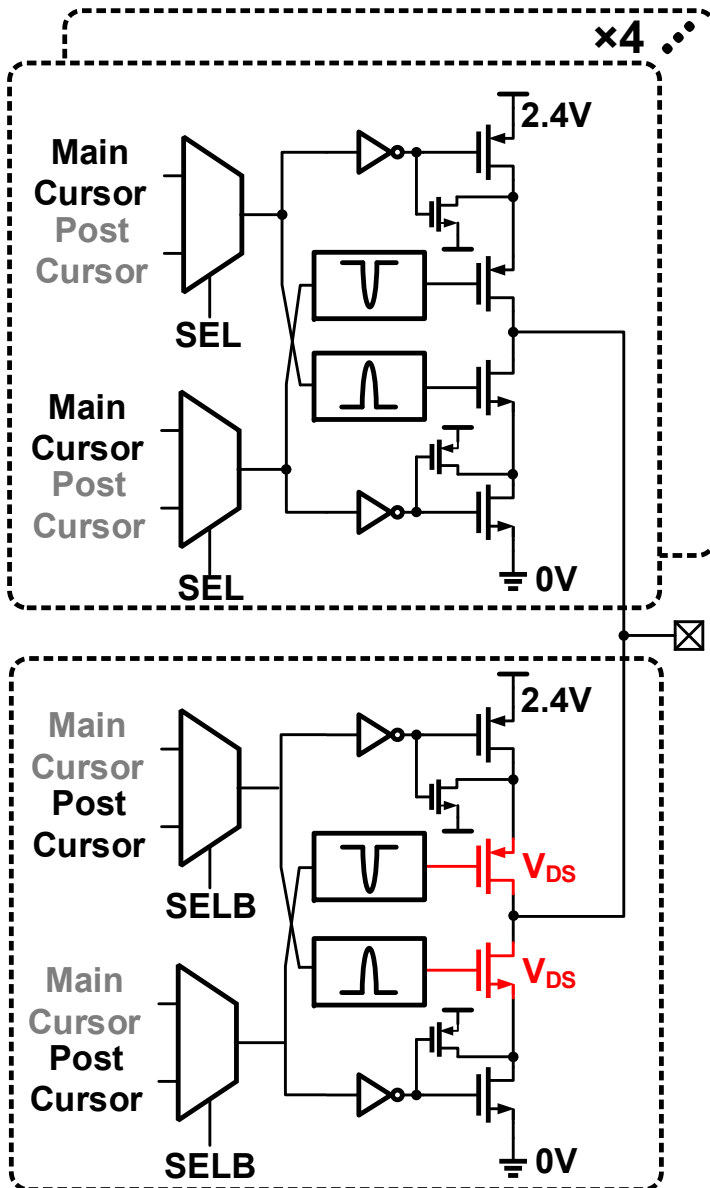
Simulated AC-Response of Output Passive Network



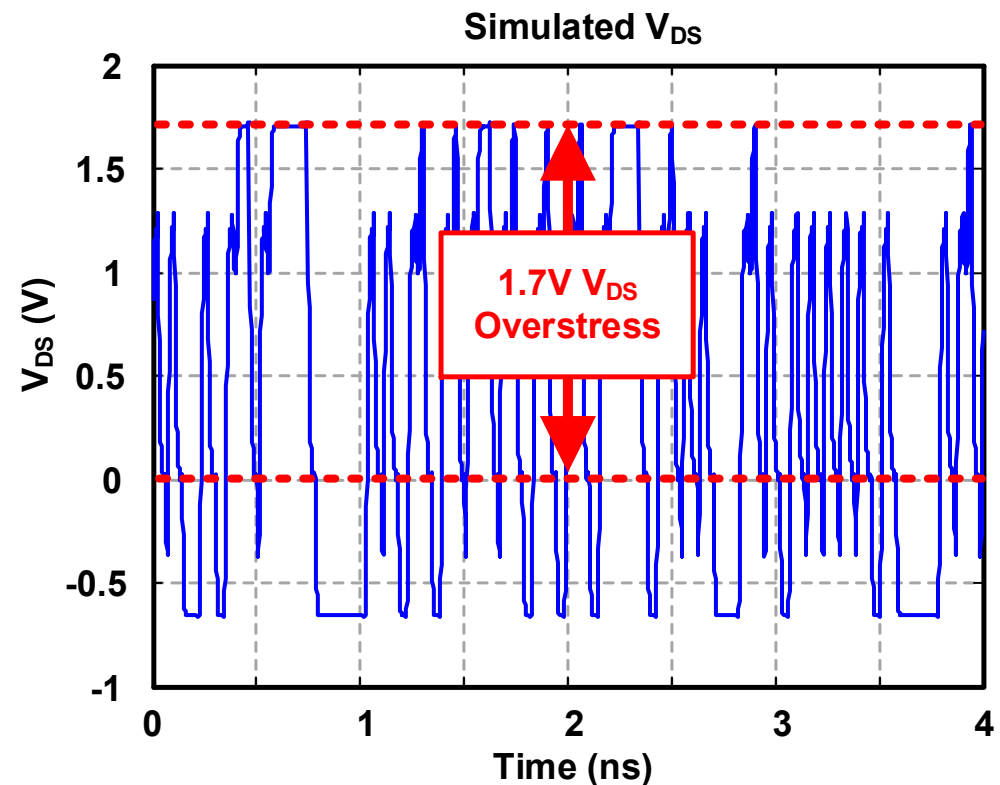
- $C_C = 3\text{pF} \rightarrow 4.4\text{V}$ differential swing
- $Z_S < 30\Omega$ to minimize low-pass attenuation
- High-pass cut-off: $< 10\text{MHz}$

Conventional Segmented Output Driver

Segmented Output Stage

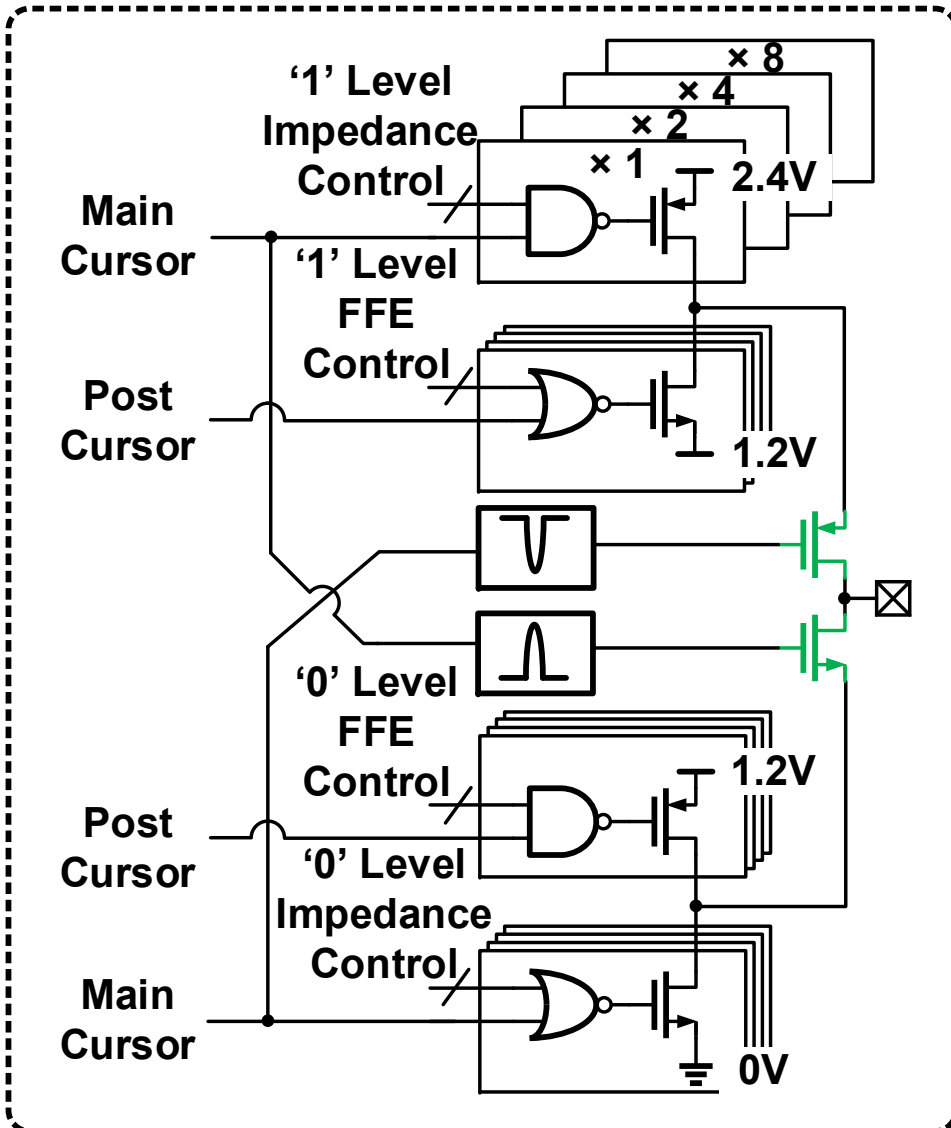


- Cascode transistors suffer V_{DS} overstress
- Large parasitic capacitance due to segmented design



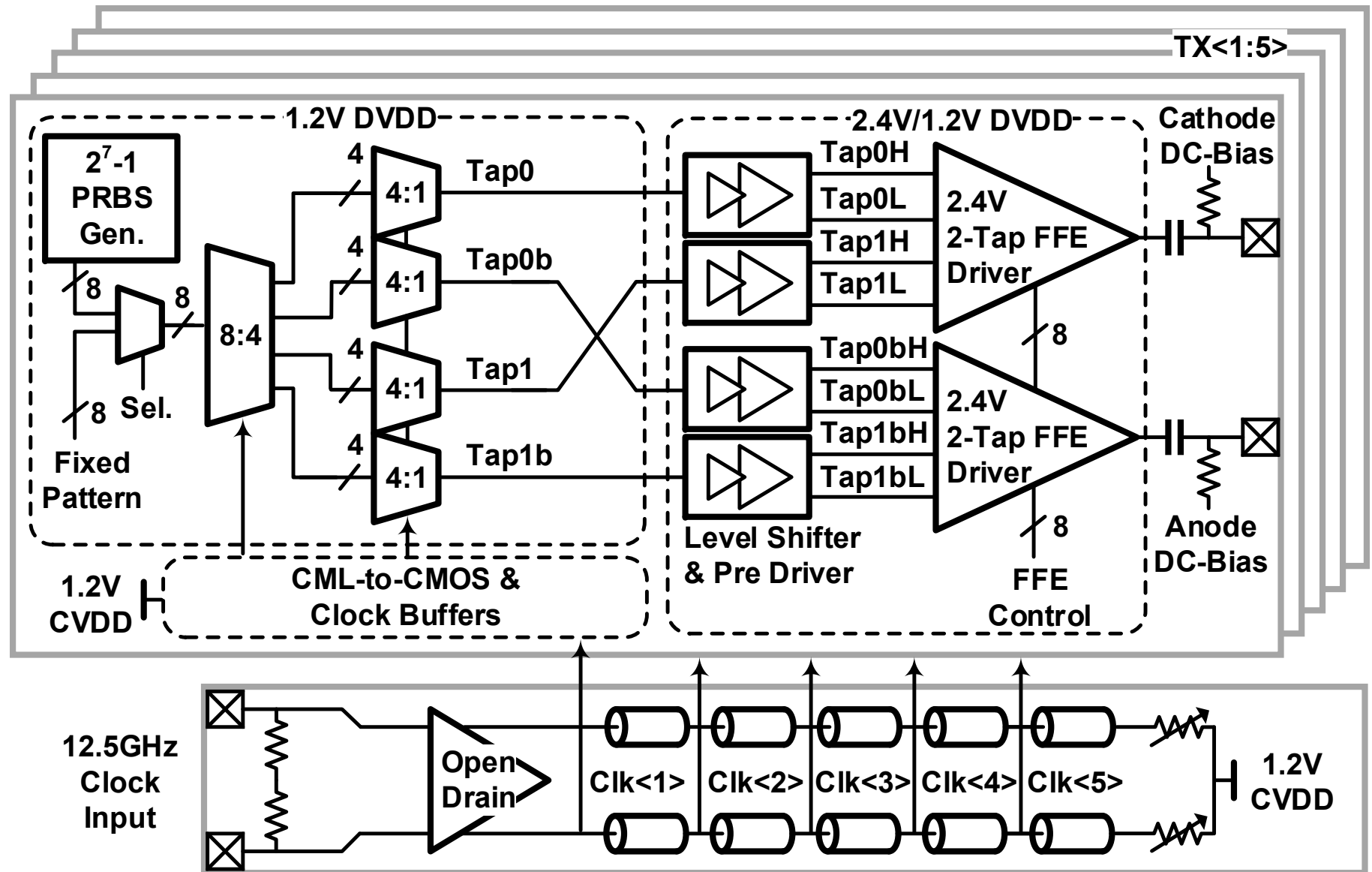
Proposed 2-Tap FFE Output Driver

Merged Output Stage

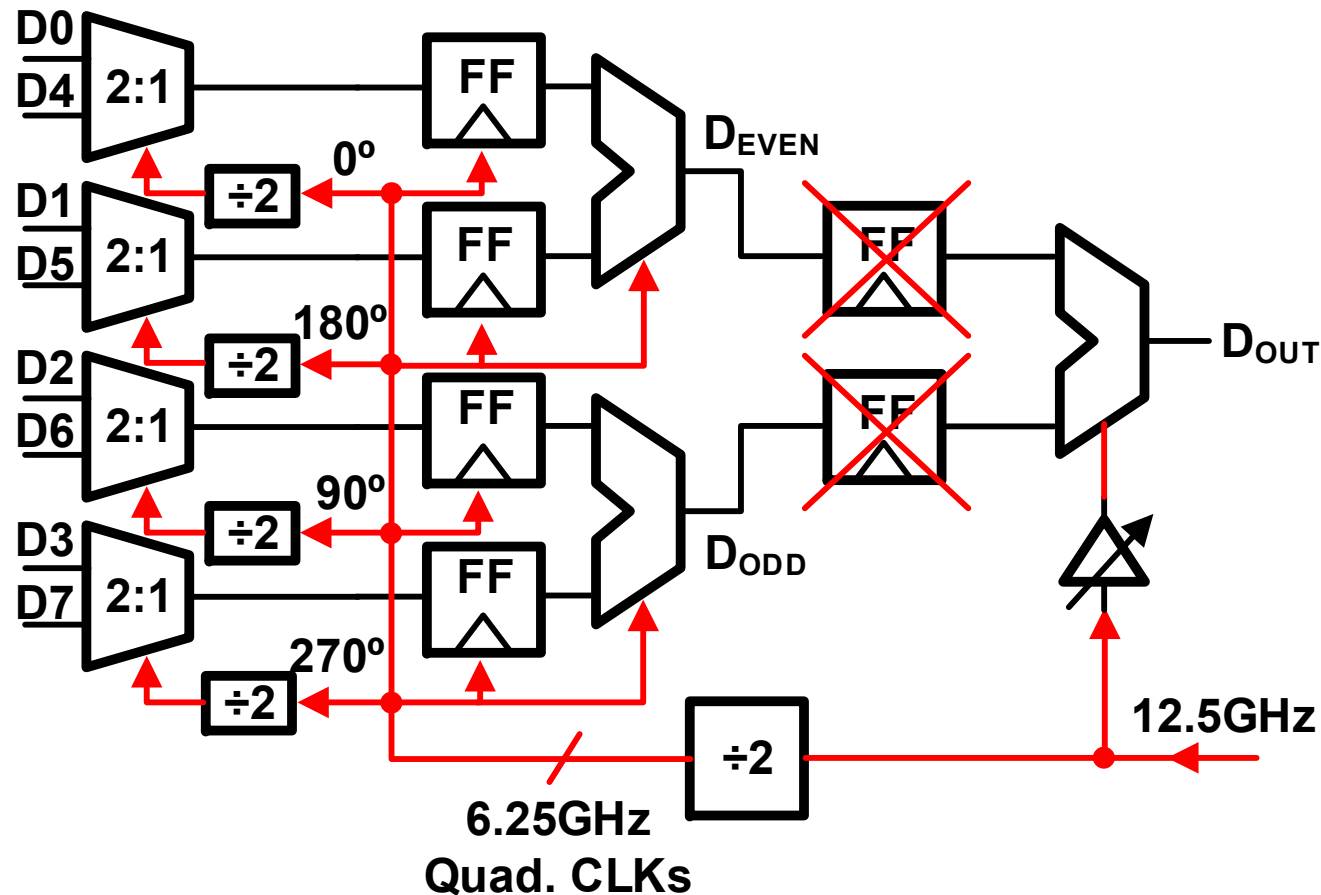


- Merged cascode transistors
- No V_{DS} overstress
- Reduced parasitics and area
- Independent '1'-Level and '0'-Level FFE coefficients

Transmitter Architecture

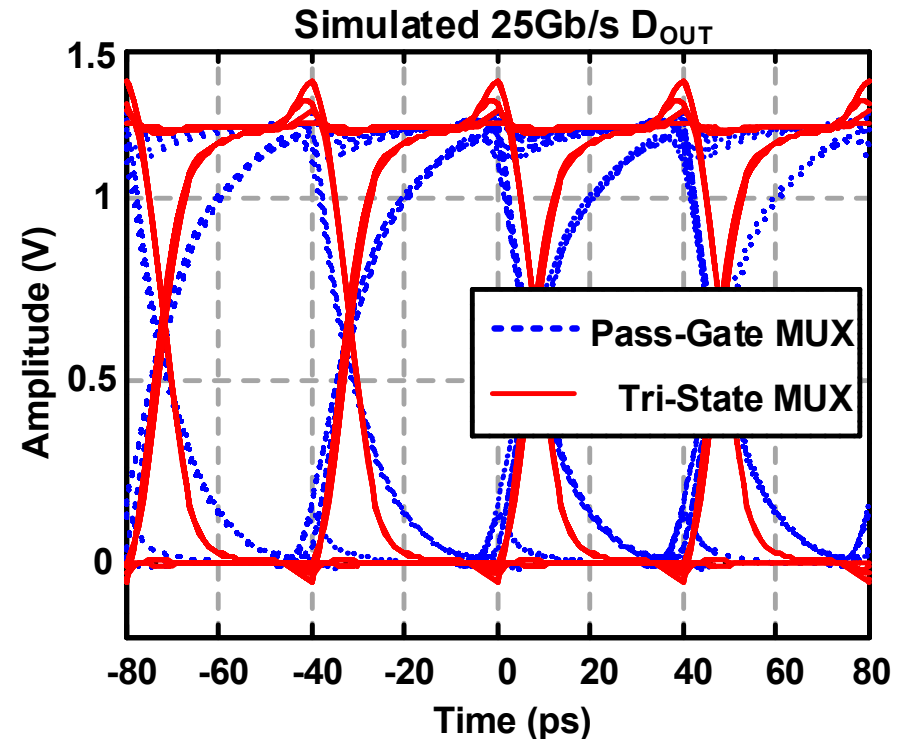
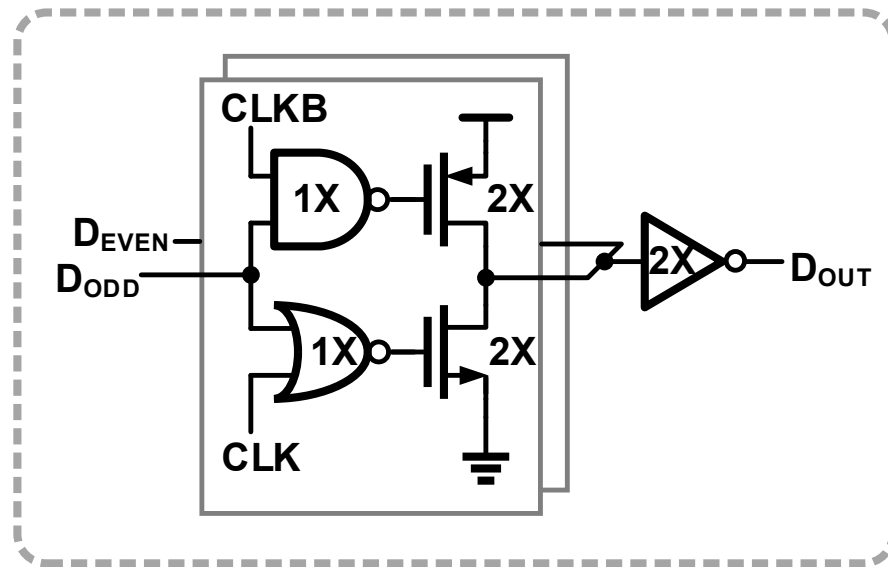


25Gb/s 8:1 CMOS Serializer



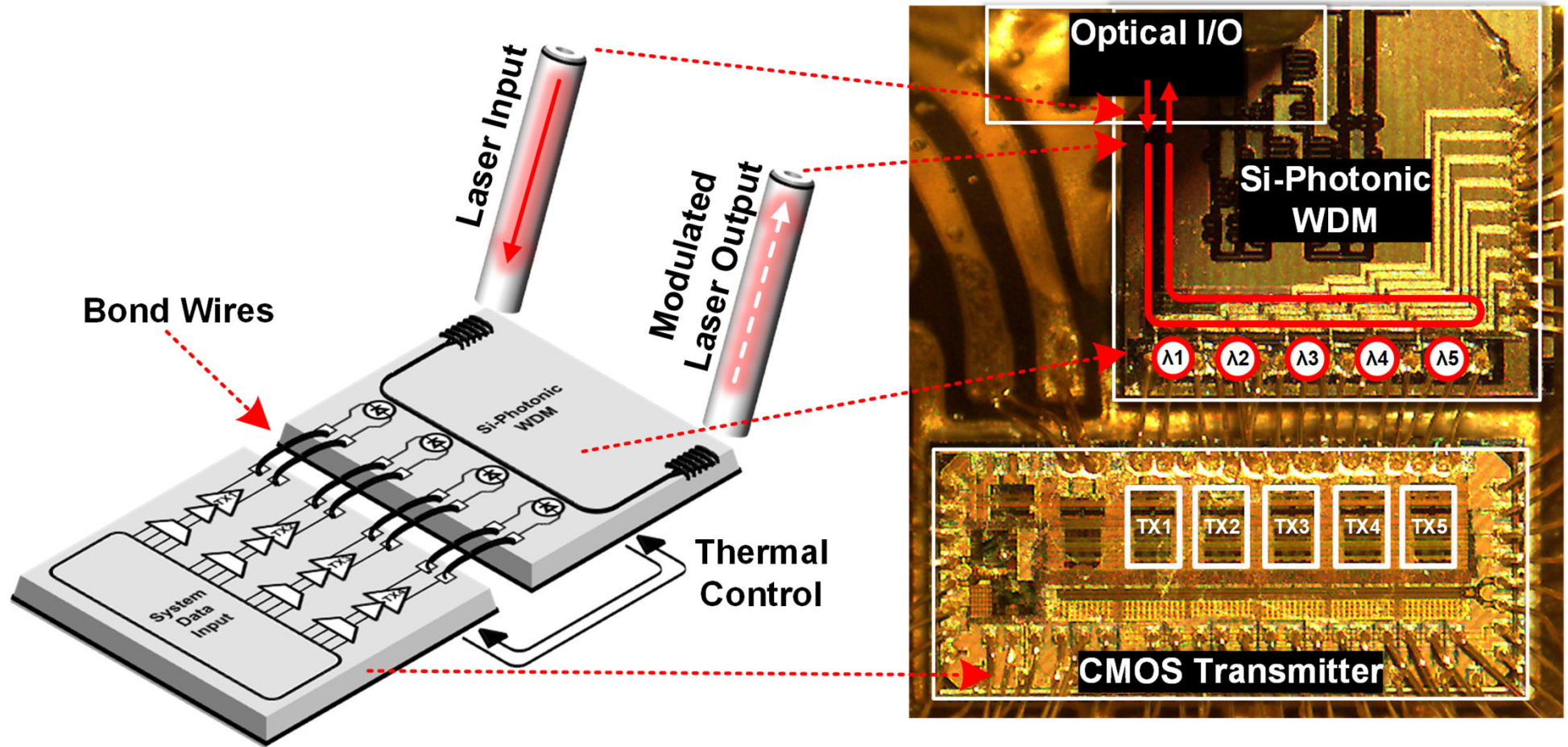
- Quadrature quarter-rate architecture eliminates high-speed retiming before final 2:1 MUX

25Gb/s 8:1 CMOS Serializer



- Tri-state inverter-based 2:1 MUX for fast edge rate

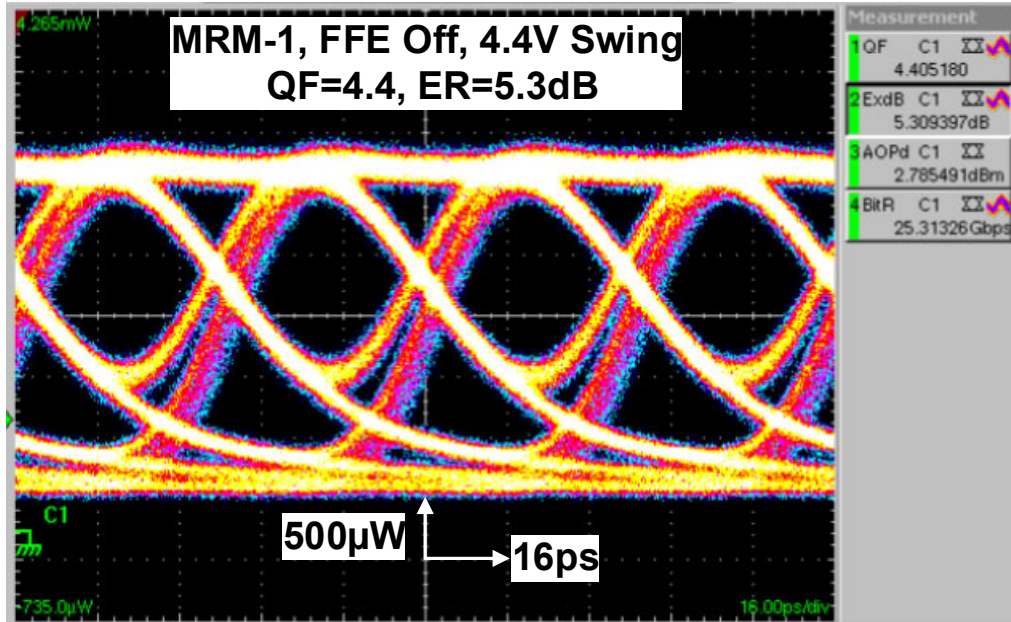
Heterogeneous Integration



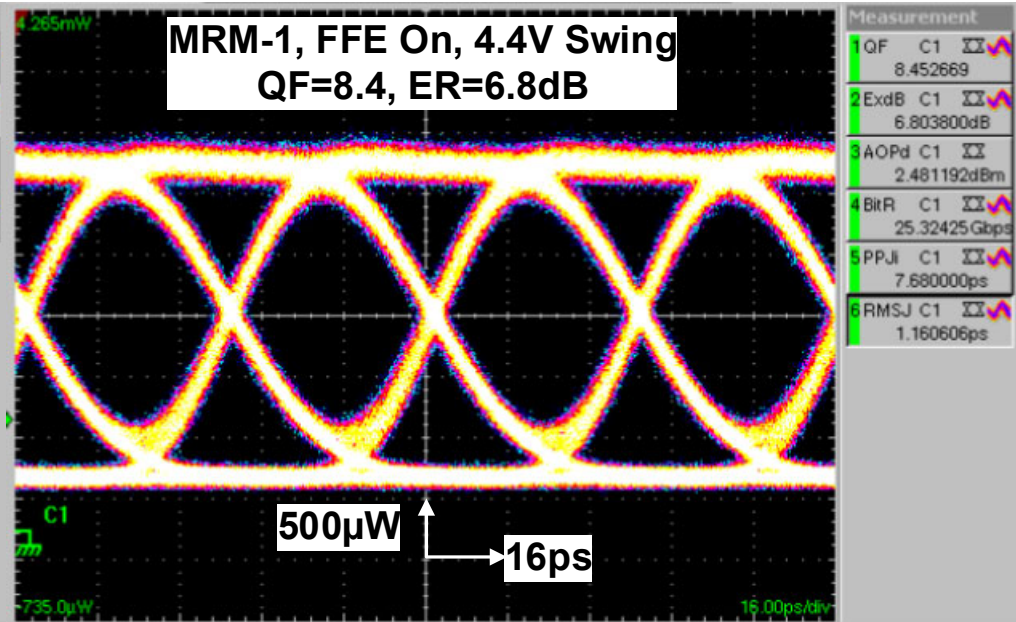
- Hybrid CMOS-Photonic packaging (<0.5mm bond-wires)
- Stable optical coupling using vertically-attached fibers

25Gb/s Optical Measurement

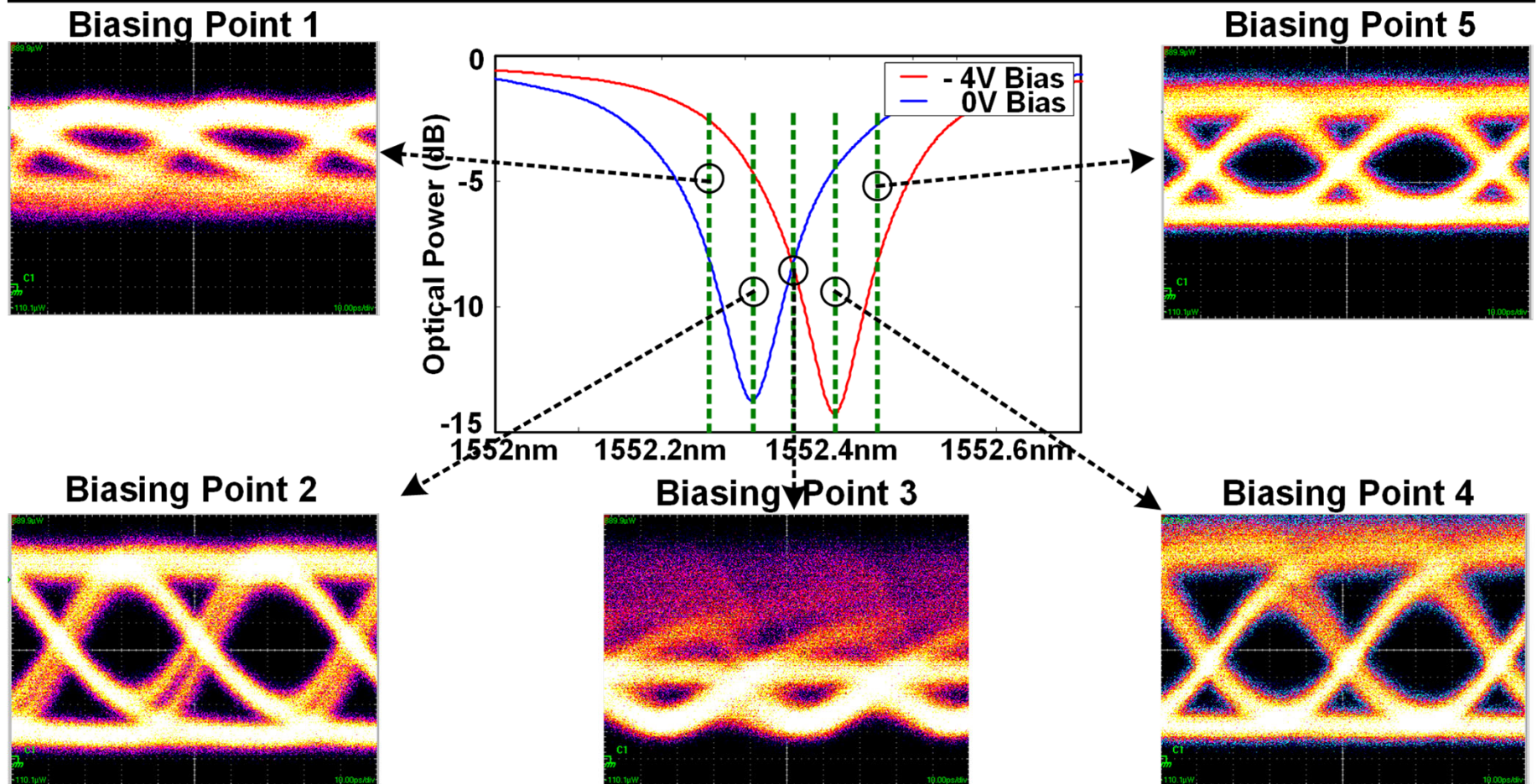
Test Channel 1 w/o FFE



Test Channel 1 w/ Asymmetric FFE

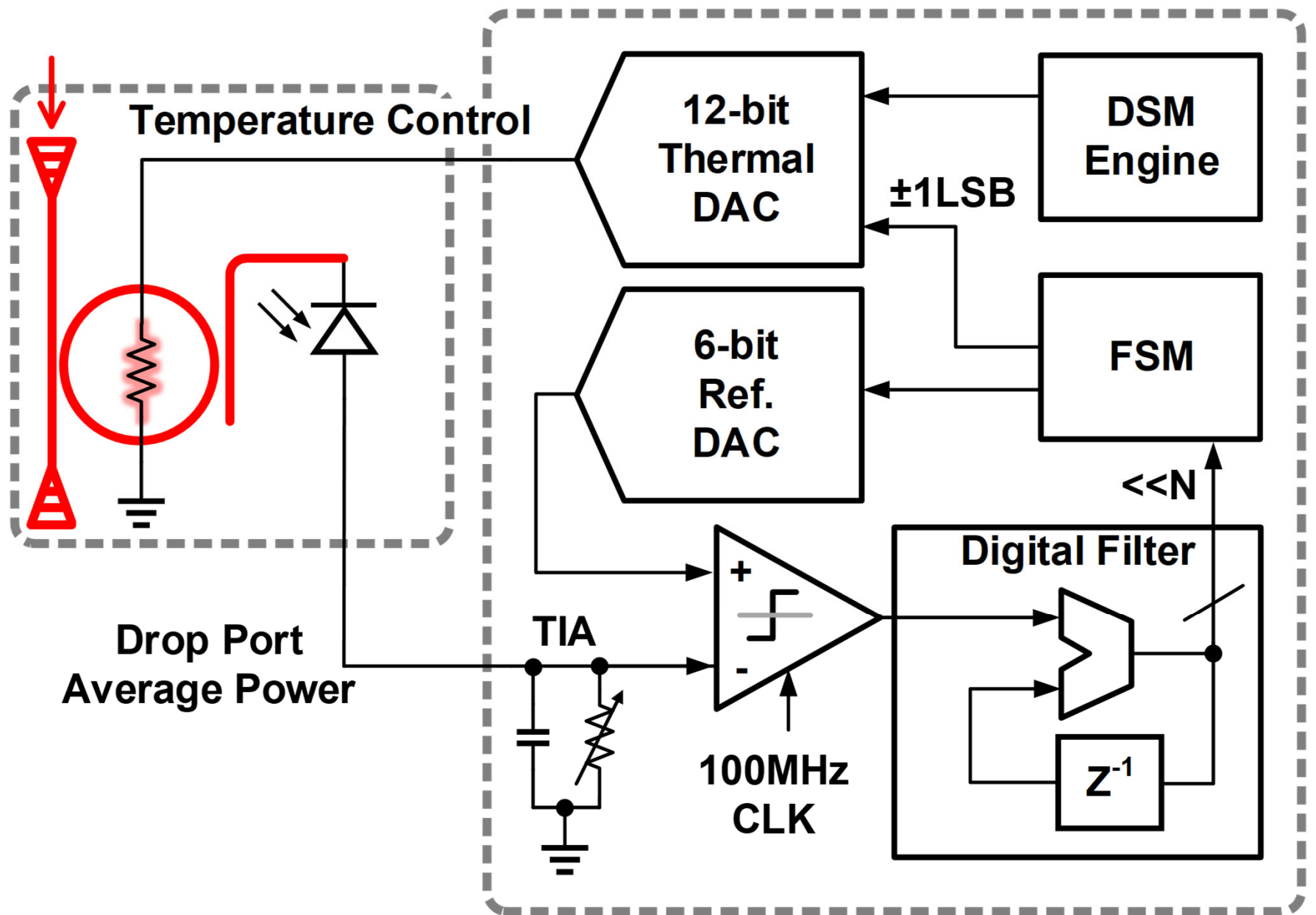


Challenge III: Wavelength Stability

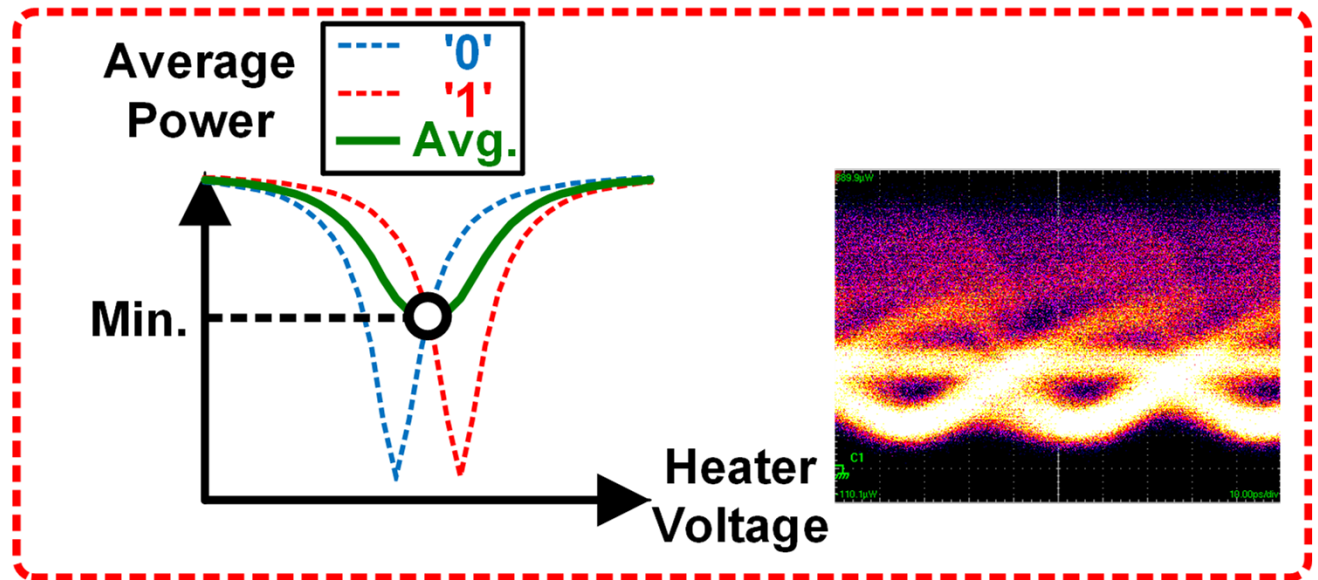
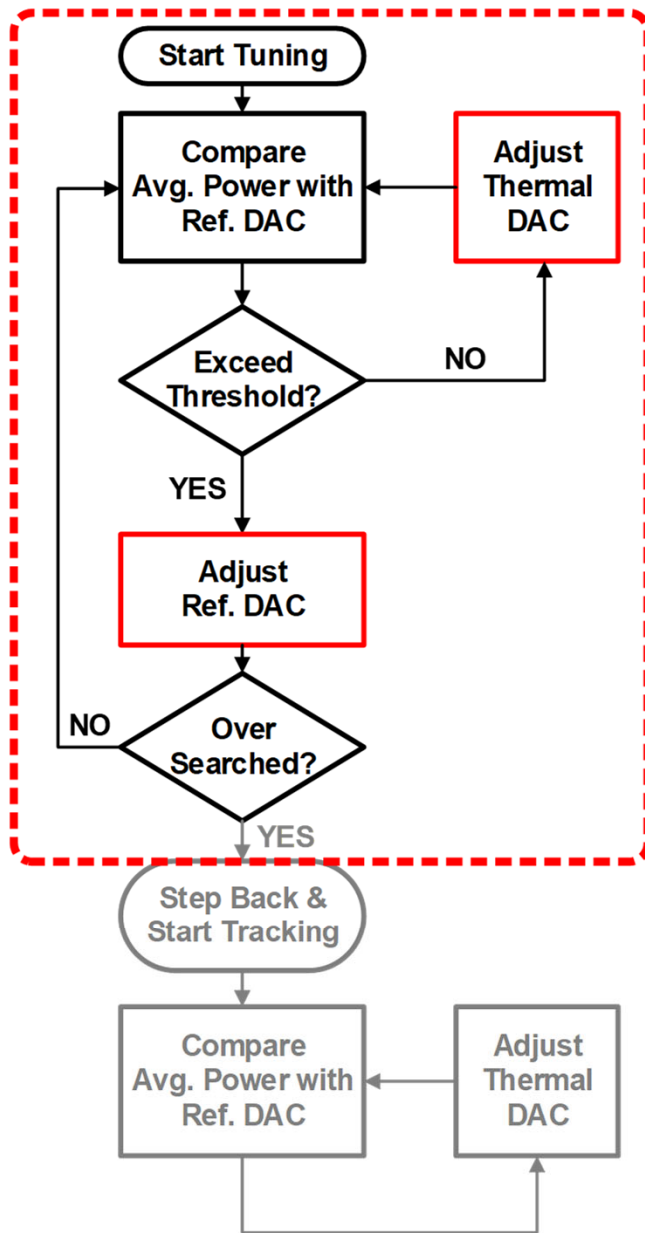


- Modulation efficiency depends strongly on wavelength
- ER degradation due to temperature fluctuation
- Closed-loop control is necessary for robust operation

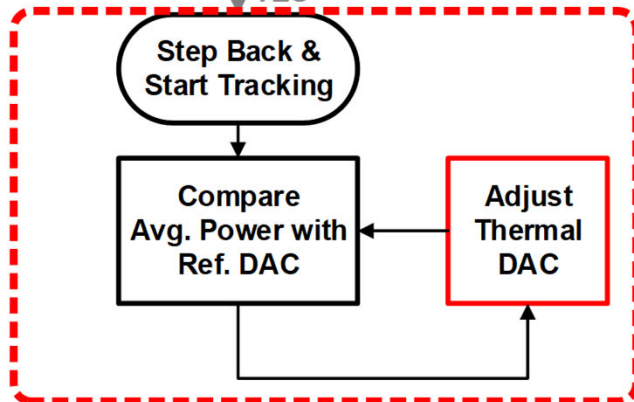
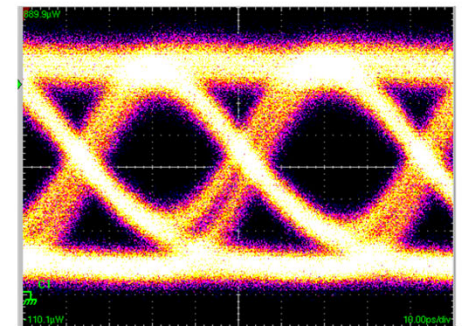
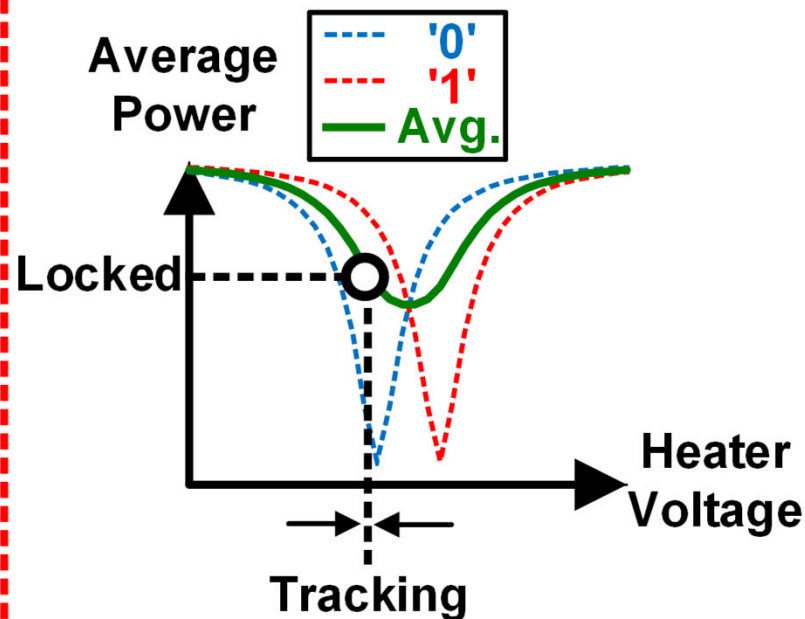
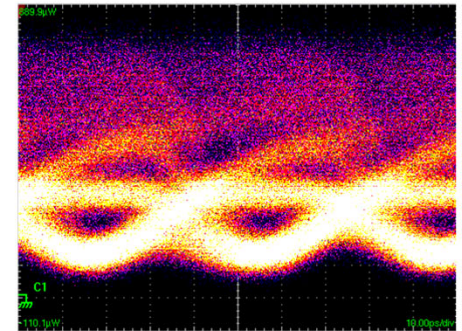
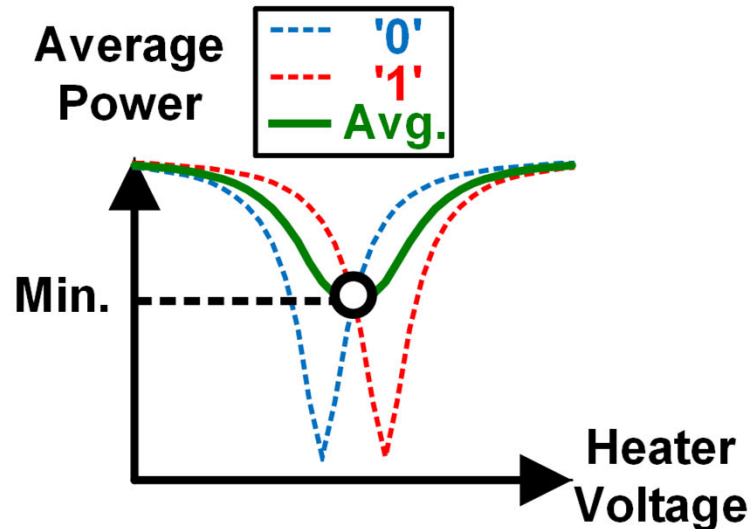
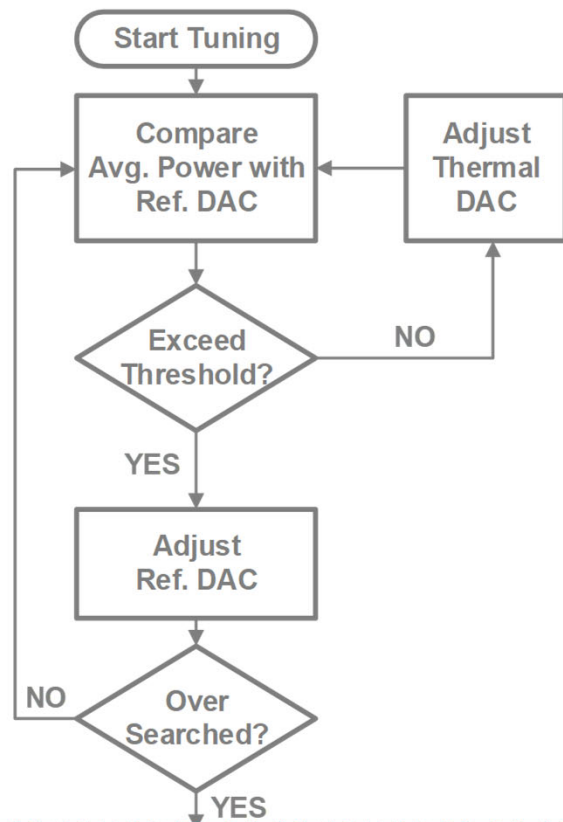
Average Power Thermal Stabilization



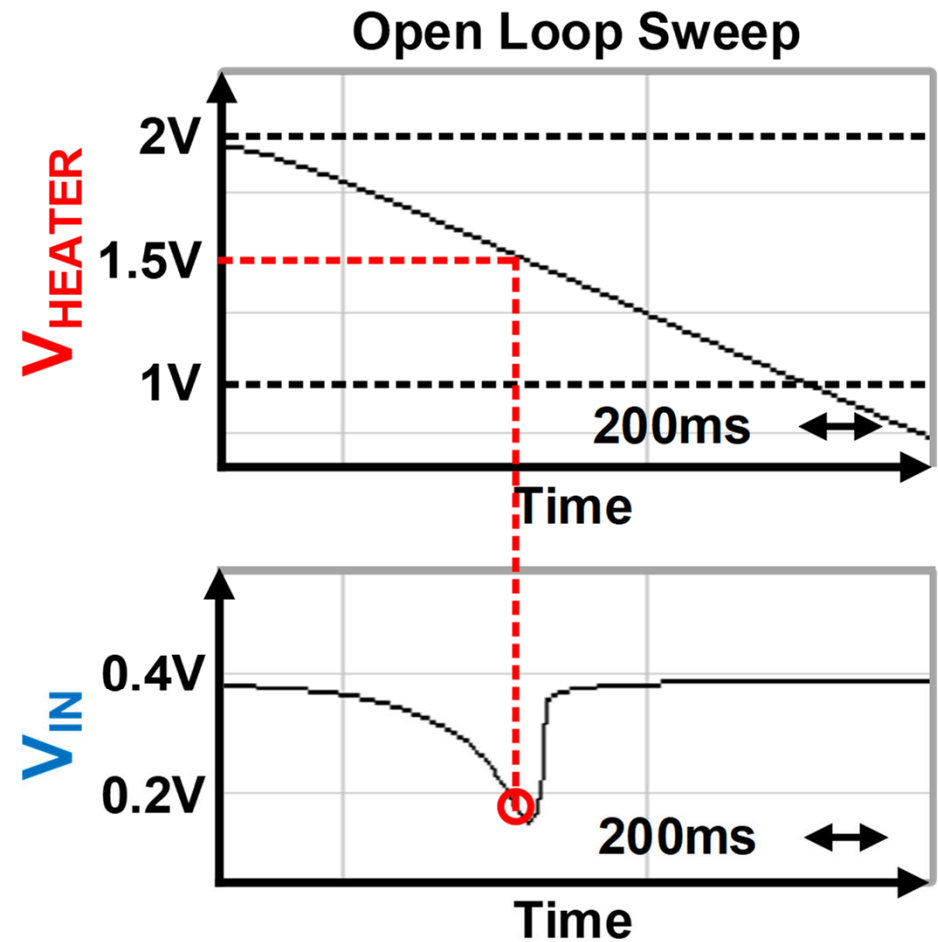
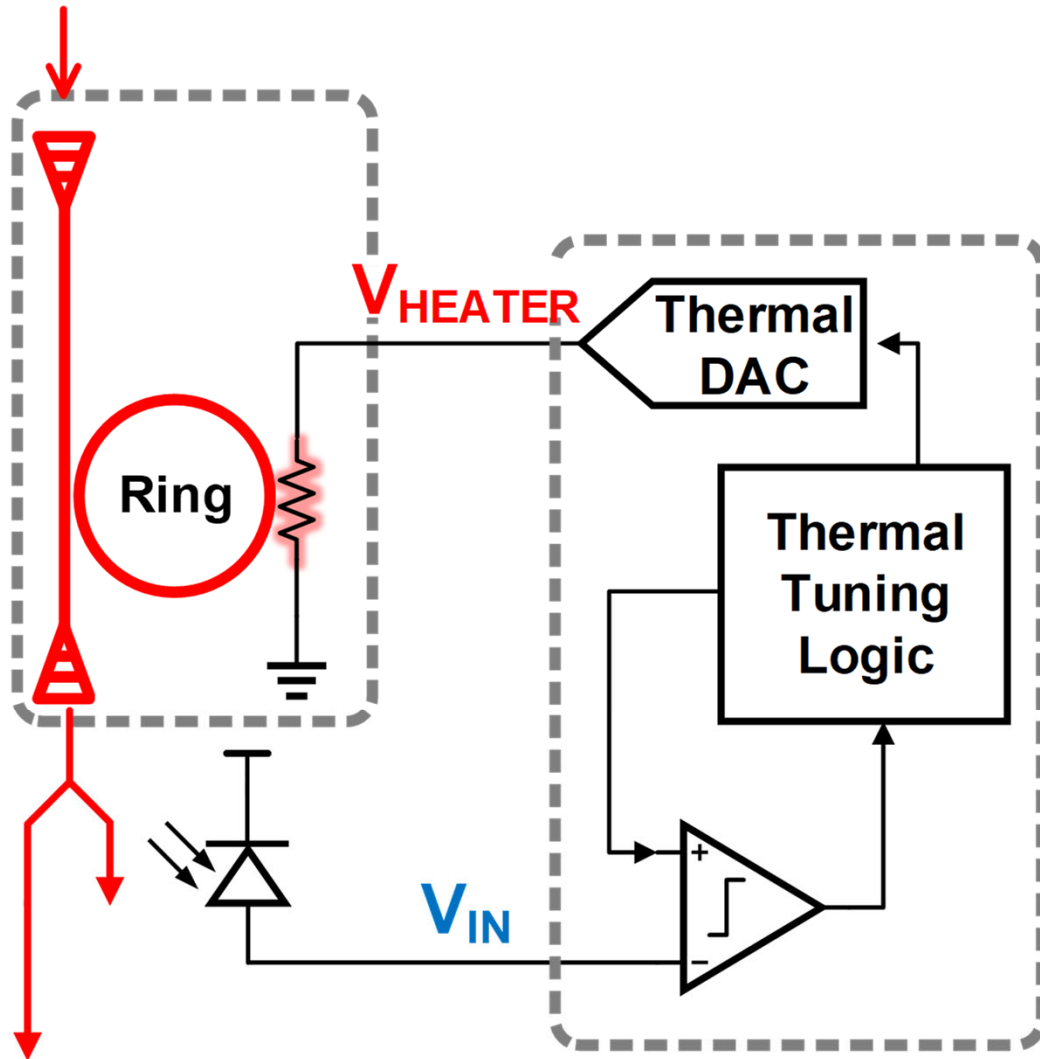
Thermal Tuning Algorithm



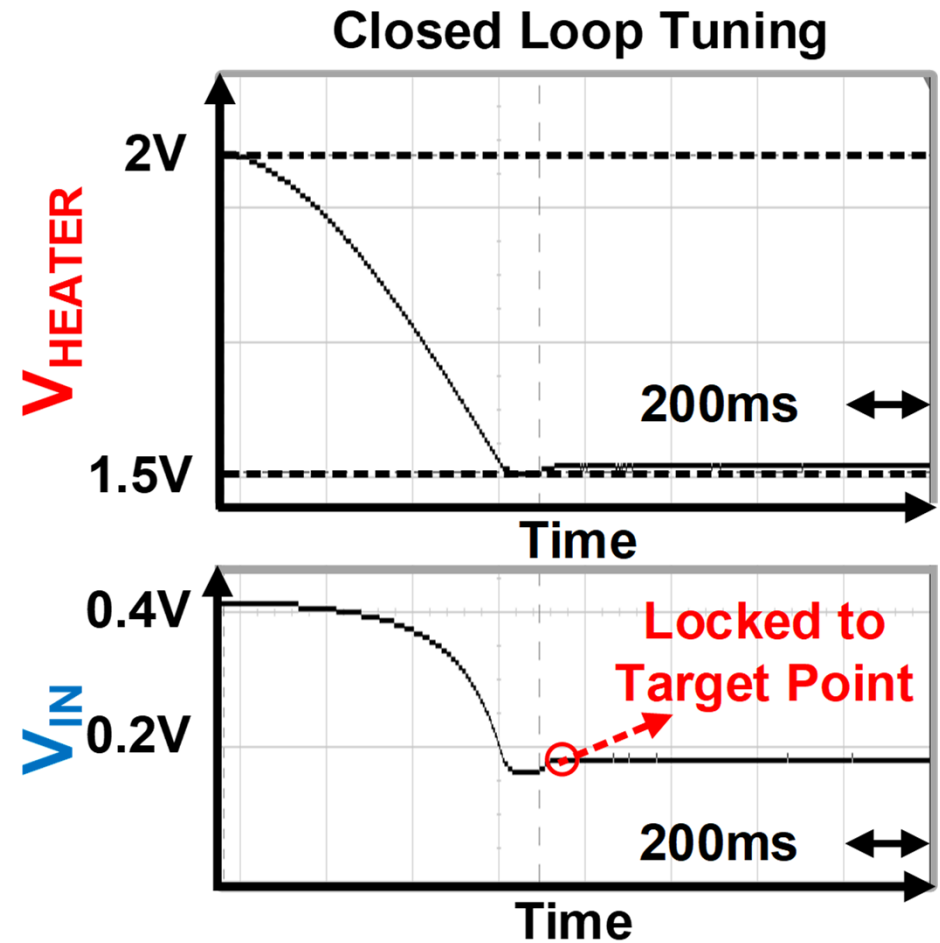
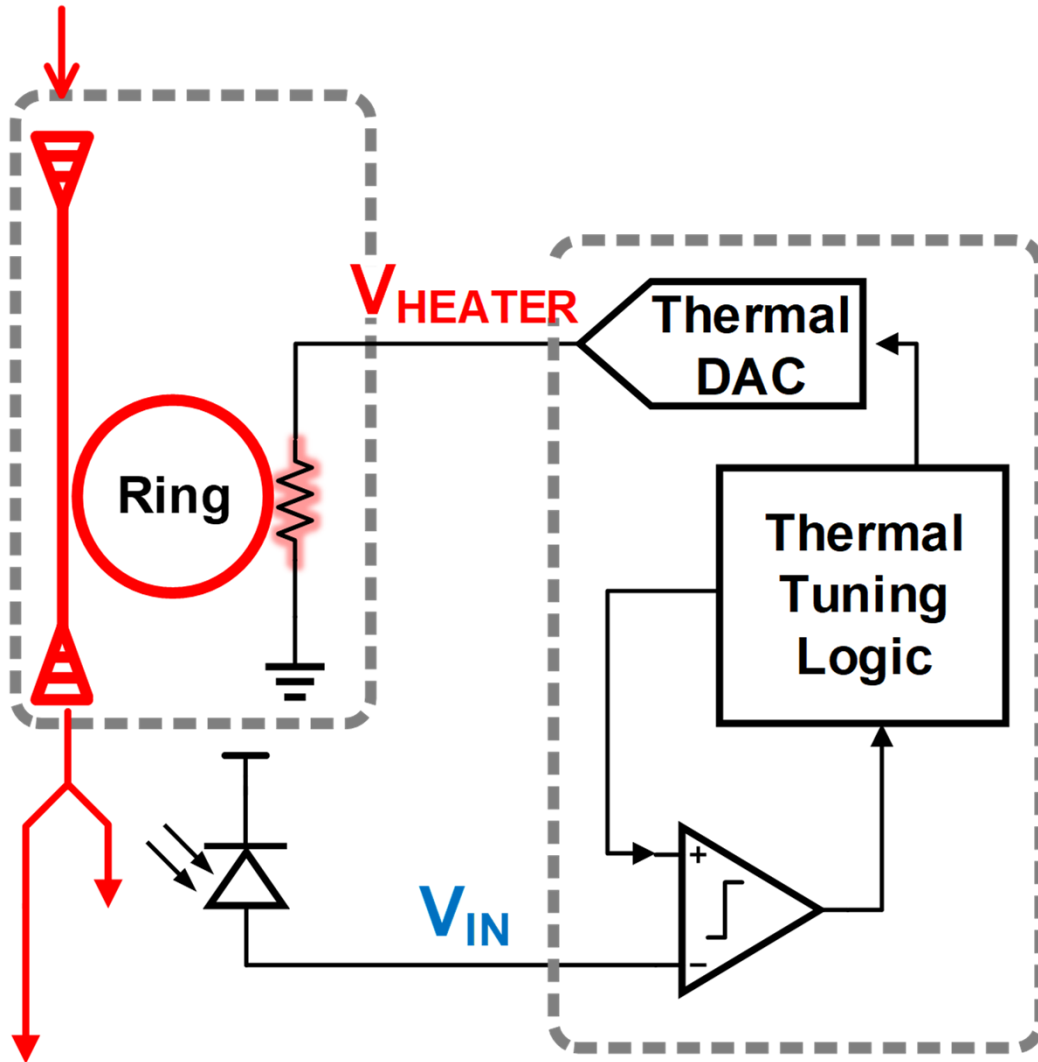
Thermal Tuning Algorithm



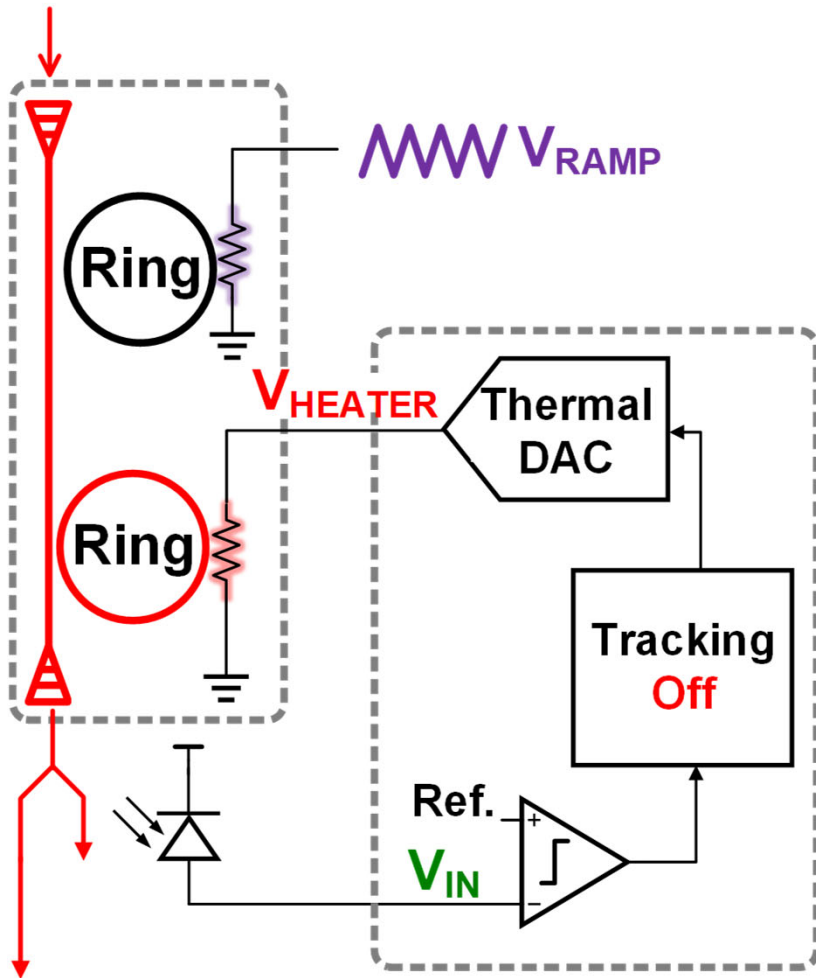
Thermal Tuning Test



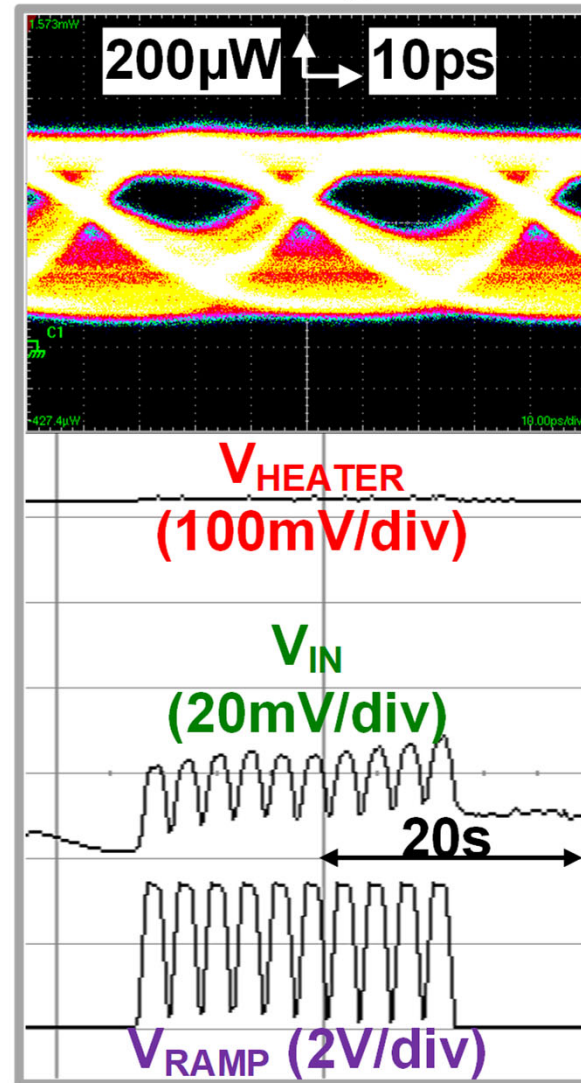
Thermal Tuning Test



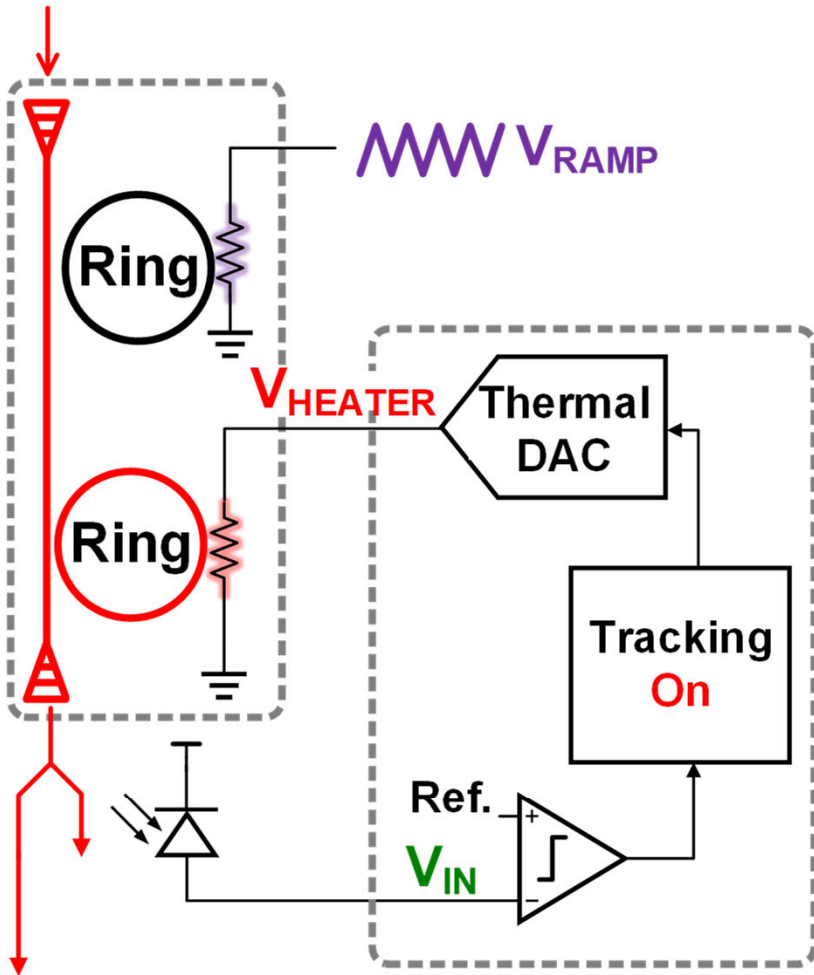
Dynamic Thermal Tracking Test



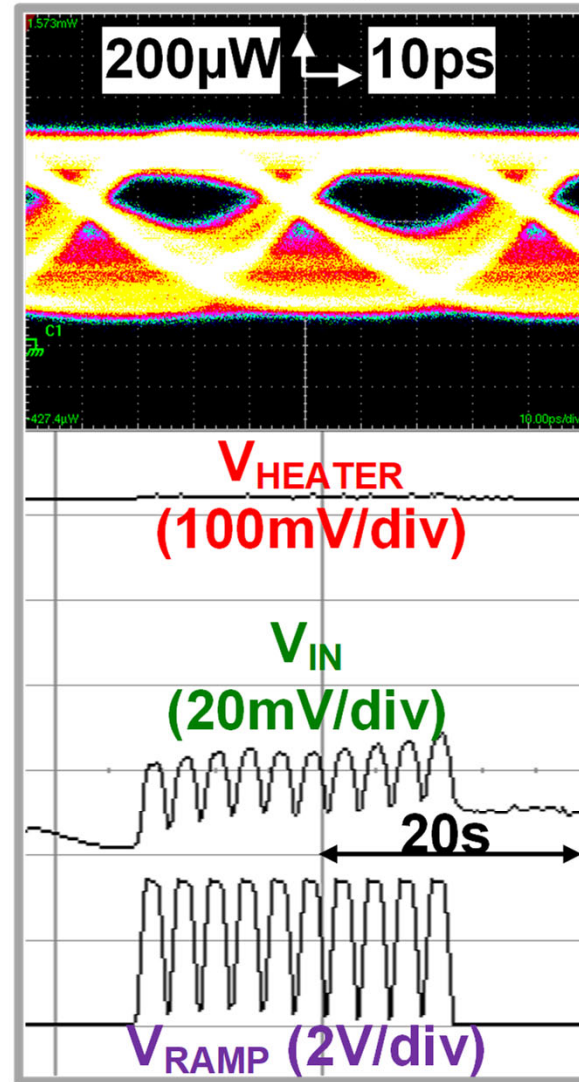
Tracking **Off**



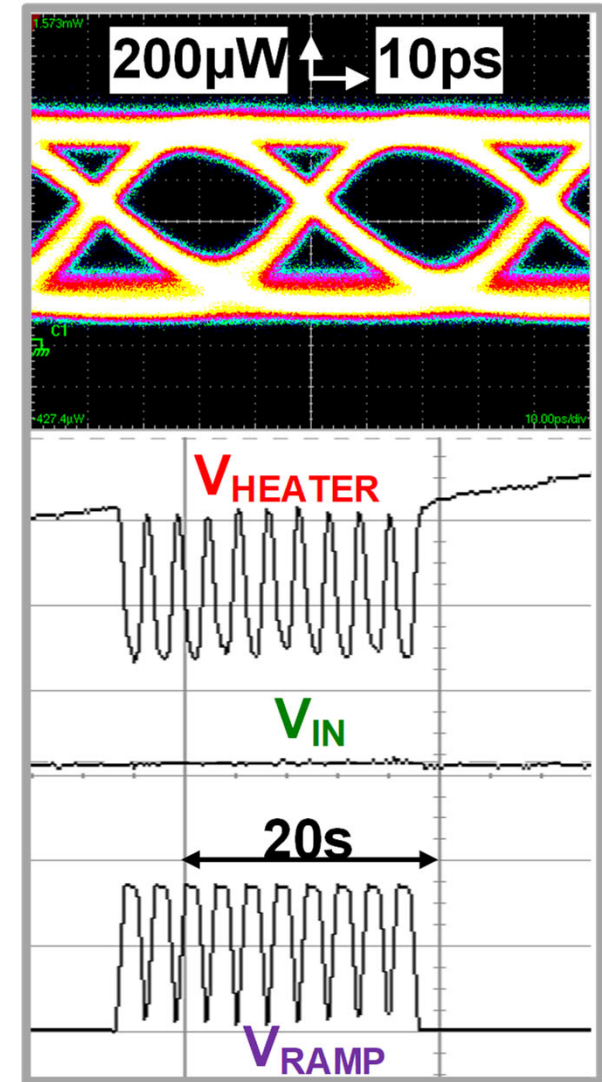
Dynamic Thermal Tracking Test



Tracking **Off**



Tracking **On**

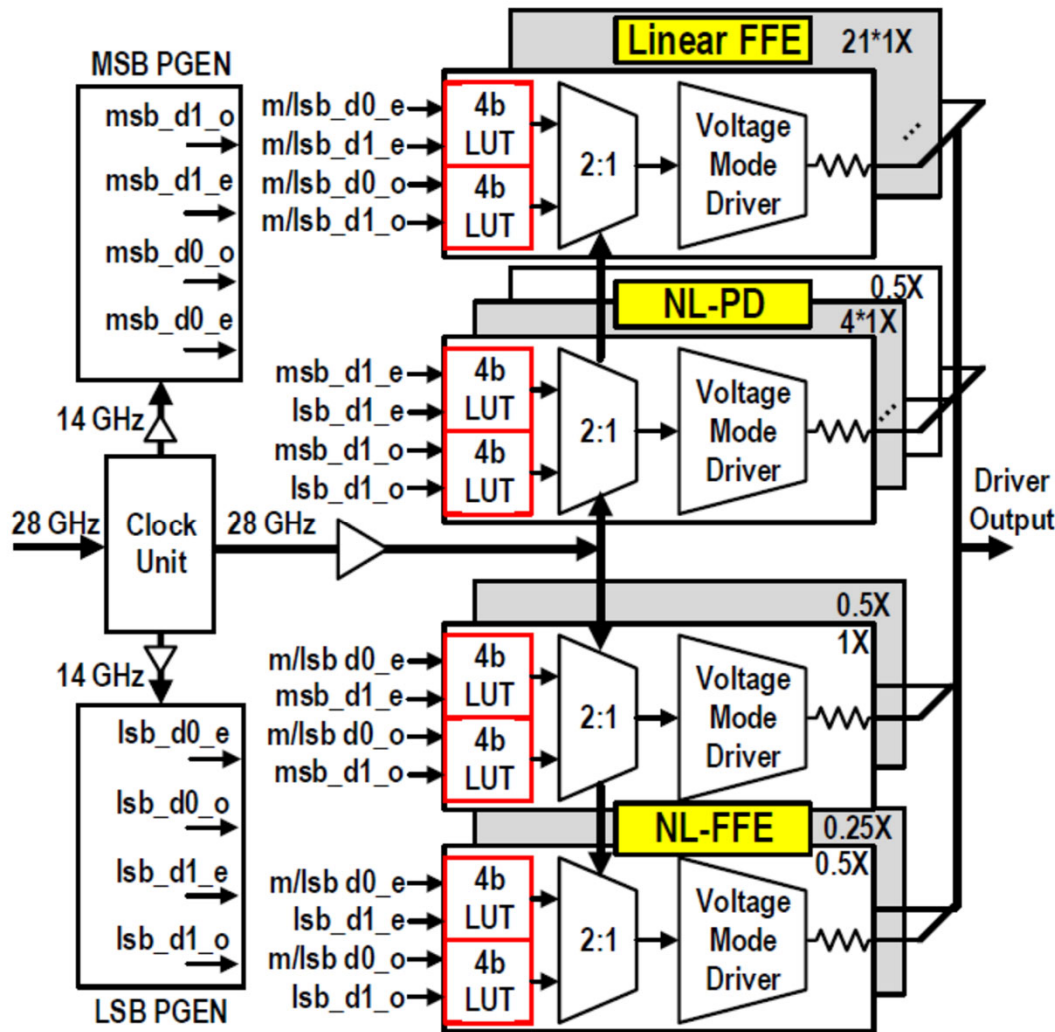


TX Performance Comparison

	Liu JSSC2012	Li ISSCC2013	Moss ISSCC2013	Buckwalter JSSC2012	This Work
Technology	40nm CMOS	65nm CMOS	45nm SOI	130nm SOI	65nm CMOS
Data Rate	10Gb/s	5Gb/s	2.5Gb/s	25Gb/s	25Gb/s
Integration	Flip Chip	Wire Bonding	Monolithic	Monolithic	Wire Bonding
MRM Type	Depletion	Injection	Injection	Depletion	Depletion
MRM Q	~15000	~8000	~4000	~13000	~5000
Supply	1V, 2V	1V, 2V	1.1V, 1.5V	+1.5V, -1.5V	1.2V, 2.4V
Channels	8	6	1	1	5
EQ	N/A	2-Tap FFE	2-Tap FFE	2-Tap FFE	2-Tap FFE
Swing	2V _{pp}	2V _{pp}	1.5V _{pp}	2.4V _{pp-diff}	4.4V _{pp-diff}
ER	7dB	12.7dB	3dB	6.5dB	7dB
MRM Stabilization	Static Thermal	Static Bias	N/A	N/A	Dynamic Thermal
Transmitter Power	*1.35mW	4.04mW	*3.07mW	*207mW	113.5mW

* w/o Serialization and Clocking

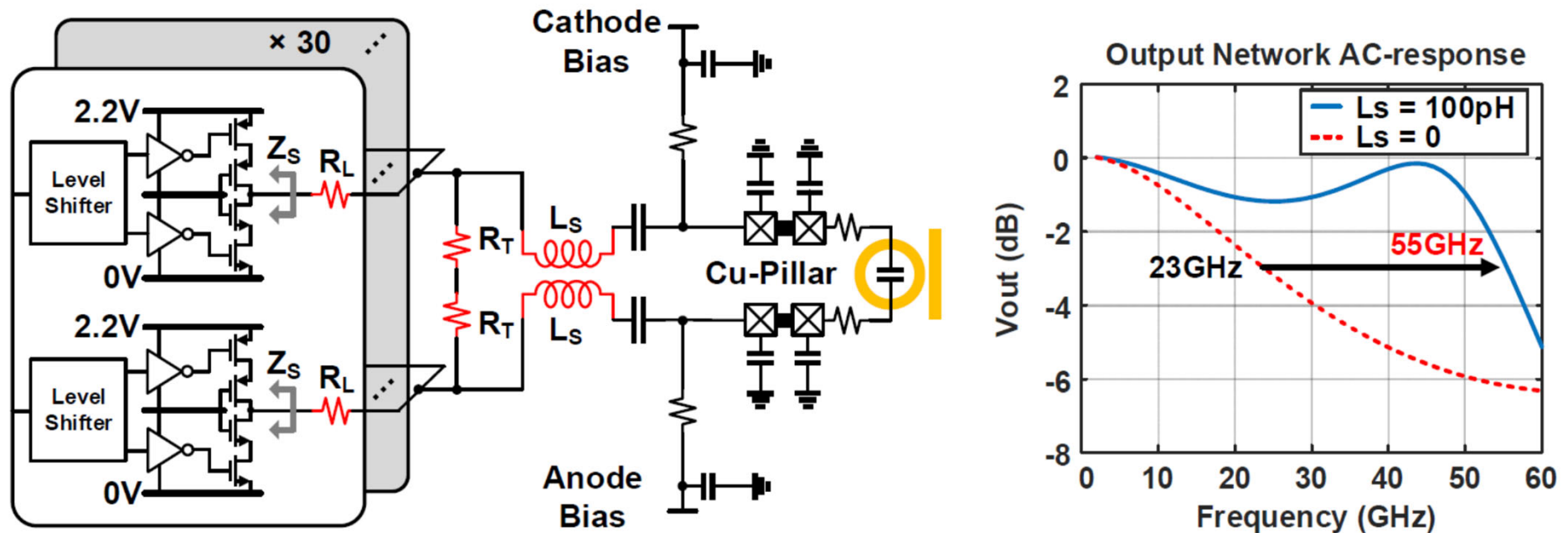
112Gb/s PAM4 Transmitter



[Li ISSCC 2020]

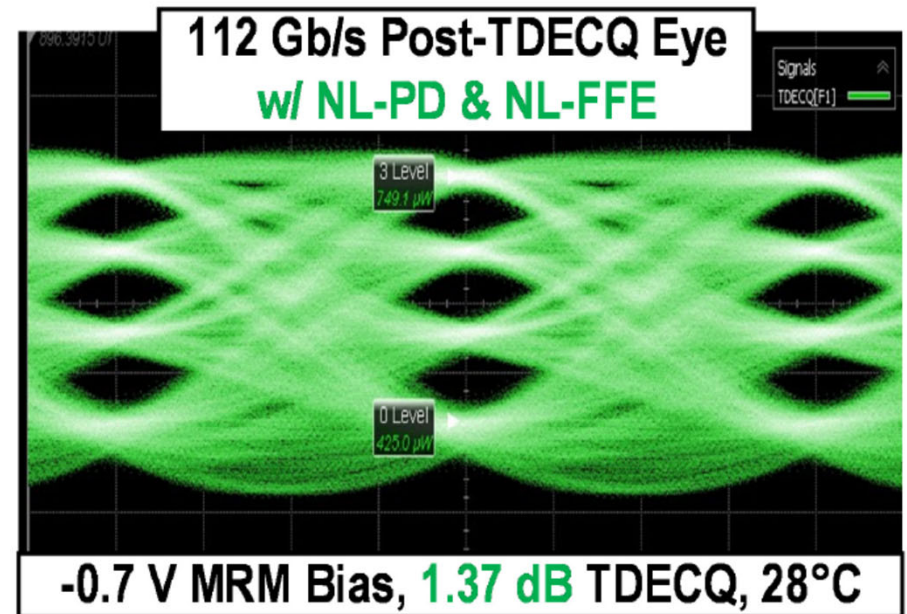
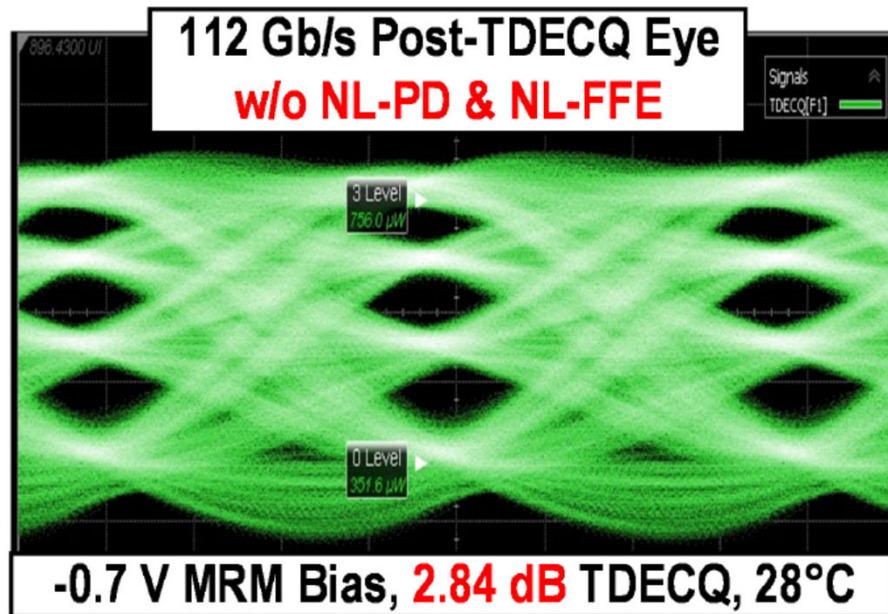
- Look-up table (LUT)
DAC-based output stage
- 2-tap linear FFE (21X slices)
- Non-linear static pre-distortion (4.5X slices)
- Nonlinear 2-tap FFE (2.25X slices)

112Gb/s DAC Output



- Differential cascade output driver w/ level shifted pre-drivers
- Per-slice series R_L and lumped shunt R_T improve linearity at the cost of reduced output swing ($3V_{ppd}$)
- Series peaking inductor provides significant bandwidth extension

112Gb/s PAM4 Eye Diagrams



- Utilizing only linear FFE results in significant eye skew and poor TDECQ
- Enabling the non-linear pre-distortion and FFE aligns the 3 eyes and improves TDECQ by ~ 1.5 dB

Next Time

- Laser Sources